Evaluating Local Structural-After-Measurement (LSAM) and Traditional Approaches for the Estimation of Complex Nonlinear Effects Among Latent Variables

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Abstract

Various methods exist to include quadratic or interaction terms involving latent variables in structural equation models (SEM). Most use a system-wide estimation approach, meaning all free parameters are estimated simultaneously. Research has investigated their performance in models with a limited number of nonlinear effects. Recently, several structural-after-measurement (SAM) approaches to handle latent quadratic and interaction terms have been proposed. In this case, estimation proceeds in two stages: first, measurement model parameters are estimated; second, structural model parameters are estimated. We present two simulation studies demonstrating that the recently proposed local SAM (LSAM) approach (Rosseel et al., 2025) performs well across a variety of conditions including distributional, reliability, and sample size manipulations. The first simulation study examined a model with three nonlinear effects under varying distributions (normal, right-skewed, uniform), reliabilities (0.4, 0.6, 0.8), and sample sizes (400, 1000). LSAM produced largely unbiased estimates with stable coverage, standard errors, and Type I error rates across most conditions. Power varied more across conditions. Performance was adversely affected under low reliability for some metrics, particularly with uniform distributions, though larger sample sizes mitigated these issues. Traditional methods (LMS, QML, and UPI) showed results consistent with prior literature. The second simulation study tested a more complex model with eight nonlinear effects under the same conditions to assess whether performance deteriorates with increased complexity. LSAM maintained adequate performance despite greater model complexity, though similar limitations emerged and power decreased under certain conditions. Traditional methods yielded more variable results, with generally poorer performance relative to Simulation 1 and LSAM.

Keywords: nonlinear effects, structural equation modeling, local structural-after-measurement, latent variables

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The estimation of interaction and quadratic effects in regression models with observed variables is a well-known procedure in behavioral and social sciences (Aiken & West, 1991; Olvera Astivia et al., 2025; Rimpler et al., 2025). In that case, an interaction is commonly calculated by the product of certain predictors, whereas a quadratic term involves the product of a predictor with itself. This allows researchers to study, for example, the dependency that one variable may have on the level of another variable, given the functional form at hand (Mize, 2019). However, when we move to structural equation models (SEM) with latent variables in the regression function (i.e., latent variable model or structural part; Bollen, 1989), this straightforward estimation procedure becomes considerably more complex. The triviality of estimating interaction and quadratic effects disappears because the scores (i.e., the subject-specific values) on the latent variables are, at the time of measurement, unobserved (Bollen & Hoyle, 2012).

Despite this complication, estimating latent interaction and quadratic terms remains important for several reasons. First, observed product terms are often less reliable than their components (Bohrnstedt & Marwell, 1978; Fisicaro & Lautenschlager, 1992), making measurement error a critical concern. SEM with latent variables addresses this issue by partitioning observed variance into latent factors and residual error, thus offering a possible advantage in estimating nonlinear effects, provided that the measurement model assumptions hold (Rhemtulla et al., 2020). Second, previous research has shown that quadratic and interaction effects may “mask” or distort one another when predictors are correlated, potentially leading to biased conclusions if not modeled simultaneously (Belzak & Bauer, 2019; Ganzach, 1997). Third, if theories are formalized (Oberauer & Lewandowsky, 2019), and they postulate nonlinear effects, suitable methods for estimating them must be available. Finally, some authors have argued that latent variable models might improve statistical power (Marsh et al., 2012), but this does not hold in general (Cham et al., 2012; Fuller, 1987).

In response to these challenges, a range of parametric and frequentist-based methods has been developed, each reflecting successive attempts to overcome the limitations identified in earlier approaches (Algina & Moulder, 2001; Boker et al., 2023; Bollen, 1995; Jöreskog & Yan, 1996; Kelava et al., 2014; Kenny & Judd, 1984; Klein & Moosbrugger, 2000; Klein & Muthén, 2007; Marsh et al., 2004; Mooijaart & Bentler, 2010; Mooijaart & Satorra, 2012; Wall & Amemiya, 2000, 2001, 2003). These methods differ considerably, however, both in their practical accessibility and in the assumptions required for their estimators to exhibit desirable properties (Kelava & Brandt, 2022). Their performance has been examined systematically in simulation studies across a wide range of conditions (e.g., Brandt et al., 2014, 2020; Cham et al., 2012; Lonati et al., 2024; Marsh et al., 2004). Importantly, most approaches adopt a joint ("system-wide") estimation strategy, in which all model parameters are estimated simultaneously. This includes the most widely implemented techniques: Latent Moderated Structural Equations (LMS), Quasi-Maximum Likelihood (QML), and Product Indicator (PI) methods.

In the PI approach, products of observed indicators serve as manifest indicators of latent nonlinear terms (interactions or quadratics) that enter the structural model (Kenny & Judd, 1984). Early constrained specifications imposed algebraic relations tying PI measurement to the linear measurement model, securing identification but restricting models to assumptions often violated by the nonnormality of indicator products (Kelava et al., 2011). Subsequent approaches removed most constraints, where relevant to this article is the Unconstrained PI (UPI) (Marsh et al., 2004) that was extended to models with both interaction and quadratic terms (Brandt et al., 2014; Kelava & Brandt, 2009). Research has also explored optimal strategies for constructing product indicators as well as centering alternatives, including mean-centering, residual-centering, and double mean-centering (Marsh et al., 2004; Schoemann & Jorgensen, 2021; Slupphaug et al., 2024).

The LMS and QML address normality violations without creating measurement models for nonlinear terms (Kelava et al., 2011; Klein & Moosbrugger, 2000; Klein & Muthén, 2007). Indeed, the LMS approximates nonnormal distributions using mixtures of normals, applying an expectation–maximization (EM) algorithm to numerically integrate weighted multivariate distributions (Klein & Moosbrugger, 2000). The QML offers a computationally efficient alternative by replacing nonnormal distributions with normal approximations preserving the same mean and variance, allowing estimation via a quasi log-likelihood with minimal efficiency loss (Kelava & Brandt, 2022; Klein & Muthén, 2007). Further extensions of LMS have also been proposed and evaluated (e.g., Cheung & Lau, 2017; Lonati et al., 2024; Shen et al., 2025).

Apart from system-wide estimation, several structural-after-measurement (SAM) approaches were proposed in the literature for estimating nonlinear effects (Cox & Kelcey, 2021; Ng & Chan, 2020; Wall & Amemiya, 2000, 2003). These methods follow the same practice when all variables are observed, using factor scores to form quadratic and interaction terms. However, factor scores retain measurement error, amplified in their products (Bohrnstedt & Marwell, 1978), which SAM methods explicitly account for in the structural model. Recently, Rosseel et al. (2025) proposed a local SAM (LSAM) approach for estimating latent interaction and quadratic effects. This approach is closely connected to the more general two-stage method of moments (2SMM) (Wall & Amemiya, 2000, 2003). In Rosseel et al. (2025), a small simulation study was conducted to evaluate the LSAM approach, which was limited to examining bias and variability of point estimates under a small set of conditions. Therefore, one aim of this paper is to conduct a thorough evaluation of LSAM under a wider set of conditions and performance criteria, such as analyzing standard errors (SEs), coverage, empirical Type I error rate, and local power (Irmer et al., 2024). A second goal is to compare LSAM's performance to that of LMS, UPI, and QML, while also providing updated benchmarks for these methods. Several considerations guide the design of these evaluations, which are outlined after the next section.

The rest of this paper is organized as follows. We first present an overview of the approaches for estimating latent interaction and quadratic effects, together with their distributional assumptions and known performance characteristics. We then outline the goals and contributions of the present research, followed by the simulation design, computational details, and performance measures before reporting results and concluding with a general discussion.

The General SEM With Latent Variables Framework

Let m be the number of latent variables and p the number of observed indicators. We assume the observed data are complete and continuous. Importantly, we use LISREL all-y notation. However, because the literature on LMS and QML traditionally uses standard LISREL notation, we will occasionally differentiate between exogenous $\boldsymbol{\eta_x}$ and endogenous $\boldsymbol{\eta_y}$ when referring to those methods. This distinction is not needed for the UPI approach.

The structural model can be written as:

$$\boldsymbol{\eta} = \boldsymbol{\alpha} + \mathbf{B}\boldsymbol{\eta} + \boldsymbol{\zeta}, \tag{1}$$

where $\boldsymbol{\eta} \in \mathbb{R}^m$ is a latent random vector, $\boldsymbol{\alpha} \in \mathbb{R}^m$ the intercepts, and $\mathbf{B} \in \mathbb{R}^{m \times m}$ a matrix with regression coefficients, where $\operatorname{diag}(\mathbf{B}) = \mathbf{0}$ and $(\mathbf{I} - \mathbf{B})$ assumed to be invertible by design. For the disturbance $\boldsymbol{\zeta} \in \mathbb{R}^m$, we assume $\mathbb{E}(\boldsymbol{\zeta}) = \mathbf{0}$ and write $\operatorname{Var}(\boldsymbol{\zeta}) = \boldsymbol{\Psi}$. Note that no distributional form is assumed here. Thus, we obtain:

$$\begin{aligned} \mathbb{E}(\boldsymbol{\eta}) &= (\mathbf{I} - \mathbf{B})^{-1}\boldsymbol{\alpha} \\ \operatorname{Var}(\boldsymbol{\eta}) &= (\mathbf{I} - \mathbf{B})^{-1}\boldsymbol{\Psi}\big((\mathbf{I} - \mathbf{B})^{-1}\big)^{\top}. \end{aligned}$$

The measurement model is defined as:

$$\mathbf{y} = \boldsymbol{\nu} + \boldsymbol{\Lambda}\boldsymbol{\eta} + \boldsymbol{\epsilon}, \tag{2}$$

where $\mathbf{y} \in \mathbb{R}^p$ is the indicator random vector, $\boldsymbol{\nu} \in \mathbb{R}^p$ the intercepts, $\boldsymbol{\Lambda} \in \mathbb{R}^{p \times m}$ the loadings, and $\boldsymbol{\epsilon} \in \mathbb{R}^p$ the measurement errors with the assumptions that $\mathbb{E}(\boldsymbol{\epsilon}) = \mathbf{0}$, $\operatorname{Cov}(\boldsymbol{\eta}, \boldsymbol{\epsilon}) = \mathbf{0}$, and $\mathbb{E}(\boldsymbol{\epsilon}\boldsymbol{\zeta}^{\top}) = \mathbf{0}$. We write $\operatorname{Var}(\boldsymbol{\epsilon}) = \boldsymbol{\Theta}$. The model-implied mean vector and covariance matrix of $\mathbf{y}$ are:

$$\begin{aligned} \boldsymbol{\mu} &= \boldsymbol{\nu} + \boldsymbol{\Lambda}\,\mathbb{E}(\boldsymbol{\eta}) \\ \boldsymbol{\Sigma} &= \operatorname{Var}(\mathbf{y}) = \boldsymbol{\Lambda}\operatorname{Var}(\boldsymbol{\eta})\,\boldsymbol{\Lambda}^{\top} + \boldsymbol{\Theta}. \end{aligned} \tag{3}$$

Traditionally, we collect the free parameters across the structural part $(\boldsymbol{\alpha}, \mathbf{B}, \boldsymbol{\Psi})$ and the measurement part $(\boldsymbol{\nu}, \boldsymbol{\Lambda}, \boldsymbol{\Theta})$ into $\boldsymbol{\theta}$. Given i.i.d. data vectors $\mathcal{Y} = \{\mathbf{y}_1, \ldots, \mathbf{y}_N\}$, we estimate $\boldsymbol{\theta}$ by minimizing the discrepancy between observed and model-implied moments. Thus, estimation proceeds as a joint ('system-wide') approach where all parameters are estimated jointly. Crucially, distributional assumptions are not inherent to the estimation framework itself. Normality is introduced most explicitly when one adopts a likelihood-based estimator such as normal-theory maximum likelihood (ML) (Browne, 1973; Shapiro, 2007). In that case, we assume $\boldsymbol{\zeta} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Psi})$ and $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Theta})$. Since $\boldsymbol{\eta} = (\mathbf{I} - \mathbf{B})^{-1}(\alpha + \boldsymbol{\zeta})$ is a linear transformation of $\boldsymbol{\zeta}$, this implies (e.g., Theorem 2.4.1 in Anderson, 2003):

$$\boldsymbol{\eta} \sim \mathcal{N}\Big((\mathbf{I} - \mathbf{B})^{-1}\alpha,\ (\mathbf{I} - \mathbf{B})^{-1}\boldsymbol{\Psi}\big((\mathbf{I} - \mathbf{B})^{-1}\big)^{\top}\Big)$$

and, consequently $\mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Approaches for Estimating Latent Interactions and Quadratic Terms

LSAM

The central idea of the LSAM is to use the estimated measurement-model parameters with observed moments to construct the latent mean $\mathbb{E}(\boldsymbol{\eta})$ and covariance $\mathrm{Var}(\boldsymbol{\eta})$ (Rosseel & Loh, 2024). For that, we start from the measurement model in Equation 2 and solve for $\boldsymbol{\eta}$:

$$\boldsymbol{\Lambda}\boldsymbol{\eta} = \mathbf{y} - \boldsymbol{\nu} - \boldsymbol{\epsilon}.$$

As in Rosseel and Loh (2024), we let $\mathbf{M} \in \mathbb{R}^{m \times p}$ be a mapping matrix with $\mathbf{M}\boldsymbol{\Lambda} = \mathbf{I}_m$[1]. Multiplying both sides by $\mathbf{M}$ gives:

$$\boldsymbol{\eta} = \mathbf{M}\left[\mathbf{y} - \boldsymbol{\nu} - \boldsymbol{\epsilon}\right] = \mathbf{M}(\mathbf{y} - \boldsymbol{\nu}) \ - \ \mathbf{M}\boldsymbol{\epsilon},$$

thus defining $\mathbf{f} = \mathbf{M}(\mathbf{y} - \boldsymbol{\nu}) = \boldsymbol{\eta} + \mathbf{r}$ with $\mathbf{r} = \mathbf{M}\boldsymbol{\epsilon} \in \mathbb{R}^m$, $\mathbb{E}(\mathbf{r}) = \mathbf{0}$, and $\mathrm{Var}(\mathbf{r}) = \mathbf{M}\boldsymbol{\Theta}\mathbf{M}^{\top}$. Then:

$$\mathbb{E}(\boldsymbol{\eta}) = \mathbf{M}\left[\mathbb{E}(\mathbf{y}) - \boldsymbol{\nu}\right],$$

$$\mathrm{Var}(\boldsymbol{\eta}) = \mathbf{M}\left[\mathrm{Var}(\mathbf{y}) - \boldsymbol{\Theta}\right]\mathbf{M}^{\top}.$$

[1] This left inverse exists only when $\boldsymbol{\Lambda}$ has full column rank ($p \geq m$). Moreover, if $\boldsymbol{\Lambda}$ is poorly conditioned (e.g., due to very weak loadings), computing $\mathbf{M}$ might become numerically unstable and potentially amplify measurement error, since $\mathrm{Var}(\mathbf{r}) = \mathbf{M}\boldsymbol{\Theta}\mathbf{M}^{\top}$.

However, in the case of a model with latent interaction and quadratic terms, we need an expression for $\mathbb{E}(\boldsymbol{\eta} \otimes \boldsymbol{\eta})$ and $\mathrm{Var}(\boldsymbol{\eta} \otimes \boldsymbol{\eta})$ without assuming normality for the latent variables (Rosseel et al., 2025). Define $\boldsymbol{\eta}^* = (1, \eta_1, \eta_2, \ldots, \eta_m)^T$, it holds that:

$$\mathbb{E}(\boldsymbol{\eta}^* \otimes \boldsymbol{\eta}^*) = \mathrm{vec}\big[\mathrm{Var}(\boldsymbol{\eta}^*)\big] + \mathbb{E}(\boldsymbol{\eta}^*) \otimes \mathbb{E}(\boldsymbol{\eta}^*),$$

$$\mathrm{Var}(\boldsymbol{\eta}^* \otimes \boldsymbol{\eta}^*) \approx \mathrm{Var}(\mathbf{f}^* \otimes \mathbf{f}^*) - \Big[\mathbf{Q}^* + \mathbf{K}_{m+1}\mathbf{Q}^* + \mathbf{Q}^*\mathbf{K}_{m+1}^\top + \mathbf{K}_{m+1}\mathbf{Q}^*\mathbf{K}_{m+1}^\top + \boldsymbol{\Gamma}_{22}^{*(NT)}(\mathbf{r}^*)\Big],$$

where

$$\mathbf{Q}^* = \mathrm{Var}(\boldsymbol{\eta}^*) \otimes \mathrm{Var}(\mathbf{r}^*) + \mathbb{E}(\boldsymbol{\eta}^*)\mathbb{E}(\boldsymbol{\eta}^*)^T \otimes \mathrm{Var}(\mathbf{r}^*),$$

and

$$\boldsymbol{\Gamma}_{22}^{\star(NT)}(\mathbf{r}^*) = (\mathbf{I}_{(m+1)^2} + \mathbf{K}_{m+1})(\mathrm{Var}(\mathbf{r}^*) \otimes \mathrm{Var}(\mathbf{r}^*)).$$

Here, $\mathbf{K}_{m+1}$ denotes the commutation matrix (Magnus & Neudecker, 1979), while $\boldsymbol{\Gamma}_{22}^{\star(NT)}(\mathbf{r}^*)$ represents the variance–covariance component of the Gamma matrix (see Appendix B in Rosseel et al., 2025) corresponding to the measurement error ($\mathbf{r}^*$) under the normality assumption. These estimated augmented summary statistics can then be employed in the structural regression model during the second stage of estimation.

Distributional Assumptions and Performance. Similar to the 2SMM approach, in the first stage, the LSAM inherits the CFA measurement model assumptions (Jöreskog, 1969). Distributional assumptions in the CFA stage depend on the estimator used (Rosseel & Loh, 2024). Concerning the second stage, specifically for the second-order corrections, LSAM assumes only $\mathbf{r} \sim \mathcal{N}(\mathbf{0}, \mathbf{M}\boldsymbol{\Theta}\mathbf{M}^\top)$. No distributional assumptions are imposed on the latent predictors $\boldsymbol{\eta}$. For the local two-step standard errors (SEs), a sandwich-based formula was developed to account for the uncertainty across both steps (see Appendix A). Rosseel et al. (2025) compared LSAM against UPI, LMS, 2SMM, and other SAM approaches. Under normal conditions, LSAM exhibited similar bias to all other methods and achieved MSE close to that of LMS. Under nonnormal conditions, LSAM maintained low bias, similarly to UPI, whereas LMS showed increased bias. Across all conditions, LSAM performed very similarly to 2SMM. Previous simulation studies investigating the 2SMM approach provide relevant benchmarks for other performance components (Brandt et al., 2014; Wall & Amemiya, 2000, 2003). That is, the 2SMM approach produced unbiased and efficient estimates under normal predictors, with only minor underestimation of SEs under nonnormal predictors, consistent with its robustness claims (Brandt et al., 2014; Wall & Amemiya, 2003). In fact, it showed less bias than the LMS and QML under the nonnormal condition for quadratic terms (see Table 9 in Brandt et al., 2014). With respect to Type I error rate, Brandt et al. (2014) reported that the 2SMM produced rates within the nominal level under normality with slight inflation under nonnormality. We do not comment extensively on previous power results, as these are inherently dependent on the specific sample size and effect size conditions employed in each study, but acknowledge that such results are available (see Tables 6-9 in Brandt et al., 2014). Importantly, these previous studies examined mainly cases with high reliability (e.g., approximately 0.7).

LMS and QML

Following the equations implemented in *modsem* (Jin et al., 2020; Slupphaug et al., 2024), here we let m_y be the number of endogenous latents and m_x the number of exogenous latent predictors, with $m = m_y + m_x$. Stack $\boldsymbol{\eta} = \begin{bmatrix} \boldsymbol{\eta}_y \\ \boldsymbol{\eta}_x \end{bmatrix}$ with $\boldsymbol{\eta}_y \in \mathbb{R}^{m_y}$ and $\boldsymbol{\eta}_x \in \mathbb{R}^{m_x}$, and collect the p indicators in $\mathbf{y} \in \mathbb{R}^p$. The same measurement model in Equation 2 holds using all-y notation with the following block structure:

$$\mathbf{B} = \begin{bmatrix} \mathbf{B}_{yy} & \mathbf{B}_{yx} \\ \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad \boldsymbol{\alpha} = \begin{bmatrix} \boldsymbol{\alpha}_y \\ \boldsymbol{\alpha}_x \end{bmatrix}, \quad \boldsymbol{\zeta} = \begin{bmatrix} \boldsymbol{\zeta}_y \\ \boldsymbol{\eta}_x - \boldsymbol{\alpha}_x \end{bmatrix}, \quad \boldsymbol{\Psi} = \mathrm{Var}(\boldsymbol{\zeta}) = \begin{bmatrix} \boldsymbol{\Psi}_{yy} & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Psi}_{xx} \end{bmatrix},$$

where $\mathbf{B}_{yy} \in \mathbb{R}^{m_y \times m_y}$, $\mathbf{B}_{yx} \in \mathbb{R}^{m_y \times m_x}$.

Following Slupphaug et al. (2024), we write the endogenous block explicitly,

$$\boldsymbol{\eta}_y = \boldsymbol{\alpha}_y + \mathbf{B}_{yy}\boldsymbol{\eta}_y + \mathbf{B}_{yx}\boldsymbol{\eta}_x + \left(\mathbf{I}_{m_y} \otimes \boldsymbol{\eta}_x^\top\right)\boldsymbol{\Omega}\,\boldsymbol{\eta}_x + \left(\mathbf{I}_{m_y} \otimes \boldsymbol{\eta}_x^\top\right)\boldsymbol{\Xi}\,\boldsymbol{\eta}_y + \boldsymbol{\zeta}_y, \tag{4}$$

where $\mathbf{I}_{m_y}$ is the $m_y \times m_y$ identity matrix, $\boldsymbol{\Omega} \in \mathbb{R}^{m_y m_x \times m_x}$ containing the coefficients for the interaction effects between exogenous variables, and $\boldsymbol{\Xi} \in \mathbb{R}^{m_y m_x \times m_y}$ for the interaction effects between exogenous and endogenous variables. Equation 4 can also be expressed in reduced form (Slupphaug et al., 2024)

Importantly, the distributional assumptions for LMS and QML are that $\boldsymbol{\eta}_x \sim \mathcal{N}(\boldsymbol{\alpha_x}, \boldsymbol{\Psi}_{xx})$, $\boldsymbol{\zeta}_y \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Psi}_{yy})$, and $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Theta})$ with mutual independence between predictors and disturbances. For LMS, these assumptions are required for the exact likelihood (Klein & Moosbrugger, 2000). For QML, they serve as working assumptions to derive conditional mean and variance functions (Klein & Muthén, 2007). The latter approach can still yield consistent estimates when predictors deviate from normality, as long as the conditional normality approximation is reasonable. Hence, both procedures target the same parameterization, but they differ in estimation strategy (Kelava et al., 2011). Briefly, for the LMS, let $\boldsymbol{\Psi}_{xx} = \mathbf{A}\mathbf{A}^\top$ and write $\boldsymbol{\eta}_x = \boldsymbol{\alpha_x} + \mathbf{A}\mathbf{z}$ with $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{m_x})$. Integrating over $\boldsymbol{\eta}_x$ via Gauss-Hermite quadrature yields a finite normal–mixture approximation to $f(\mathbf{y})$. Parameters are estimated by full-information ML (EM), with SEs derived from the Fisher information matrix (Klein & Moosbrugger, 2000), although robust (sandwich) SEs are also available. For the QML, a fixed full-rank linear transformation $\mathbf{T}$ of $\mathbf{y}$ is applied to isolate the nonnormal component while the remaining transformed components are (conditionally) normal. In practice, this means that only one transformed indicator per nonlinear latent criterion retains nonnormality, while the remaining $p - 1$ transformed indicators are jointly normal and serve as conditioning variables. A quasi log-likelihood is maximized and SEs were originally proposed via the sandwich estimator (Klein & Muthén, 2007); SEs based on the observed Fisher information are also available.

Distributional Assumptions and Performance. Assuming the measurement model is correctly specified, under correctly specified distributional assumptions, LMS produces estimates that are consistent, asymptotically unbiased, normally distributed, and efficient (Klein & Moosbrugger, 2000). QML is consistent and asymptotically normal, but not guaranteed to be fully efficient (Klein & Muthén, 2007). Simulations show that LMS yields unbiased estimates when $\boldsymbol{\eta}_x$ is normally distributed, but becomes biased under nonnormality (Brandt et al., 2014), with bias increasing at larger sample sizes depending on the degree of nonnormality (Cham et al., 2012). Recent work by Lonati et al. (2024), which also manipulated distributions for structural disturbances and measurement errors, found that the latter had minimal negative impact. Under normality, QML is slightly less efficient than LMS but still performs adequately (Brandt et al., 2014). When latent predictors are nonnormal, QML is theoretically more robust than LMS (Kelava et al., 2014), and although it shows some bias (Marsh et al., 2004), this bias is generally smaller than that of LMS, with differences remaining modest (Brandt et al., 2014). For Type I error rates, Brandt et al. (2014) found some inflation under nonnormality but nominal rates under normality for both, while Cham et al. (2012) reported substantial inflation for LMS under different nonnormal distributions. Similarly, Marsh et al. (2004) observed high Type I error rates for QML. For power, we also refer to previous studies (Brandt et al., 2014; Marsh et al., 2004). To the best of our knowledge, coverage results are available only for LMS, which shows deterioration under nonnormality, varying by distribution (Cham et al., 2012), with similar negative findings when nonnormality was induced through ordered-categorical indicators (Aytürk et al., 2020).

UPI

For this article, we focus on the extended UPI (Kelava & Brandt, 2009) with double mean-centering (Lin et al., 2010) and all-pairs strategy (Marsh et al., 2004). The general SEM in Equation 1–Equation 3 remains unchanged; we only augment the measurement block with products of centered indicators. For each y_i, set $z_i = y_i - \bar{y}_i$; for any relevant pair (i, j), define $p_{ij} = z_i z_j - \overline{z_i z_j}$. Let $\mathbf{p}$ collect the p_{ij} and write

$$\begin{bmatrix} \mathbf{y} \\ \mathbf{p} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\nu} \\ \mathbf{0} \end{bmatrix} + \begin{bmatrix} \boldsymbol{\Lambda} \\ \mathbf{0} \end{bmatrix} \boldsymbol{\eta} + \begin{bmatrix} \mathbf{0} \\ \boldsymbol{\Lambda}_p \end{bmatrix} f(\boldsymbol{\eta}) + \begin{bmatrix} \boldsymbol{\epsilon} \\ \boldsymbol{\epsilon}_p \end{bmatrix}.$$

where $f(\boldsymbol{\eta})$ collects the latent nonlinear terms measured by $\mathbf{p}$.

Under this extended UPI, the product–indicator block is unconstrained: $\boldsymbol{\Lambda}_p$ and $\mathrm{Var}(\boldsymbol{\epsilon}_p)$ are freely estimated. Additionally, residual covariances are freely estimated between product indicators that share a common constituent indicator (Kelava & Brandt, 2009; Marsh et al., 2004).

Distributional Assumptions and Performance. Also assuming the measurement model is correctly specified, the standard estimation approach relies on ML (Marsh et al., 2004; Yang, 1998), which requires multivariate normality for valid inference. Yet, product indicators are intrinsically nonnormal. The ML estimator itself is consistent under different distributions and achieves asymptotic normality when data follow a multivariate normal distribution (Yuan et al., 2005). Theoretical evidence indicates that such nonnormality primarily undermines the accuracy of standard errors and fit statistics (Wall & Amemiya, 2001; see Lonati et al., 2024 and references therein). Simulation studies of nonlinear effects generally show minimal parameter bias under both normal and nonnormal conditions (Brandt et al., 2014; Kelava et al., 2014; Lonati et al., 2024; Marsh et al., 2004). However, standard errors are systematically underestimated (Brandt et al., 2014; Cham et al., 2012; Kelava et al., 2014; Lonati et al., 2024; Marsh et al., 2004). Previous research has used robust SEs to alleviate this problem, but because they are based on asymptotic theory, they remain underestimated in smaller samples (Brandt et al., 2014, 2020; Cham et al., 2012; Lonati et al., 2024). Findings for Type I error rates are mixed: some studies report inflation (Brandt et al., 2014), while others find control close to nominal levels (Cham et al., 2012; Marsh et al., 2004). As before, see previous studies for power results (Brandt et al., 2014; Cham et al., 2012; Marsh et al., 2004). Generally speaking, results for UPI approaches under ordinal data lead to similar conclusions (Aytürk et al., 2020; Lodder et al., 2021). Finally, it is important to note that the UPI with double mean-centering has not been systematically examined in prior simulation work (e.g., Foldnes & Hagtvet, 2014; Lin et al., 2010).

Present Research and Contributions

First, as highlighted above, a comprehensive evaluation of LSAM beyond bias and variability of point estimates was not previously feasible because analytical SEs had not been derived. To enable this, we developed and present an analytical formula for calculating SEs that accounts for the uncertainty of both estimation steps in the LSAM approach (see Appendix A).

Second, earlier simulation work comparing LMS, QML, and UPI was limited to commercial software or error-prone implementations with a slow estimation process (Slupphaug et al., 2024). Recent developments in open-source software allow us to address these limitations (Slupphaug et al., 2024), making this also the first simulation to be based solely on open-source implementations. Additionally, simulation designs in previous work lacked the ability to generate data with exact covariance control while also allowing for a nonnormal copula. Previous simulations have nearly exclusively relied on the Vale and Maurelli (1983) method for nonnormal data generation, despite its known issues (Foldnes & Grønneberg, 2015; Olvera Astivia & Zumbo, 2015). The only exception is Lonati et al. (2024), who adopted the approach in Mair et al. (2012), which nonetheless distorts both the marginal distributions and the copula structure when enforcing the target covariance matrix (Grønneberg et al., 2022). Recent data generation tools (Grønneberg et al., 2022) allow us to overcome these limitations.

Third, the literature often suggests that the LMS, QML, and UPI face "scalability" limitations as higher-order terms increase (Kelava et al., 2011; Rosseel et al., 2025), though the extent remains unclear. On the other hand, the LSAM, in principle, should be less affected (Rosseel et al., 2025)[2]. Here, we interpret scalability broadly to encompass not only degradation in statistical performance (as measured by specific evaluation criteria), but also increases in computation time and convergence failures. This may explain why previous literature focused on elementary interaction models (EIM) (Jaccard & Wan, 1995), with few studies examining models with up to three nonlinear terms under a limited amount of conditions (Brandt et al., 2014; Kelava et al., 2014; Klein & Muthén, 2007). In fact, little work has addressed statistical models with multiple endogenous variables for LMS and QML (Jin et al., 2020), requiring specific programming modifications (Slupphaug et al., 2024). However, researchers are estimating (or trying to estimate) much more complex models in practice (Drum et al., 2017; Hamilton et al., 2024; Li et al., 2019). To address this, we consider models with three and eight nonlinear effects among latent variables. We note, however, that LMS was excluded from the latter case due to computational constraints. This will be further discussed as it is related to the scalability issue.

[2] Following one of the previous sections, for LMS, the dimensionality of the numerical integration via Gauss-Hermite quadrature grows with the number of exogenous predictors involved in nonlinear terms, substantially increasing computational cost. For QML, while less affected than LMS, estimation complexity still increases as the conditional mean and variance functions expand. For UPI, additional nonlinear terms require additional product indicators, increasing the number of parameters to estimate. For LSAM, the measurement model estimation remains unchanged regardless of the number of nonlinear terms — these only enter in the second-stage structural regression on augmented summary statistics.

Last, while some simulation design factors and performance metrics considered here have been examined for some approaches, they have not been systematically investigated for others. Specifically, as discussed, for UPI with double mean-centering, no systematic simulation evaluation has been conducted. For QML, to the best of our knowledge, coverage has never been reported, and Type I error findings remain mixed across methods. Additionally, the effect of different levels of reliability has not been extensively investigated. Furthermore, the separate manipulation of measurement error and structural disturbance distributions has only been investigated in Lonati et al. (2024), which we extend here.

Simulation Design and Performance Evaluation

In this section, we present the simulation design and performance metrics for the two simulation studies. The information presented here follows the suggestions from previous articles that investigated the quality of simulation studies (Morris et al., 2019; Pawel et al., 2025; Siepe et al., 2024; White et al., 2024).

Simulation Conditions

Both simulations used a 2 (sample sizes) × 3 (reliability levels) × 3 (latent predictor distributions) × 2 (measurement error distributions) × 2 (structural disturbance distributions) factorial design, yielding 72 conditions. Simulation 1 evaluated UPI, LMS, QML, and LSAM; Simulation 2 excluded LMS due to computational constraints[3]. To assess empirical Type I error rates and local power, we tested two specifications: (a) linear model data generation with full model estimation (assessing Type I error), and (b) full nonlinear model for both generation and estimation (assessing power). This yielded 144 conditions per simulation, each replicated 1,000 times.

Sample size was selected as a design factor because the estimators under comparison rely

[3] For illustration purposes, with *microbenchmark* (Mersmann, 2024) (version 1.5.0) on the same operating system as detailed in Reproducibility and Computational Details, using the dataset for the first iteration within the first condition for Simulation 2, it took approximately 29 minutes (1729.536 seconds) to fit a single model with LMS with 32 nodes (see *modsem* documentation for more information about nodes).

on fundamentally different estimation frameworks which may exhibit differential sensitivity to sample size due to their distinct asymptotic properties, finite-sample behavior, and number of parameters to be estimated (Kelava et al., 2011; Klein & Moosbrugger, 2000; Klein & Muthén, 2007; Rosseel et al., 2025). Sample sizes of 400 and 1000 represented moderate to large samples. We selected 400 to examine performance below the suggested 500-threshold minimum for complex latent interaction models (Brandt et al., 2020; Wall et al., 2012) while remaining realistic for applied research. The 1000 sample size allowed evaluation of behavior approaching asymptotic properties and aligned with previous simulations (e.g., Cham et al., 2012; Lonati et al., 2024; Wall & Amemiya, 2003).

As mentioned, the effect of different levels of reliability has not been as extensively investigated in prior simulation studies, yet it is particularly relevant for latent variable models where measurement conditions directly affect estimation (Bollen, 1989) and measurement error may be amplified in product terms (Bohrnstedt & Marwell, 1978). Previous simulations typically examined reliability above 0.7 (Brandt et al., 2014; Kelava et al., 2014; Wall & Amemiya, 2003); two studies contrasted high versus low reliability (Brandt et al., 2020; Lonati et al., 2024), but in Brandt et al. (2020) the effect of reliability was conditional on structural misspecifications. To extend on their findings, we include three reliability levels (0.4, 0.6, 0.8—termed low, moderate, and high).

Last, the three distributional conditions—normal, right-skewed (i.e., skewness ≈ 2, excess kurtosis ≈ 6), and uniform—address concerns about nonnormality that are both empirically and methodologically motivated. Non-normal (univariate and multivariate) predictors commonly emerge in psychological research (Cain et al., 2017; Micceri, 1989), and nonnormality of latent predictors has been a guiding concern in the methodological development of this literature (Cham et al., 2012; Kelava et al., 2011; Klein & Moosbrugger, 2000; Klein & Muthén, 2007; Lonati et al., 2024; Marsh et al., 2012; Rosseel et al., 2025; Schoemann & Jorgensen, 2021). The right-skewed condition reflects distributions observed in empirical settings, while the uniform distribution, though less commonly encountered in practice, was included for comparison with previous simulation studies and to represent symmetric nonnormality (Lonati et al., 2024; Wall & Amemiya, 2003). With respect to the variation of measurement error and structural disturbance distributions, the assumption of normality for measurement errors may be particularly relevant for the LSAM, given its second-order corrections assumption. Plus, as highlighted, this manipulation was only recently investigated in Lonati et al. (2024) but not for all approaches under comparison here (see p. 4 where the authors give examples of how nonnormality may arise in each of these sources).

Data Generation

Appendix B describes the complete data generation in detail. We followed a similar procedure as in previous studies (Cox & Kelcey, 2021; Ng & Chan, 2020; Rosseel et al., 2025). In the baseline condition, all stochastic components were normal: exogenous latent predictors, measurement errors, and structural disturbances. Under this baseline, structural parameters, structural residual variances, and measurement error variances were jointly calibrated to achieve $R^2 \approx 0.30$ and reliability $\rho \approx \{0.4, 0.6, 0.8\}$. Individual nonlinear effects each accounted for approximately 1-2% of the total variance in the dependent latent variables, consistent with prior simulation studies (Brandt et al., 2014; Lonati et al., 2024; Marsh et al., 2004) (see Figure 1 for the population models).

For Simulation 1, the dependent latent variable was generated according to the same structural model as in Brandt et al. (2014):

$$\eta_3 = \beta_{30} + \beta_{31}\eta_1 + \beta_{32}\eta_2 + \beta_{33}(\eta_1\eta_2) + \beta_{34}\eta_1^2 + \beta_{35}\eta_2^2 + \zeta_3$$

with parameters $\beta_{30} = -0.255$, $\beta_{31} = \beta_{32} = 0.316$, $\beta_{33} = 0.139$, $\beta_{34} = \beta_{35} = 0.101$, and $\mathrm{Var}(\zeta_3) = \psi_{33}$ (see Appendix B). For the linear model comparison, nonlinear terms were omitted (i.e., $\beta_{33} = \beta_{34} = \beta_{35} = 0$).

For Simulation 2, we targeted a more complex model to address the scalability concerns outlined previously. The dependent latent variables were generated as follows:

$$\eta_4 = \beta_{40} + \beta_{41}\eta_1 + \beta_{42}\eta_2 + \beta_{43}\eta_3 + \beta_{44}(\eta_1\eta_2) + \beta_{45}(\eta_1\eta_3) + \beta_{46}\eta_1^2 + \beta_{47}\eta_2^2 + \zeta_4$$

$$\eta_5 = \beta_{50} + \beta_{51}\eta_4 + \beta_{52}\eta_1 + \beta_{53}\eta_2 + \beta_{54}\eta_3 + \beta_{55}(\eta_1\eta_4) + \beta_{56}(\eta_2\eta_4) + \beta_{57}\eta_1^2 + \beta_{58}\eta_3^2 + \zeta_5$$

with parameters $\beta_{40} = -0.215$, $\beta_{50} = -0.160$, $\beta_{41} = \beta_{42} = \beta_{43} = 0.20$, $\beta_{44} = \beta_{45} = 0.11$, $\beta_{46} = \beta_{47} = 0.08$; $\beta_{51} = \beta_{52} = \beta_{53} = \beta_{54} = 0.16$, $\beta_{55} = \beta_{56} = 0.08$, $\beta_{57} = \beta_{58} = 0.06$, and $\mathrm{Var}(\zeta_4) = \psi_{44}$, $\mathrm{Var}(\zeta_5) = \psi_{55}$ (see Appendix B). For the linear model comparison, all interaction and quadratic terms were also omitted.

The exogenous latent predictors were generated as normal, right-skewed, or uniform using the Vine-To-Anything (VITA) simulation method with regular vine copulas and specified covariance matrices (Foldnes & Grønneberg, 2015; Grønneberg et al., 2022). Second, measurement errors were generated as either normal, $\epsilon_j \sim \mathcal{N}(0, \theta_{jj})$, or right-skewed, $\epsilon_j \sim \mathrm{Exp}^*(0, \theta_{jj})$, where j indexes observed indicators[4]. Third, structural disturbances were similarly generated as either normal, $\zeta_k \sim \mathcal{N}(0, \psi_{kk})$, or right-skewed, $\zeta_k \sim \mathrm{Exp}^*(0, \psi_{kk})$, where k indexes dependent latent variables.

Nonconvergence and Outliers

Before analyzing the data, we identified nonconvergent solutions as those where the point estimates, standard errors, p-values, or confidence interval bounds for the structural parameters of interest were not produced or contained undefined or infinite values. Following Lonati et al. (2024), we excluded the entire iteration from the analysis across estimation approaches as nonconvergent solutions signal challenging iterations for all methods[5]. We calculated

[4] $\mathrm{Exp}^*(0, \theta_{jj})$ denotes a centered and scaled exponential distribution with mean 0 and variance θ_{jj}, defined as $\sqrt{\theta_{jj}}(X - 1)$ where $X \sim \mathrm{Exp}(1)$. Note that $\mathrm{Exp}(1) \equiv \mathrm{Gamma}(1, 1)$, which is the same base distribution used for the right-skewed exogenous latent predictors.

[5] Convergent solutions may still be implausible or incomplete, for example when estimates for other parameters in the model are not obtained, when negative variances occur in $\mathbf{\Theta}$ (the measurement error covariance matrix), or when $\mathbf{\Psi}$ (the structural disturbance covariance matrix) is not positive semi-definite. We tracked these situations and obtained qualitatively identical results when excluding them from the analysis (results are available on the GitHub repository), with one worthy exception that will be clarified in the Results section.

convergence rate as $\text{Convergence}_{rate} = N_{\text{complete}}/N_{\text{total}} \times 100\%$, where N_{complete} represents successfully converged iterations with complete output for that specific approach and N_{total} represents all attempted iterations for the same approach.

Similar to Lonati et al. (2024) again, we identified outliers using the box-plot method applied to iterations where all methods successfully converged. For each method-parameter combination, quartiles and interquartile ranges were calculated from this common set of converged iterations, and point estimates outside $[Q_1 - 3 \times \text{IQR}, Q_3 + 3 \times \text{IQR}]$ (where $\text{IQR} = Q_3 - Q_1$) were flagged as outliers. Following the same exclusion logic as for convergence failures, we excluded the entire iteration from the analysis across all estimation approaches when any method produced an outlier point estimate for the main or nonlinear effects. To complement the possible arbitrary threshold selection, we also computed median-based before outlier removal (Brandt et al., 2020). In addition, following Lai and Hsiao (2022), we computed trimmed-mean–based estimates as a further robustness check.

Performance Measures

Method performance was assessed through multiple metrics following the formulas from the *simhelpers* package (Joshi & Pustejovsky, 2025) (version 0.3.1), with the exception of the median-based metrics that were based on Brandt et al. (2020) and trimmed mean-based on Lai and Hsiao (2022). Following the *simhelpers* documentation and references therein, we also quantified the Monte Carlo Standard Errors (MCSE) simulation uncertainty for most of the reported performance measures using the jack-knife technique. Generally speaking, the measures are classified in terms of absolute and relative criteria measures, standard error assessment, hypothesis testing, and confidence intervals. In the main text, we only report and discuss the metrics reported in Table 1 for the interaction and quadratic effects. For the more extensive set of measures and estimates, we refer the reader to the online supplemental materials (see Table S1).

Reproducibility and Computational Details

The simulation was conducted on a Mac Studio with an Apple M4 Max chip (with 14 cores and 36 GB of RAM), running macOS Sequoia 15.5. All analyses were performed in R (R

Core Team, 2025) (version 4.4.3). Parallel processing was implemented through the *doParalell* package (version 1.0.17) (Corporation & Weston, 2022), with *doRNG* (version 1.8.6.2) (Gaujoux, 2025) to ensure independent and reproducible random number streams. For the UPI with double mean-centering, LMS, and QML approach, the *modsem* package was used (Slupphaug et al., 2024) (version 1.0.12) with robust (sandwich) SEs. For the LSAM approach, the *lavaan* package was used (Rosseel, 2012) (version 0.6.21.2400). In this case, we estimated the latent factors with a single measurement block as opposed to separately to maintain some similarity with the other approaches (Rosseel & Loh, 2024). Importantly, apart from the aforementioned, we used the default settings for the functions provided in the packages[6]. For generating data, we used the packages *covsim* (version 1.1.0) (Grønneberg et al., 2022) and *rvinecopulib* (version 0.7.3.1.0) (Nagler & Vatter, 2025). Figures were produced with *ggplot2* (version 3.5.2) (Wickham, 2016). Finally, this paper was written with *apaquarto* (Schneider, 2024) (see GitHub repository for more information on reproducing the manuscript and analyses).

Results for Simulation Studies

The results reported here focus primarily on conditions where nonlinear effects were present. This section is organized as follows: we first present the main results for Simulation 1 and 2 under normal measurement errors and structural disturbances, followed by a separate subsection examining how nonnormal measurement errors and structural disturbances affect results relative to those main analyses results.

Main Results: Simulation 1

Convergence Issues and Outliers

Convergence problems were generally rare. The main exception was UPI, which showed reduced convergence under moderate sample size, low reliability, and uniform latent predictors.

[6] This is important to acknowledge because there are arguments for the functions used that, in principle, could be manipulated to obtain better results. For instance, one could change the maximum steps for the M-step in the EM algorithm for the LMS. However, these manipulations lead to additional researcher degrees of freedom and hence were avoided.

In this scenario, convergence rates were 75.1% with the full model and 73% with the linear model. LMS and QML converged well in general, with the lowest rates in the same setting being 97.1% and 92.2% (full model), and 96.8% and 90.4% (linear model), respectively. LSAM consistently converged, with rates of 100% across all conditions.

Outlier rates followed a comparable pattern. The most pronounced rates again occurred under moderate sample size, low reliability, and uniform latent predictors. UPI produced the highest outlier frequencies, reaching 14.59% with the full model and 16.59% with the linear model. For the same condition, LSAM also yielded the most outliers (8.89% with the linear model and 8.5% with the full model). In contrast, LMS and QML generated virtually none, with their highest rates reaching only 3.97% and 2.27% (both full model) in the same scenario as the other approaches. Detailed results for convergence and outliers are provided in Appendix C (see Table C1).

Relative Bias and RMSE When Nonlinear Effects Were Present

Figures 2 and 3 present relative bias and RMSE across distributional conditions at a moderate sample size (N = 400). Following Brandt et al. (2014), values exceeding 0.10 (10%) were considered biased. Under normal distributions, LSAM showed stable performance across reliability levels for all nonlinear effects. Bias emerged only for the interaction term at low reliability for the other approaches, with LMS and QML also biased under medium reliability. At larger sample sizes, LSAM and UPI were unbiased even at low reliability, whereas LMS and QML continued to show bias under low and medium reliability. With right-skewed distributions, LSAM displayed bias for the interaction term only under low reliability, while UPI showed no bias for all nonlinear effects. In contrast, LMS and QML exhibited severe negative bias for interactions and positive bias for quadratic effects across reliability levels. At larger sample sizes, LSAM and UPI were unbiased across conditions, but LMS and QML remained biased. Under uniform distributions, LSAM was unbiased at all reliability levels. UPI showed bias for some nonlinear effects at low and medium reliability, whereas LMS and QML produced unbiased interaction estimates but consistent negative bias for quadratic effects throughout reliability levels. These last patterns remained qualitatively unchanged at larger sample sizes (see Figure S3 in supplemental materials).

For relative RMSE, all methods performed similarly under normality, with minor differences at low reliability where LMS and QML showed the lowest values. Under right-skewed distributions, LSAM yielded the lowest RMSE, though the advantage was modest and diminished further at high reliability. For uniform distributions, LMS and QML produced substantially lower RMSE, but performance across methods converged as reliability increased.

Efficiency and Accuracy of SEs When Nonlinear Effects Were Present

Figure 4 presents SE/SD ratios for right-skewed and uniform conditions across sample sizes, with exact values shown when they fall outside the y-axis range. Following Brandt et al. (2014), ratios below 0.90 (underestimation) or above 1.10 (overestimation) indicate bias. Under normality, LSAM displayed one small overestimation at low reliability and moderate sample size. In contrast, UPI produced substantial overestimation across all nonlinear effects under the same condition (see Figure S6 in supplemental materials), while LMS and QML yielded extremely accurate ratios. For right-skewed distributions, LSAM showed some overestimation at low reliability with moderate sample sizes. UPI consistently overestimated under low reliability. We also note that in some cases an underestimation is observed. Under uniform distributions, LSAM and UPI overestimated SEs for all nonlinear effects at low reliability[7]. UPI additionally showed a negligible underestimation at medium reliability and moderate sample sizes. LMS and QML remained stable across basically all conditions. Larger sample size improved all results, except for the UPI under the uniform and low reliability condition. Last, we also see a slight underestimation for one quadratic term at high reliability for all approaches in the right-skewed scenario.

Coverage for When Nonlinear Effects Were Present

Figure 5 displays coverage results for right-skewed and uniform conditions across sample sizes, also with exact values shown when they fall outside the y-axis range. Coverage between

[7] For LSAM, we noted this overestimation arose from a small number of clear outliers in the SEs. Although these values were retained in the analysis, they are identified and displayed in supplemental materials (see Figure S1).

93% and 97% was considered acceptable. Although, previous literature considered above 90% as an acceptable performance (Aytürk et al., 2020; Cham et al., 2012). LSAM slightly exceeded this range at low reliability and moderate sample size, but not at larger samples, across all distributions. UPI also showed similar results, but still with some exceeding values with a large sample for some distributions. Under normality, LMS and QML performed within the acceptable range across reliability and sample sizes (see Figure S7 in supplemental materials). reliability. For right-skewed distributions, however, LMS and QML exhibited substantial undercoverage. Larger sample sizes did not improve coverage for these methods, while higher reliability reduced—but did not eliminate—the amount of undercoverage. Under uniform distributions, LMS and QML showed undercoverage exclusively for the quadratic effects, with the same remark previously highlighted.

Type I Error Rate

Figure 6 presents Type I error rates at large sample sizes across distributions, with values between 2.5–7.5% considered acceptable. Under normality, LSAM, LMS, and QML remained within the acceptable range across reliability levels, though UPI fell below this range at low reliability. For right-skewed distributions, LSAM and UPI stayed within acceptable bounds, whereas LMS and QML showed severe inflation for quadratic effects across all reliability levels. Under uniform distributions, LSAM and UPI generally performed well, with values falling below the acceptable range only at low reliability. In contrast, LMS and QML yielded consistently well-controlled error rates in this case. Results at moderate sample sizes were qualitatively similar, though LMS and QML showed somewhat less inflation under right-skewness. Moreover, Type I error rates for LSAM and UPI fell below the acceptable range at low reliability across distributions (see Figure S8 in supplemental materials).

Empirical Power

Given the inflated Type I error rates and biased results observed for some approaches, power results should be interpreted with caution (Morris et al., 2019). That said, we restrict our conclusions to only specific conditions. At large sample sizes, LSAM reached power levels near or above 80% under medium and high reliability across all distributions for most nonlinear effects. The main exception was under the uniform distribution, where quadratic effects remained underpowered. The other approaches showed broadly similar patterns in the same conditions. Under right-skewness, however, LMS and QML diverged extremely with low power only for the interaction term under medium reliability (see Figure S9 in supplemental materials).

Absolute Bias, RMSE, SEs, and Coverage When Nonlinear Effects Were Absent

Bias patterns resembled those observed when nonlinear effects were present, though with reduced magnitude. A notable difference emerged under the uniform distribution: LMS and QML produced unbiased quadratic estimates. RMSE results showed no substantive differences beyond reduced magnitude. The accuracy of SEs was generally comparable. Coverage improved somewhat when nonlinear effects were absent. Specifically, LMS and QML showed higher coverage under right-skewed distributions, though still falling below the desirable range for quadratic effects. Under the uniform distribution, however, coverage for quadratic effects improved substantially for both LMS and QML (see Figures S2, S4, S6, and S7 in supplemental materials).

Summary of Simulation 1

With the exception of low reliability conditions, LSAM demonstrated adequate performance across all evaluated metrics. LSAM generally yielded unbiased estimates, whereas QML and LMS exhibited systematic bias: positive and negative bias for nonlinear effects under right-skewed distributions, and negative bias for quadratic effects under uniform distributions. UPI also produced largely unbiased results. For SEs, LMS and QML were the most efficient, although LSAM achieved comparable accuracy at medium and high reliability across all distributions, particularly with large sample sizes. Coverage rates remained relatively stable for LSAM and UPI, but not for LMS and QML. Type I error rates fell within the acceptable range for LSAM and UPI, except under uniform conditions with low reliability. By contrast, LMS and QML displayed marked inflation under right-skewed distributions. Finally, empirical power was generally highest for LMS and QML, though this finding must be interpreted cautiously given their inflated Type I error and biased results. When Type I error was controlled, all approaches achieved desirable power levels for most effects, with the exception of quadratic effects under the uniform distribution, which remained underpowered.

Main Results: Simulation 2

Convergence Issues and Outliers

Convergence problems were most severe for UPI, particularly with a moderate sample, low reliability, and uniform predictors. Rates fell to 59.6% for both full model and linear model. QML converged more often under these conditions but still at suboptimal levels (79.7% with the full model; 80.9% with the linear model). LSAM converged 100% under all conditions. For normal and right-skewed predictors, all approaches had a convergence rate above 96%.

Outlier rates also varied substantially across methods. UPI again performed worst under the condition of uniform predictors with moderate sample size and low reliability, reaching 36.2% with the linear model and 38.81% with the full model. QML showed much lower outlier rates overall, with the highest occurring under right-skewed predictors (2.47% with the linear model), followed by uniform predictors (4.72% with the full model). LSAM's highest outlier rates also occurred outside normality, with uniform predictors in the full model (5.75%) and right-skewed predictors in the linear model (7.52%). We note the final sample size for the condition with moderate sample size, low reliability, and uniform distribution is relatively small (see Table C2 in Appendix C for more information).

Relative Bias and RMSE When Nonlinear Effects Were Present

Figures 7 and 8 present relative bias and RMSE for all nonlinear effects across distributions with a moderate sample size. Under normality, results were mixed. At low reliability, LSAM showed bias for some nonlinear effects, which disappeared at medium reliability. UPI and QML also showed bias at low reliability, but only QML continued to exhibit bias at medium reliability. With right-skewed distributions, LSAM was biased for one nonlinear effects at low and medium reliability. UPI only yielded biased estimates at low reliability. QML, however, showed substantial bias affecting most nonlinear effects across all reliability levels. Under the uniform distribution, LSAM showed biased results only at low reliability. UPI showed substantial bias for most nonlinear effects at low reliability, which was reduced at medium reliability. QML displayed bias for some effects at low reliability, improving with increasing reliability, yet all quadratic terms remained biased across reliability levels. At larger sample sizes, both LSAM and UPI no longer resulted in many of the biases observed under moderate sample sizes (see Figures S11, S19, and S27 in supplemental materials).

Relative RMSE was comparable across methods under normality and right-skewness. Under uniform predictors, QML yielded substantially lower RMSE than LSAM and UPI at low reliability, with results converging at medium reliability (see Figures S13, S21, and S29 in supplemental materials for a large sample size).

Efficiency and Accuracy of SEs When Nonlinear Effects Were Present

Figure 9 displays SE/SD ratios across distributions at a moderate sample size. Under normal distributions, LSAM and QML produced ratios within the acceptable range across reliability levels, whereas UPI overestimated for several nonlinear effects at low reliability, improving for most effects at medium reliability. With right-skewed distributions, LSAM and QML ratios were consistently within the desirable range, but UPI ratios were overestimated at low reliability. Under uniform distributions, all approaches yielded overestimated ratios for some effects at low reliability. However, the magnitude is the largest for the UPI approach. Results generally improved with larger samples, although the UPI approach remained overestimated for most effects (see Figures S14, S22, and S30 in supplemental materials).

Coverage When Nonlinear Effects Were Present

Figure 10 presents coverage results for a moderate sample size, with exact values shown when they fall outside the y-axis range. Under normality, LSAM generally achieved acceptable coverage, with some overcoverage at low reliability. QML maintained coverage within the desirable range across all nonlinear effects, whereas UPI showed overcoverage for most effects at low reliability. With right-skewed distributions, LSAM again produced acceptable coverage, apart from one slight overcoverage at low reliability. In contrast, QML yielded inadequate coverage, reaching acceptable levels only at high reliability for some effects. Under uniform distributions, LSAM and UPI showed a similar pattern as in the other conditions, with overcoverage at low reliability. Note, however, that UPI with a medium reliability resulted in some overcoverage. For QML, coverage for interaction terms was adequate, while quadratic terms remained under-covered at both low and medium reliability. Patterns were qualitatively similar at larger sample sizes (see Figures S15, S23, and S31 in supplemental materials).

Type I Error Rate

Figure 11 presents Type I error rates for a large sample size under right-skewed and uniform distributions. Under normality, LSAM and QML approaches maintained rates within the nominal 5% level across reliability levels. This was not the case for the UPI approach (see Figure S16 in the supplemental materials). Under right-skewed conditions, LSAM demonstrated well-controlled Type I error rates even at low reliability, whereas QML showed substantial inflation for quadratic effects at low and medium reliability. In this case, the UPI approach results fell below the acceptable range for multiple nonlinear effects. Under uniform distributions, LSAM showed some values below this acceptable range under low reliability for two quadratic terms. This was even more problematic for the UPI with all nonlinear effects, while for the QML the rate was acceptable (see Figures S16, S24, and S32 in supplemental materials for moderate sample size).

Empirical Power

Following the same previous remarks, under normal distributions, power only exceeded 80% with large samples and high reliability, primarily for the nonlinear effects in the structural equation for η_4. Results under right-skewed conditions yielded elevated power rates under the same setting. In contrast, under uniform distributions, power remained substantially lower for all nonlinear effects, even with large samples and high reliability, with the exception of the two interaction terms in η_4 (see Figures S17, S25, and S33 in supplemental materials).

Absolute Bias, RMSE, SEs, and Coverage When Nonlinear Effects Were Absent

The most notable differences emerged across conditions were reduced bias values in the case of the QML approach for certain nonlinear effects were observed throughout distributions.

SE/SD ratios were highly similar. Coverage improved for all approaches under normal and uniform conditions. Under nonnormality, however, QML continued to exhibit undercoverage mainly with large sample sizes. RMSE remained similar (see Figures S10, S12, S14, S15, S18, S20, S22, S23, S26, S28, S30, and S31 in supplemental materials).

Summary of Simulation 2

Across most conditions, LSAM produced largely unbiased estimates, accurate SEs, and acceptable coverage, except under low reliability, where some bias and overcoverage appeared. UPI was hampered by convergence failures and bias under uniform predictors, along with overestimated SEs, particularly under uniform and right-skewed distributions with low or medium reliability. QML was efficient and often yielded accurate SEs, but displayed persistent bias for several nonlinear effects under right-skewed and uniform distributions, inflated Type I error rates under right-skewness, and undercoverage of nonlinear effects—especially quadratic terms—under both right-skewed and uniform predictors, most notably at low and medium reliability. Relative RMSE differences among methods were modest: QML was occasionally more efficient under uniform predictors, LSAM and UPI performed well under right-skewness, and all methods performed similarly at medium reliability across distributions. For empirical power, acceptable levels ($\geq$ 80%) were attained only under normality with large samples and high reliability, primarily for nonlinear effects in the structural equation for η_4. Under uniform predictors, power was substantially lower for all nonlinear effects, except the two interaction terms in η_4.

Additional Distributional Conditions: Measurement Error and Structural Disturbance

Results are presented as dumbbell plots comparing normal (main results) to nonnormal (i.e., right-skewed) conditions for measurement errors and structural disturbance[8].

[8] A natural concern may be that the right-skewed measurement errors and structural disturbances can lead to additional convergence failures and outliers, thus the effective sample sizes may differ between conditions. To account for this, we computed the MCSE of the difference. This is not reported here but further clarified in the online supplemental materials.

Simulation 1

With respect to convergence, right-skewed measurement errors and structural disturbances had the most negative consequences on the cases with low reliability, especially with uniform latent variables. For this reason, we restrict the comparison here to cases with medium and high reliability. Similarly, for outliers, the same conditions yielded the worst results. In fact, in the condition with uniform latent variables, low reliability, and right-skewed measurement errors, convergence rates for the UPI approach were extremely low. We also note that the right-skewed measurement errors, in contrast to the structural disturbances, were more frequently associated with convergence and outliers (see Table C1).

For the relative bias, focusing on moderate sample sizes, right-skewed measurement error with medium reliability lead to substantial worsening for all approaches, especially in the case with uniform latent predictors. However, under high reliability, conclusions stay largely in agreement with main results (see Figure 12). Note, however, the direction of the bias is different for interaction and quadratic effects. For relative RMSE, under moderate sample sizes, the largest difference is for the UPI approach with medium reliability and right-skewed measurement error (see Figure 13), where the the values increased. For the SE/SD ratio, results are largely similar, with the most notable difference for the UPI under right-skewed measurement error and medium reliability. In this case, we see an overestimation with right-skewed and uniform latent variables (see Figure 14). All of these results are not different under a large sample size. For coverage, results yielded the most substantial differences under medium reliability. Indeed, with right-skewed measurement error and latent variables, the coverage for LMS and QML actually improved. This is rather unexpected. While for right-skewed measurement error and uniform latent variables, most approaches also yielded lower coverage results except the UPI where this was mild (see Figure 15). Note that a moderate sample size yielded the same conclusions. Results for Type I error followed the behavior from coverage.

Simulation 2

Due to the conclusions being largely the same as what happened with the model in Simulation 1, we do not show any figures in the manuscript (see online supplemental materials SX). Convergence and outliers yielded similar results as well. One notable exception is that QML showed substantial convergence issues when latent variables where uniformly distributed, right-skewed measurement errors and under low reliability.

For the relative bias, similar trends are observed as in Simulation 1. Right-skewed measurement error with medium reliability lead to substantial worsening for all approaches, primarily under normal and uniform latent predictors. Under high reliability, results remain largely in agreement with the main results (see Figures SX). For the relative RMSE, under moderate sample size and medium reliability, with right-skewed measurement error, we see some increase for LSAM and UPI under right-skewed latent predictors, while under uniform predictors only UPI. Again, results with high reliability remain the same as main results (see Figures SX). For the SE/SD ratio, we see a large difference throughout the latent predictors' distributions (see Figures SX). In this case, the most relevant are with medium reliability and right-skewed latent predictors. For coverage, similar patterns as in Simulation 1 are observed.

Discussion

The analysis of nonlinear effects with latent variables requires estimation procedures that yield reliable results for inference and hypothesis testing. Most existing approaches rely on distributional assumptions that are frequently violated in applied research (Cain et al., 2017). In this study, we compared the recently proposed LSAM with commonly used methods across two simulation studies, evaluating their performance under different distributional, sample size, reliability conditions, and levels of model complexity. Thus, because SAM approaches for nonlinear effects have long been underrepresented in the literature (Rosseel & Loh, 2024) and until recently were not implemented in widely accessible software, demonstrating that a SAM procedure can provide reliable results is important for both methodological development and applied practice.

Performance of LSAM

LSAM performed well across most evaluation metrics, offering a balance of accuracy and computational feasibility. Under medium and high reliability, LSAM produced unbiased estimates, adequate SEs, controlled Type I error rates, and maintained stable coverage even under nonnormal conditions. It also achieved acceptable power ($\geq$ 80%) in large samples with adequate reliability in case of normal and right-skewed distributions for all nonlinear effects, while under uniform distributions quadratic terms were underpowered. Convergence rates were high, but with relatively larger outlier estimates under low reliability conditions. Taken together, these findings position LSAM as a promising estimator that combines reasonable statistical performance with improved usability and software support. At the same time, limitations specifically under moderate sample sizes and low reliability for some performance metrics highlight the need for cautious application.

Relation to Previous Findings

With respect to Simulation 1, the bias results for LSAM were comparable to the 2SMM results reported in previous studies (Brandt et al., 2014; Rosseel et al., 2025), with largely unbiased estimates across conditions that manipulated the distribution of latent variables. LMS and QML showed a negative bias for the interaction effect and a positive bias for the quadratic effects under right-skewed distributions, consistent with Brandt et al. (2014) and Rosseel et al. (2025) as well. Under uniform settings, interaction terms remained unbiased, while quadratic effects were biased. Notably, Lonati et al. (2024) found unbiased LMS estimates for a single interaction under uniform distributions, whereas including quadratic effects in the current study led to substantial bias for those terms—highlighting that estimator performance depends critically on the number and configuration of nonlinear effects. This extends the results of Brandt et al. (2014), which show deteriorating performance as the amount of nonlinear effects increased in the nonnormal condition. UPI produced largely unbiased estimates with a few exceptions, aligning with earlier results (Brandt et al., 2014; Cham et al., 2012; Marsh et al., 2004). SE/SD ratio results for LMS and QML were consistent with Brandt et al. (2014), showing higher efficiency relative to UPI. For UPI, the direction and magnitude of standard error bias has been shown to depend on sample size and type of nonnormality, a pattern consistent with the present findings (Brandt et al., 2014; Cham et al., 2012; Lonati et al., 2024; Marsh et al., 2004). Interestingly, in the present study, which employed robust SEs, these tended to be overestimated. This pattern, however, reverses if not using robust SEs[^We included the results for non-robust SEs for the interested reader in the GitHub repository]. This finding underscores the importance of carefully considering the choice of standard error estimator, as the correction for one source of bias may introduce another. Type I error rates were generally controlled for LSAM under a large sample size, as with the 2SMM approach, whereas LMS and QML showed inflation under right-skewed conditions, consistent with Brandt et al. (2014). For UPI, Type I error rates were not inflated, consistent with previous studies (Cham et al., 2012; Marsh et al., 2004), despite Brandt et al. (2014) observing otherwise. LSAM and UPI also maintained more stable coverage under nonnormal conditions compared to LMS (Cham et al., 2012) and QML. For empirical power, LSAM achieved levels at or above 80% only with medium or high reliability under large samples for normal and right-skewed distributions for some nonlinear effects, as with 2SMM in Brandt et al. (2014); UPI displayed a similar pattern. LMS and QML reached power above 80% even under low reliability, as in Brandt et al. (2014), but these results must be interpreted cautiously given their inflated Type I error rates. Interestingly, we also observed a tendency to underestimate the main effects under right-skewed distributions for LMS and QML with a linear model (see supplemental materials). This finding is difficult to compare with previous studies, as they focused solely on nonlinear effects and did not always report results for main effects. However, a similar pattern is visible in Brandt et al. (2014) (see Table 7).

To our knowledge, this is the first study to examine more than three nonlinear effects simultaneously, highlighting scalability challenges that arise under realistic levels of model complexity. Indeed, in Simulation 2, UPI struggled across multiple metrics compared to Simulation 1, whereas LSAM and QML remained largely consistent with their earlier performance. However, in specific conditions, QML and UPI had substantial convergence issues.

Importantly, these challenges are not only statistical but also practical, since their impact depends on how efficiently estimation methods can be implemented in software. That said, computation time also provided additional insights, although such comparisons would be contingent on implementation details (e.g., C++ vs. R) rather than reflecting inherent properties of the statistical methods themselves (see supplemental materials for computation time). In this case, the implementation of LMS and QML in C++ within *modsem* (Slupphaug et al., 2024) has substantially reduced computational demands, making previously infeasible analyses more realistic. Nevertheless, LMS remains computationally intensive, and estimation time increases rapidly with the number of nonlinear effects, so, in this sense, scalability continues to pose practical limitations even in optimized implementations. We also highlight that for the UPI approach, the number of parameters substantailly increases with more complex models, which induces more convergence and outlier issues. However, the previous limitation that demanded a manual model specification is bypassed here. Therefore, taken together, these software developments enhance the feasibility of nonlinear latent variable modeling while underscoring the need for continued caution regarding estimator assumptions, computational burden, and measurement conditions.

Limitations and Future Research

As with any simulation study, the present findings are conditional on the specific design factors and assumptions considered. We assumed that nonnormality in the indicators stemmed solely from the latent predictor variables. In practice, nonnormality may also involve measurement error or other structural sources (Lonati et al., 2024). In addition, the sample sizes studied may not reflect the small samples often encountered in applied research, limiting generalizability—particularly for UPI. We also focused exclusively on distributional violations and did not examine structural misspecifications (Brandt et al., 2020). The robustness of LSAM and related approaches to such violations remains an important direction for future research.

Moreover, the broader literature has explored various incremental methodological modifications to address estimation under nonnormality. For example, the reliability-corrected single-indicator LMS approach (Cheung & Lau, 2017; Su et al., 2019) was proposed to mitigate sensitivity to distributional violations, although Lonati et al. (2024) found improvements mainly in small samples. Other studies have investigated alternative ways of forming product indicators (Marsh et al., 2004) and compared Wald z-tests with likelihood ratio tests in terms of Type I error and empirical power, noting their only asymptotic equivalence (Buse, 1982; Cham et al., 2012). Regarding robust SEs, simulation evidence suggests that these adjustments yield limited improvements (Brandt et al., 2014, 2020; Cham et al., 2012; Lonati et al., 2024; Marsh et al., 2004), particularly in small samples, and our results corroborate this finding[9]. Thus, future work should clarify the scope and limits of such robustness corrections. Finally, several alternative approaches were not explored (Boker et al., 2023; Bollen, 1995; Kelava et al., 2014), some partly due to limited software implementations (Kelava et al., 2014; Mooijaart & Bentler, 2010).

With respect to specific estimation choices, future research should clarify whether bounded estimation procedures may improve UPI performance under small samples and specific distributions (De Jonckere & Rosseel, 2022). The number of indicators per latent variable also warrants investigation, as it has contrasting implications across methods: while additional indicators increase the parameter burden for UPI, they should in principle improve measurement quality for LSAM. Another important avenue for future work is evaluating whether alpha-correction methods (Bogaert et al., 2023) or separating measurement blocks within models enhance the estimation of nonlinear effects under LSAM (Rosseel & Loh, 2024). Last, the influence of ordinal and missing data should also be investigated for the LSAM (**cite?**).

General Recommendations for Practitioners

As noted in previous studies (Brandt et al., 2014, 2020; Marsh et al., 2004), no single estimator uniformly outperforms the others across all conditions. Performance depends on the interplay between distributional assumptions, sample size, reliability, and model complexity. Our recommendations should therefore not be generalized beyond the conditions examined here, and researchers should evaluate which factors are most relevant to their specific application. That said,

[9] For the interested reader, simulation data with robust SEs are available on the GitHub repository.

our results, taken together with the existing literature, do point to several considerations that can inform estimator selection.

First, the normality of the latent predictors is a critical assumption that differentiates the estimators most clearly. LMS and QML assume normally distributed latent predictors and, when this assumption holds, they offer the best performance generally speaking. However, when it is violated, both can produce biased estimates, poor coverage, and consequently inflated Type I error rates, particularly for quadratic effects. Since this assumption is not directly testable (see Lonati et al., 2024 for an indirect diagnostic), researchers who suspect or cannot rule out nonnormality in their latent predictors should favor LSAM. The UPI is also an alternative but one must be aware of the issues with convergence and efficiency that is dependent on the type of nonnormality. Relatedly, it is important to recognize that estimator performance is not all or nothing—different metrics may favor different approaches within the same condition. For instance, LSAM produced small biases under nonnormal conditions with moderate and low reliability, but its SEs tended to be overestimated in those same scenarios. Conversely, LMS and QML showed the opposite pattern: higher efficiency in SE estimation but at the cost of biased point estimates and compromised Type I error control. Thus, researchers should weigh which performance aspects are most consequential for their research question—whether accurate point estimation, precise interval estimation, or controlled error rates take priority.

Second, model complexity plays an important role. With increasing numbers of nonlinear effects, UPI exhibited convergence difficulties, while QML also showed elevated nonconvergence rates under some conditions. LSAM, by contrast, converged consistently across all conditions in both simulation studies. LMS was excluded from the more complex model due to computational constraints, which itself underscores a practical scalability limitation. Researchers fitting models with many nonlinear terms should be aware that LSAM and QML currently offer the most feasible options, with LSAM being more robust to convergence failures and QML being more efficient when its distributional assumptions are met.

Finally, adequate measurement quality remains essential regardless of which estimator is chosen. All approaches struggled under low reliability, consistent with previous findings (Brandt et al., 2020). In practice, ensuring high-quality indicators with sufficient reliability should be a prerequisite before attempting to estimate nonlinear latent variable effects.

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Table 1

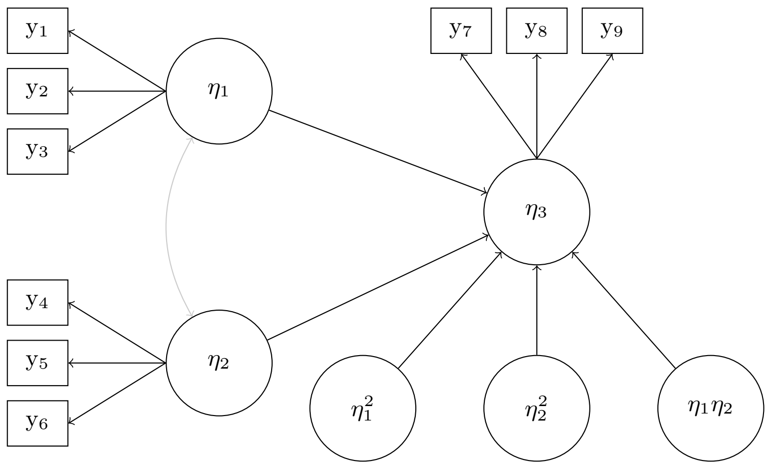
Subset of the performance measures for evaluating parameter estimates in the simulation studies

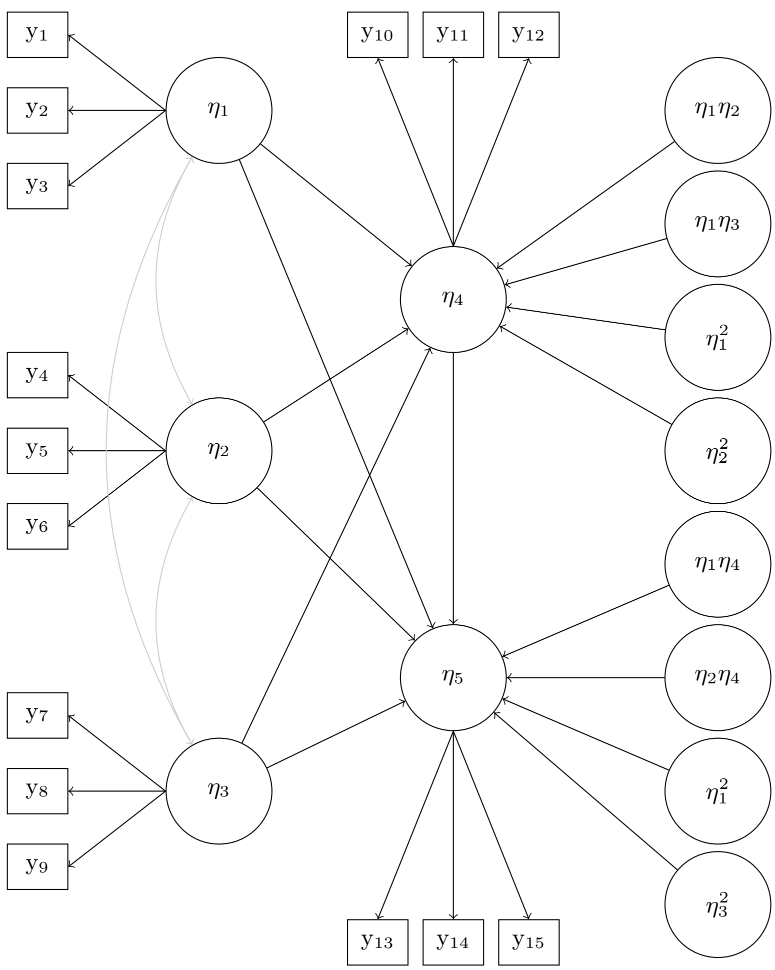
Performance Measure	Empirical Calculation	MCSE
Absolute Criteria		
Absolute Bias	$\bar{B} - \beta$	$\sqrt{S_B^2/K}$
MSE	$\frac{1}{K}\sum_{k=1}^{K}(B_k - \beta)^2$	$\sqrt{\frac{1}{K}[S_B^4(k_B - 1) + 4S_B^2 g_B(\bar{B} - \beta) + 4S_B^2(\bar{B} - \beta)^2]}$
RMSE	$\sqrt{\frac{1}{K}\sum_{k=1}^{K}(B_k - \beta)^2}$	$\sqrt{\frac{K-1}{K}\sum_{j=1}^{K}(\text{RMSE}_{(j)} - \text{RMSE})^2}$
Relative Criteria		
Relative Bias	$(\frac{\bar{B}}{\beta}) - 1$	$\sqrt{S_B^2/(K\beta^2)}$
Relative MSE	$\frac{(\bar{B}-\beta)^2+S_B^2}{\beta^2}$	$\sqrt{\frac{1}{K\beta^4}[S_B^4(k_B - 1) + 4S_B^2 g_B(\bar{B} - \beta) + 4S_B^2(\bar{B} - \beta)^2]}$
Relative RMSE	$\sqrt{\frac{(\bar{B}-\beta)^2+S_B^2}{\beta^2}}$	$\sqrt{\frac{K-1}{K}\sum_{j=1}^{K}(\text{rRMSE}_{(j)} - \text{rRMSE})^2}$
Standard Error		
Mean SE	$\overline{SE} = \frac{1}{K}\sum_{k=1}^{K}\widehat{SE}_k$	—
SE/SD Ratio	$\overline{SE}/S_B$	—
Hypothesis Testing		
Rejection Rate	$r_\alpha = \frac{1}{K}\sum_{k=1}^{K} I(P_k < \alpha)$	$\sqrt{r_\alpha(1 - r_\alpha)/K}$
Confidence Intervals		
Coverage	$c_\beta = \frac{1}{K}\sum_{k=1}^{K} I(A_k \leq \beta \leq B_k)$	$\sqrt{c_\beta(1 - c_\beta)/K}$

Note. B = estimator for parameter β; K = number of replications producing estimates $B_1, ..., B_K$; $\bar{B} = (1/K)\sum_{k=1}^{K} B_k$ = sample mean; $S_B^2 = (1/(K-1))\sum_{k=1}^{K}(B_k - \bar{B})^2$ = sample variance; $S_B = \sqrt{S_B^2}$ = sample standard deviation; k_B = standardized kurtosis; g_B = standardized skewness; $I(\cdot)$ = indicator function; P_k = p-value from replication k; α = significance level; A_k, B_k = CI endpoints; r_α and c_β denote rejection rate and coverage, respectively. MCSE = Monte Carlo Standard Error, dashes indicate MCSEs not computed. MSE = Mean Squared Error. RMSE = Root Mean Squared Error.

Figure 1

Data Generating Model for the simulation studies. Top: Model for Simulation 1. Bottom: Model for Simulation 2.

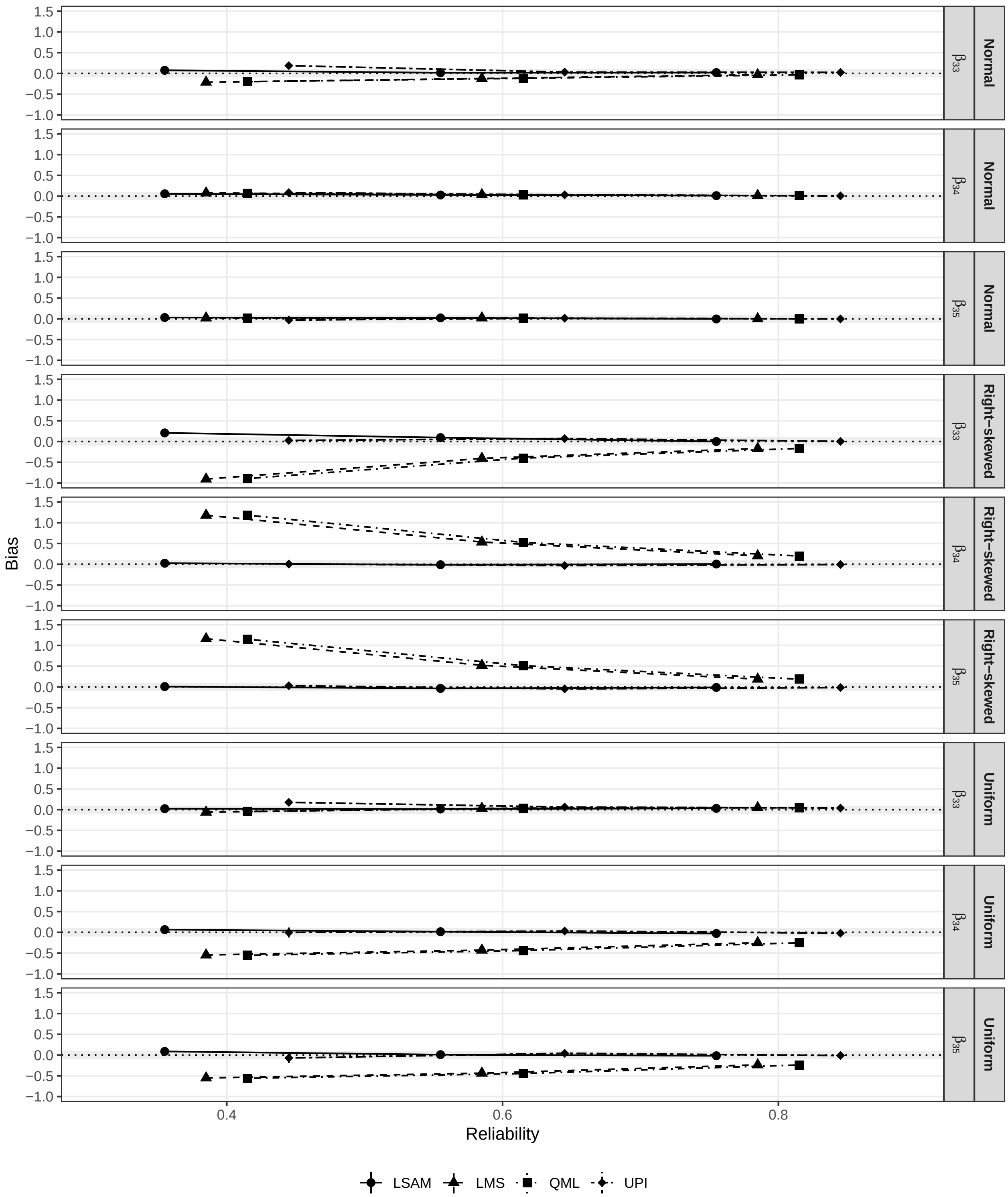




Note. Residual variances and covariances and intercepts are not shown for clarity.

Figure 2

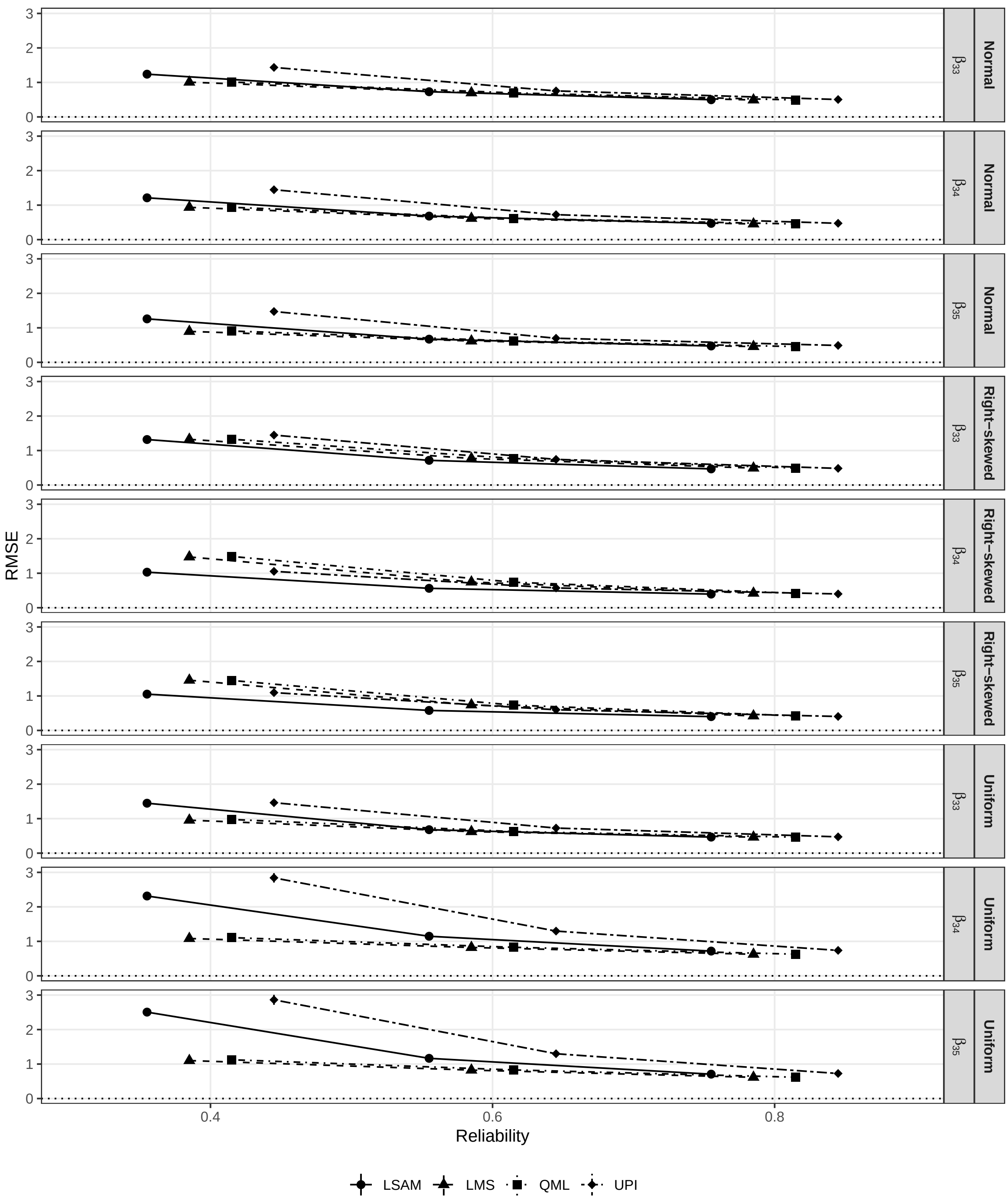
Relative bias for Simulation 1 with varying distributions and reliability levels (N = 400).



Note. Shaded regions represent acceptable deviations. Vertical bars represent the MCSEs.

Figure 3

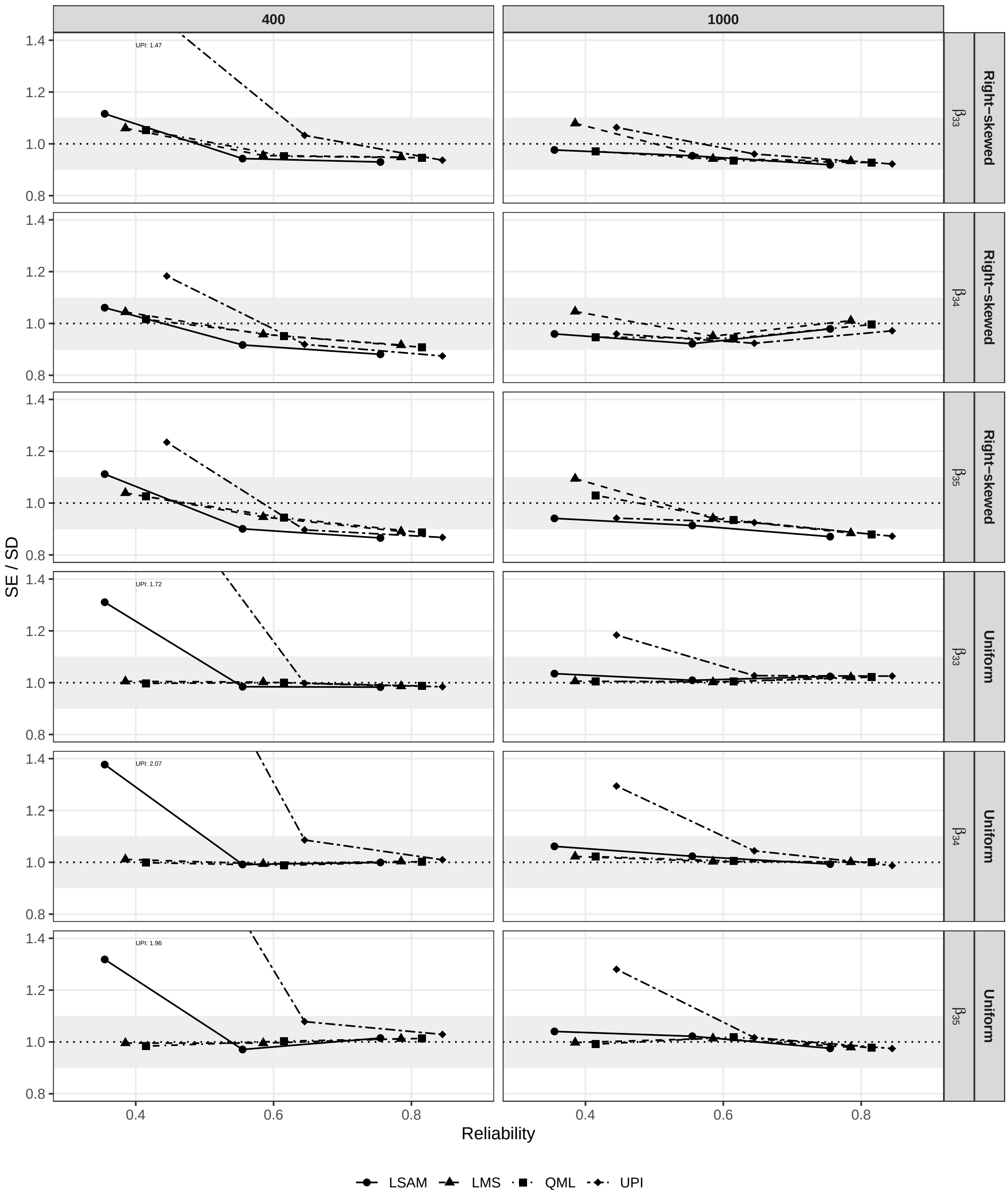
Relative RMSE for Simulation 1 with varying distributions and reliability levels (N = 400).



Note. Shaded regions represent acceptable deviations. Vertical bars represent the MCSEs.

Figure 4

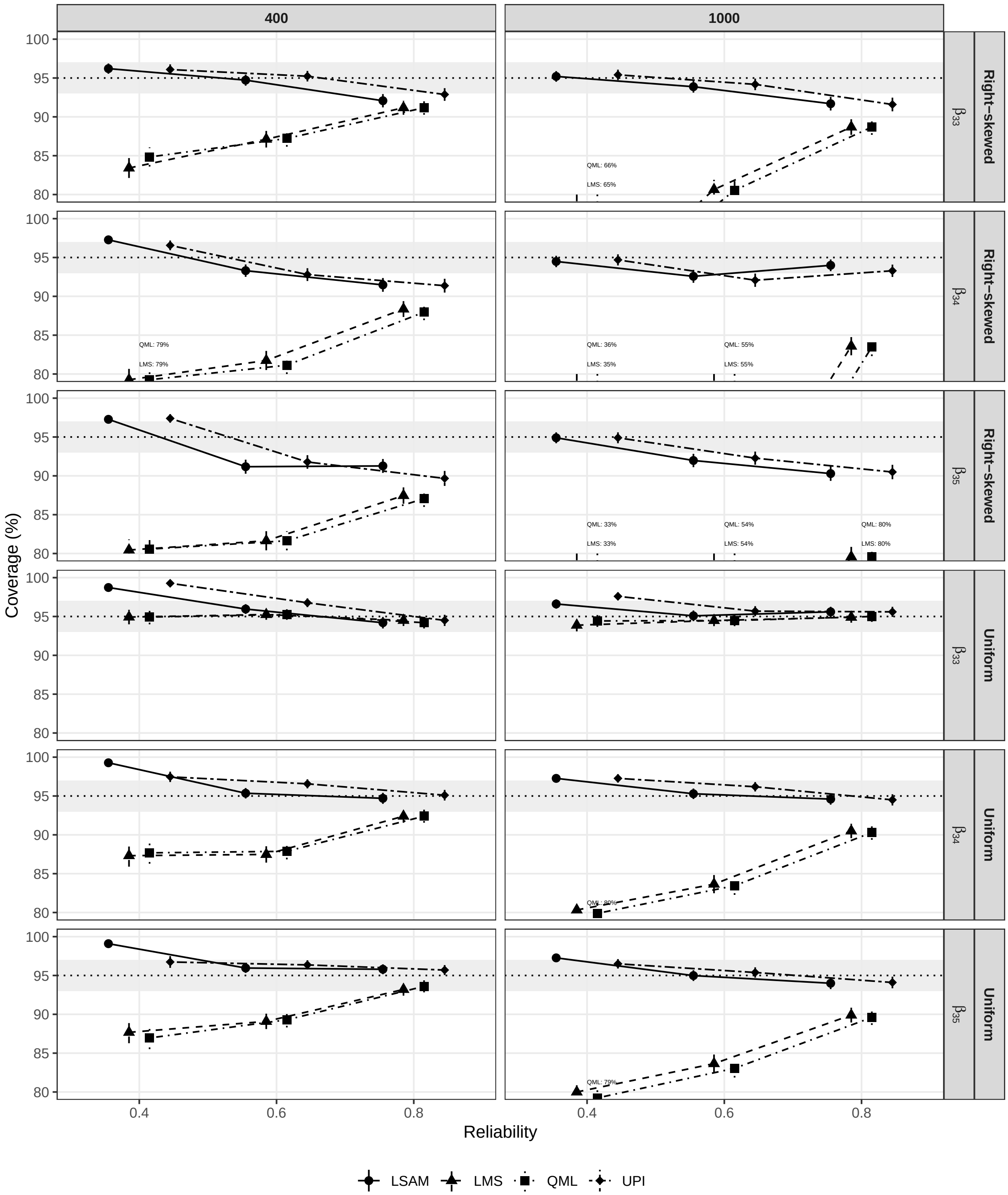
SE/SD Ratio for Simulation 1 with varying sample sizes and reliability levels (Right-skewed and uniform conditions).



Note. Shaded regions represent acceptable deviations.

Figure 5

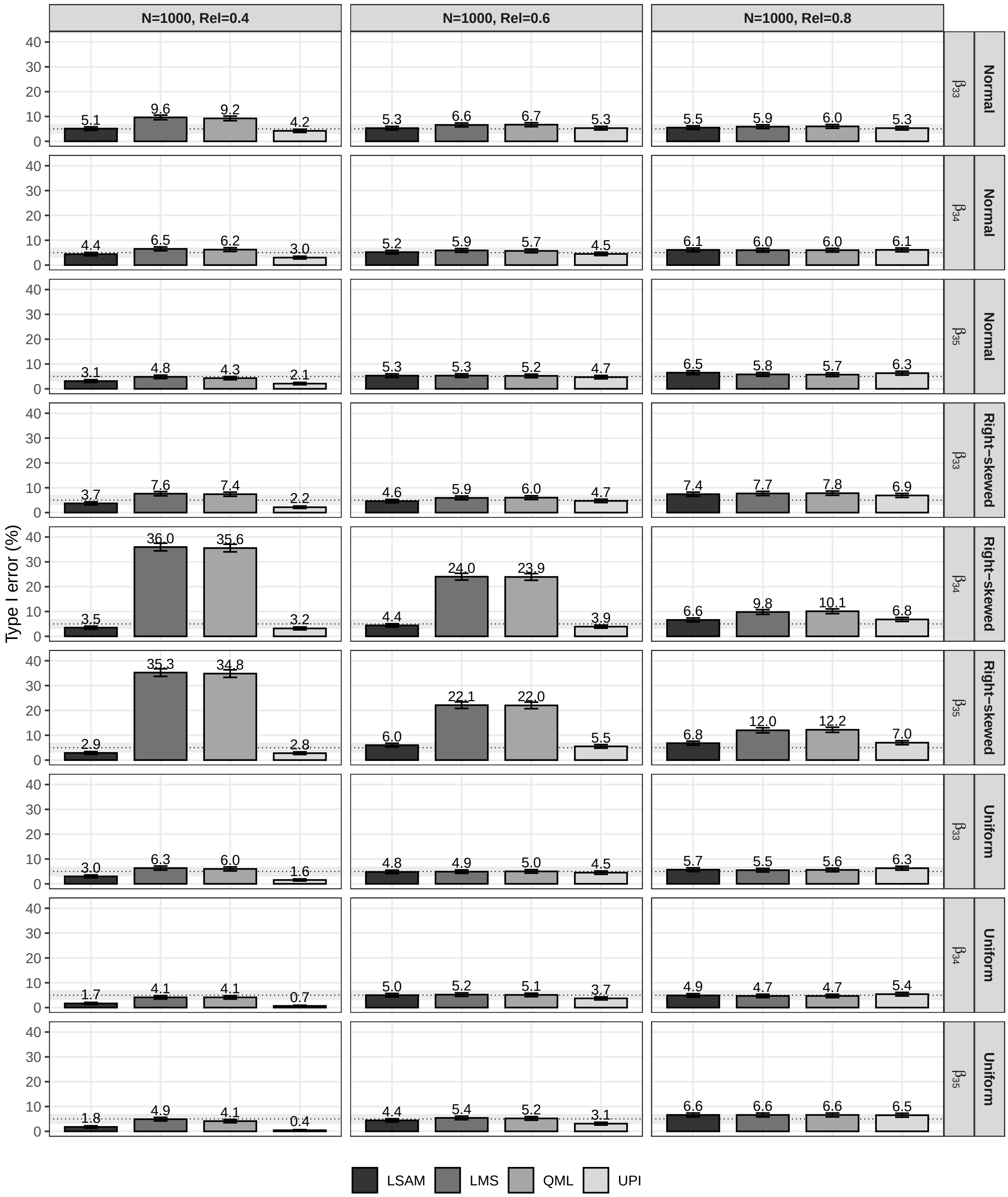
Coverage for Simulation 1 varying distributions, reliability levels, and sample sizes.



Note. Shaded regions represent acceptable deviations. Vertical bars represent the MCSEs.

Figure 6

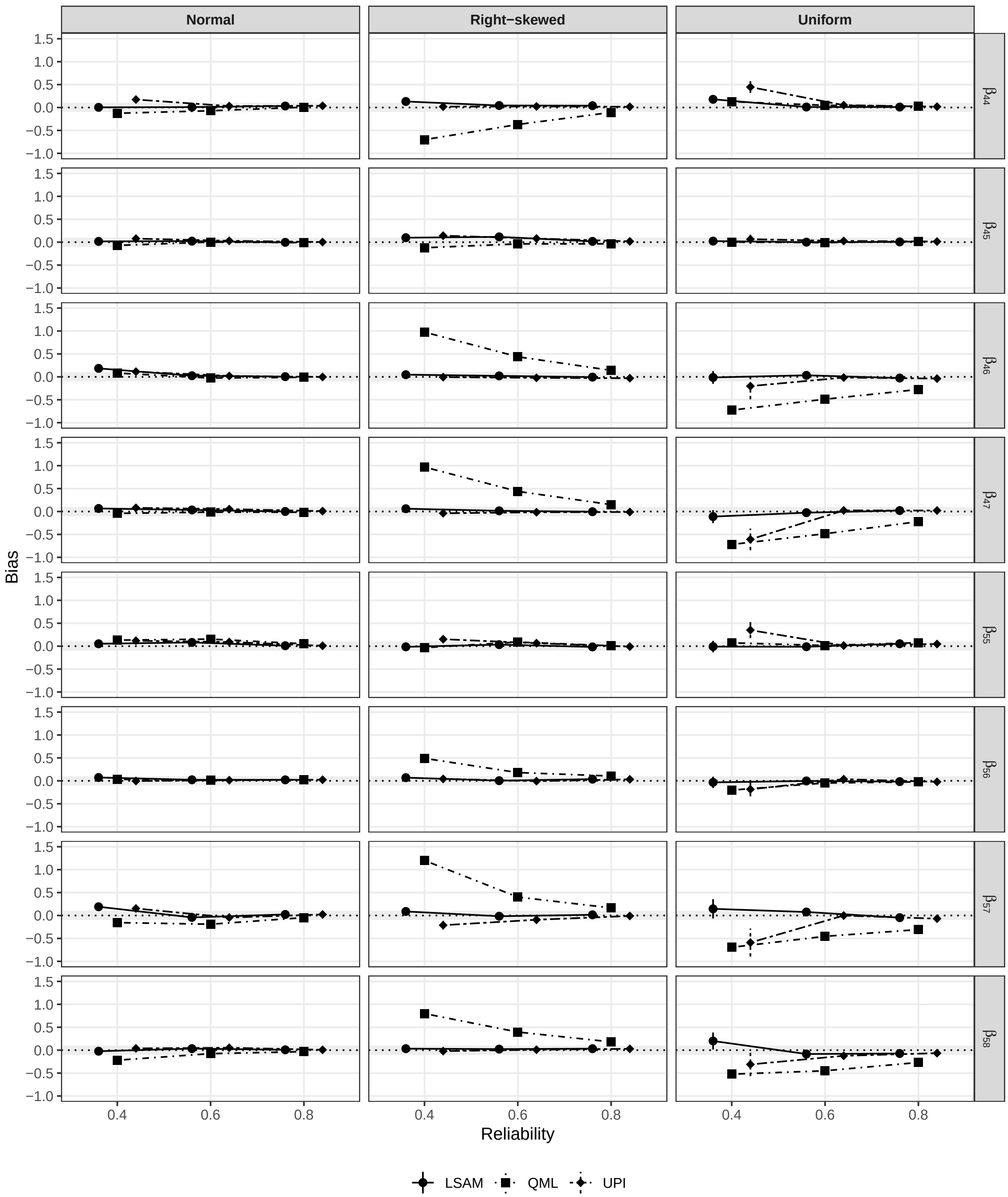
Type I error rate for Simulation 1 varying distributions and reliability levels (N = 1000).



Note. Shaded regions represent acceptable deviations. Errors bars represent the MCSE.

Figure 7

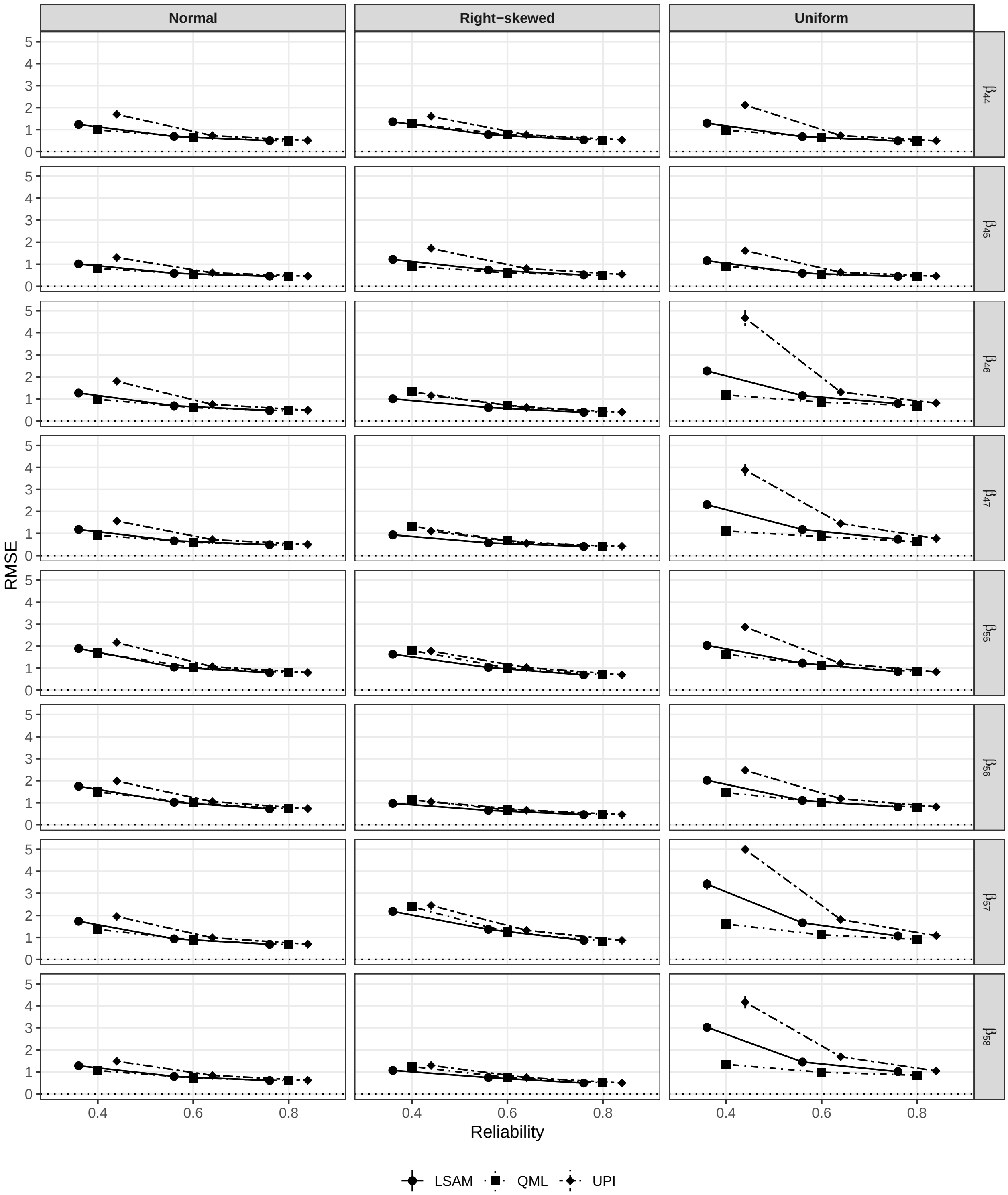
Relative bias for Simulation 2 with varying distributions and reliability levels (N = 400).



Note. Shaded regions represent acceptable deviations. Vertical bars represent the MCSEs.

Figure 8

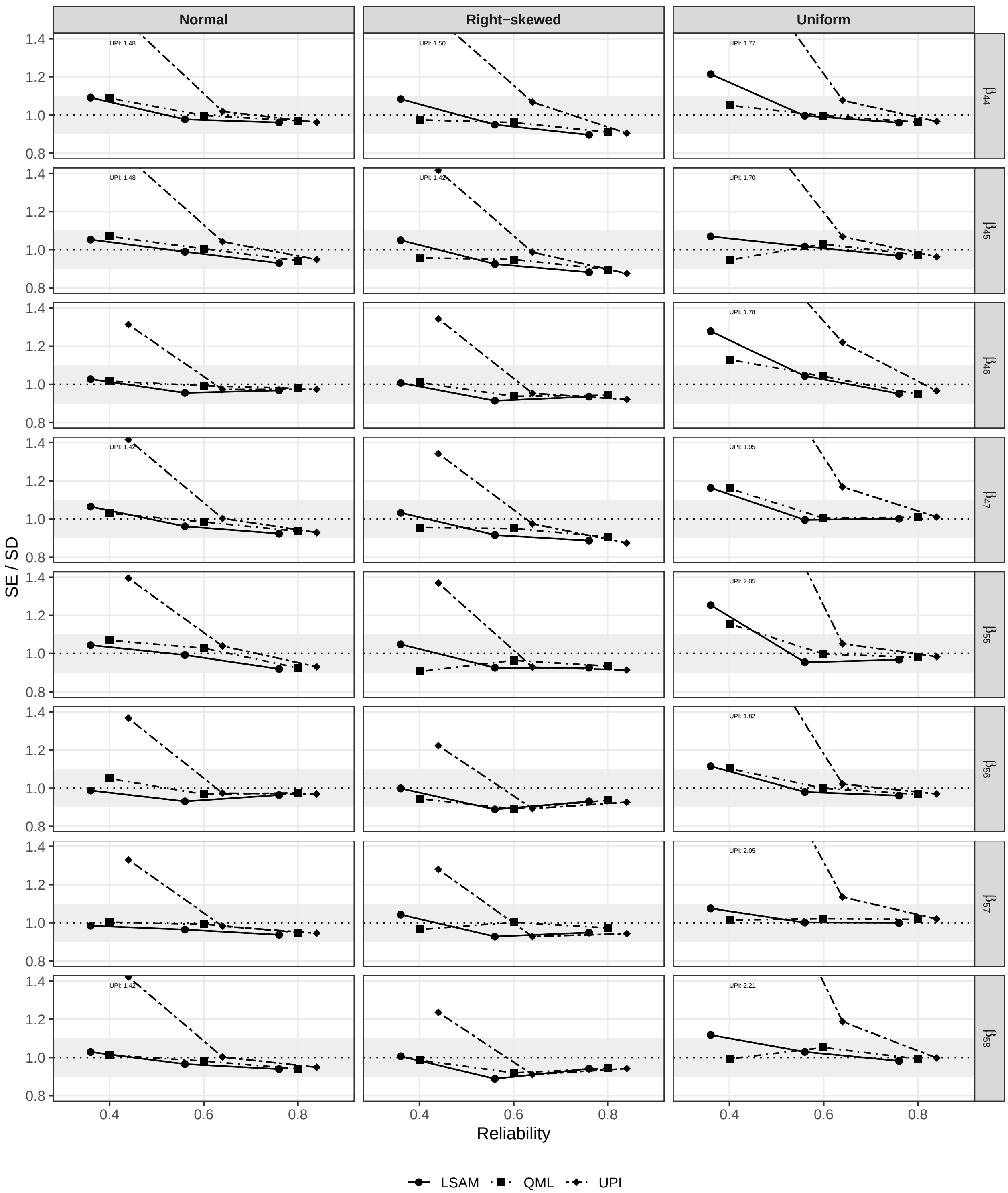
Relative RMSE for Simulation 2 with varying distributions and reliability levels (N = 400).



Note. Shaded regions represent acceptable deviations. Vertical bars represent the MCSEs.

Figure 9

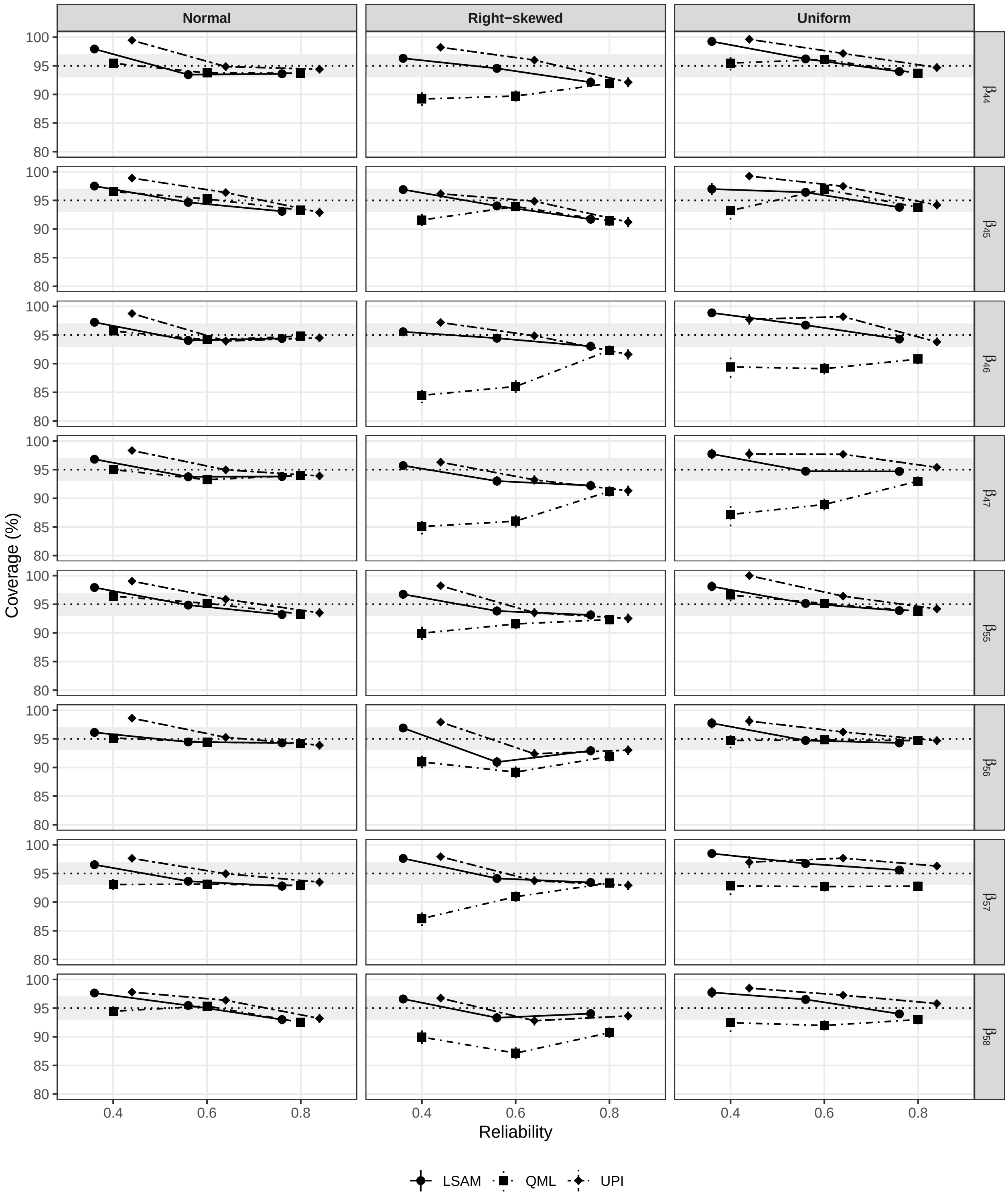
SE/SD Ratio for Simulation 2 with varying distributions and reliability levels (N = 400).



Note. Shaded regions represent acceptable deviations.

Figure 10

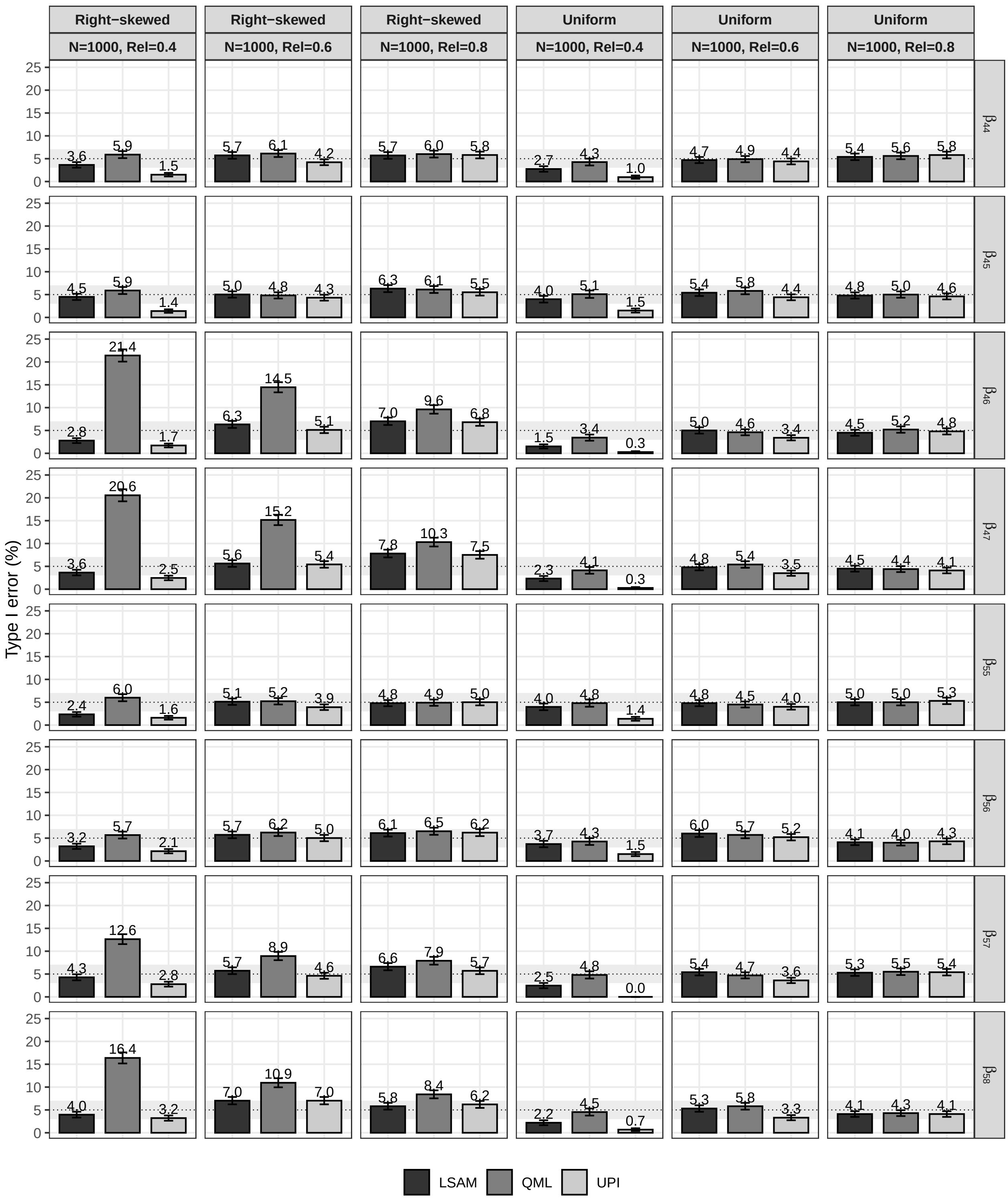
Coverage for Simulation 2 with varying distributions and reliability levels (N = 400).



Note. Shaded regions represent acceptable deviations. Vertical bars represent the MCSEs.

Figure 11

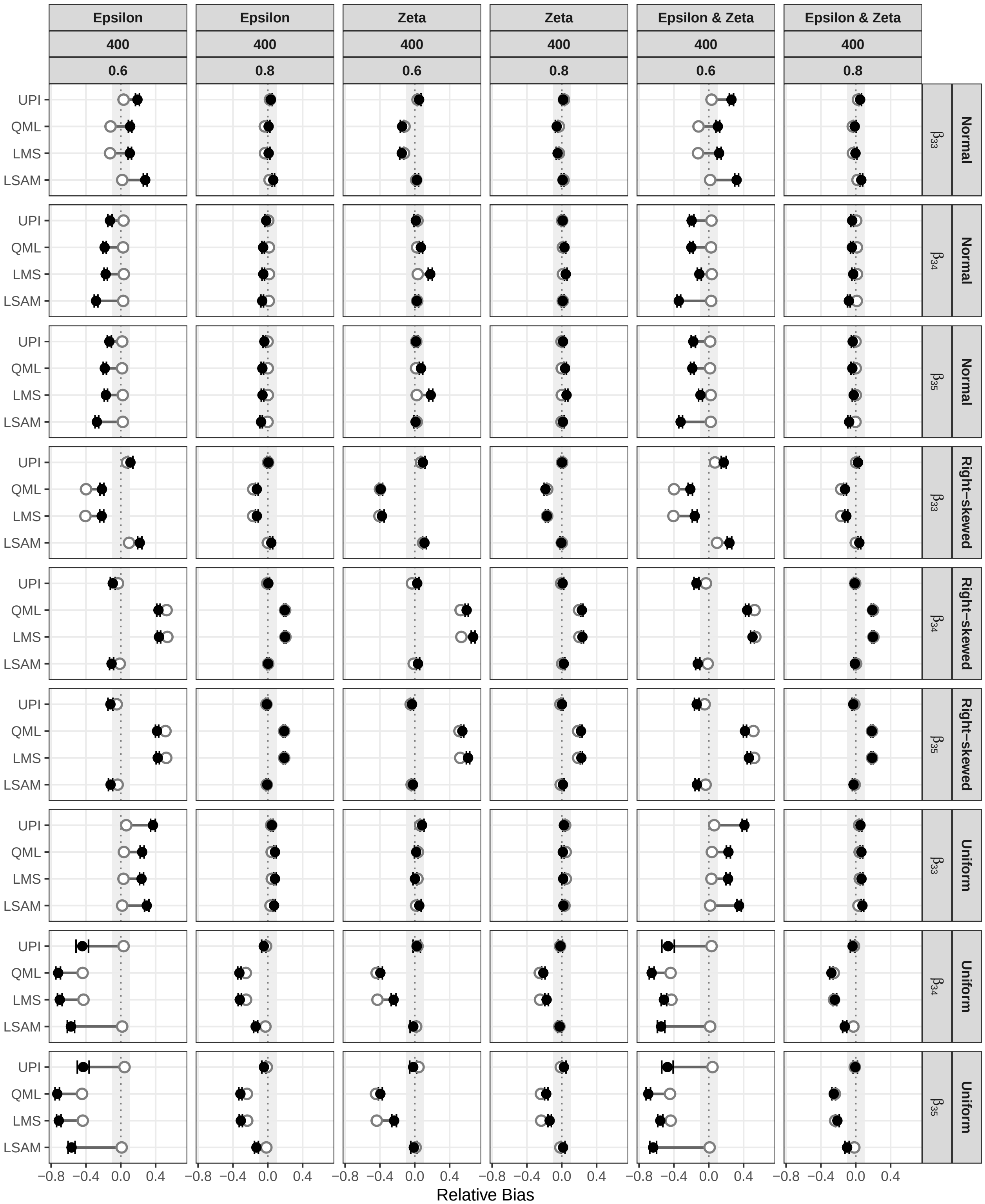
Type I error rate for Simulation 2 with varying distributions and reliability levels (N = 1000).



Note. Shaded regions represent acceptable deviations. Error bars represent the MCSE.

Figure 12

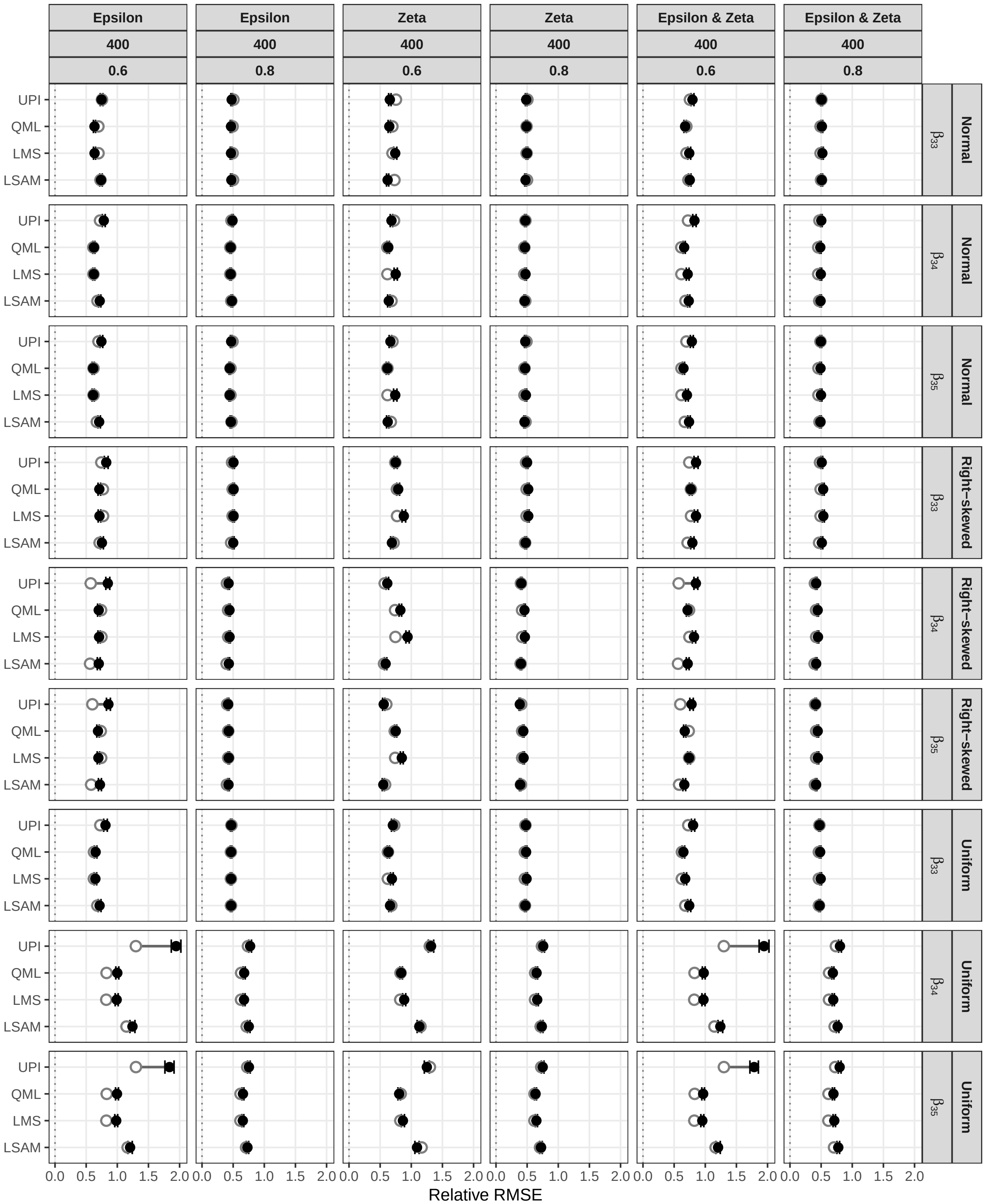
Additional distributional conditions: Relative bias for Simulation 1 (N = 400).



Note. White dots represent results under normal measurement error and structural disturbance, while black dots represent nonnormal conditions. Shaded region indicates acceptable range. Error bars show MCSE around nonnormal estimates.

Figure 13

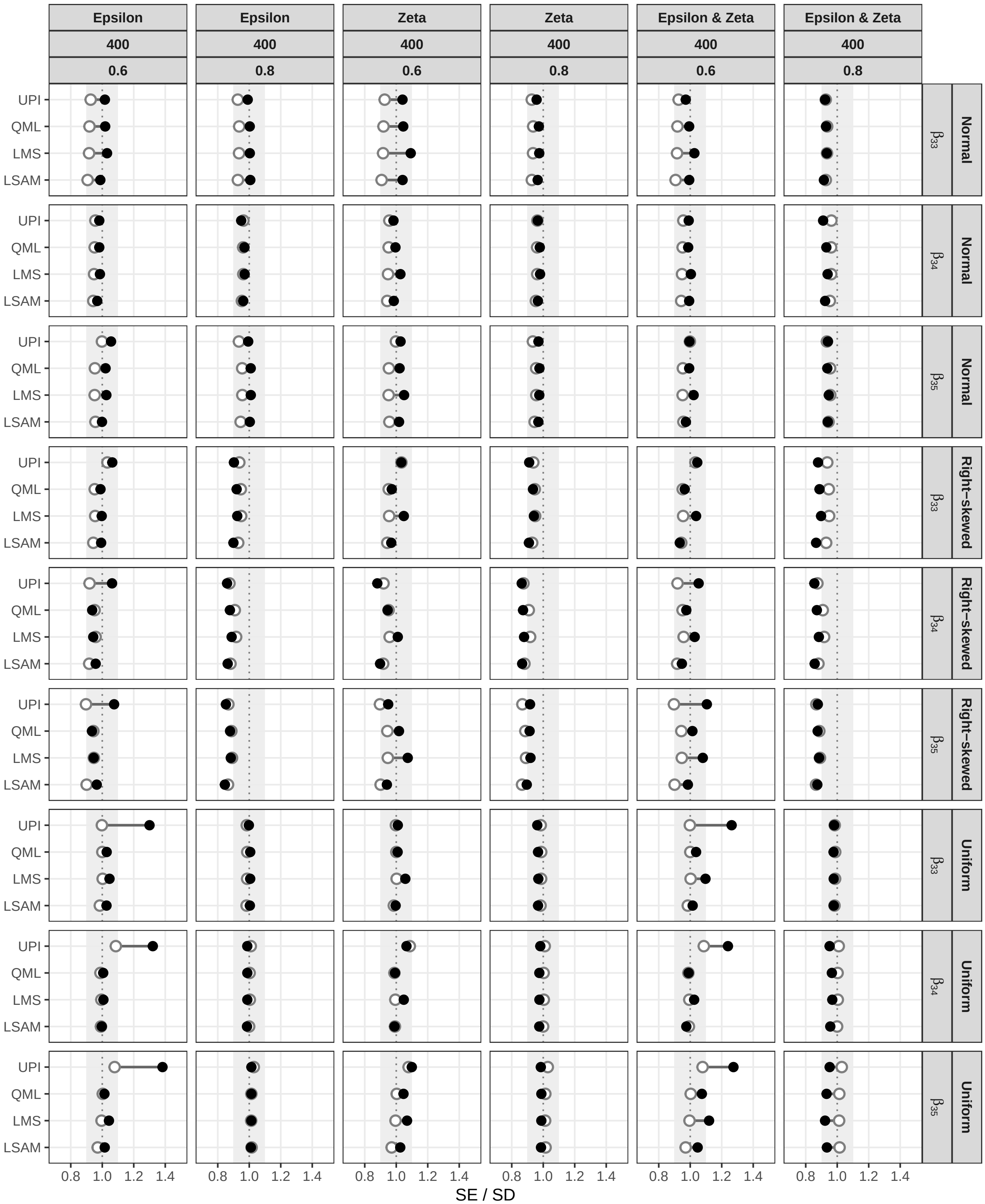
Additional distributional conditions: Relative RMSE for Simulation 1 (N = 400).



Note. White dots represent results under normal measurement error and structural disturbance, while black dots represent nonnormal conditions. Error bars show MCSE around nonnormal estimates.

Figure 14

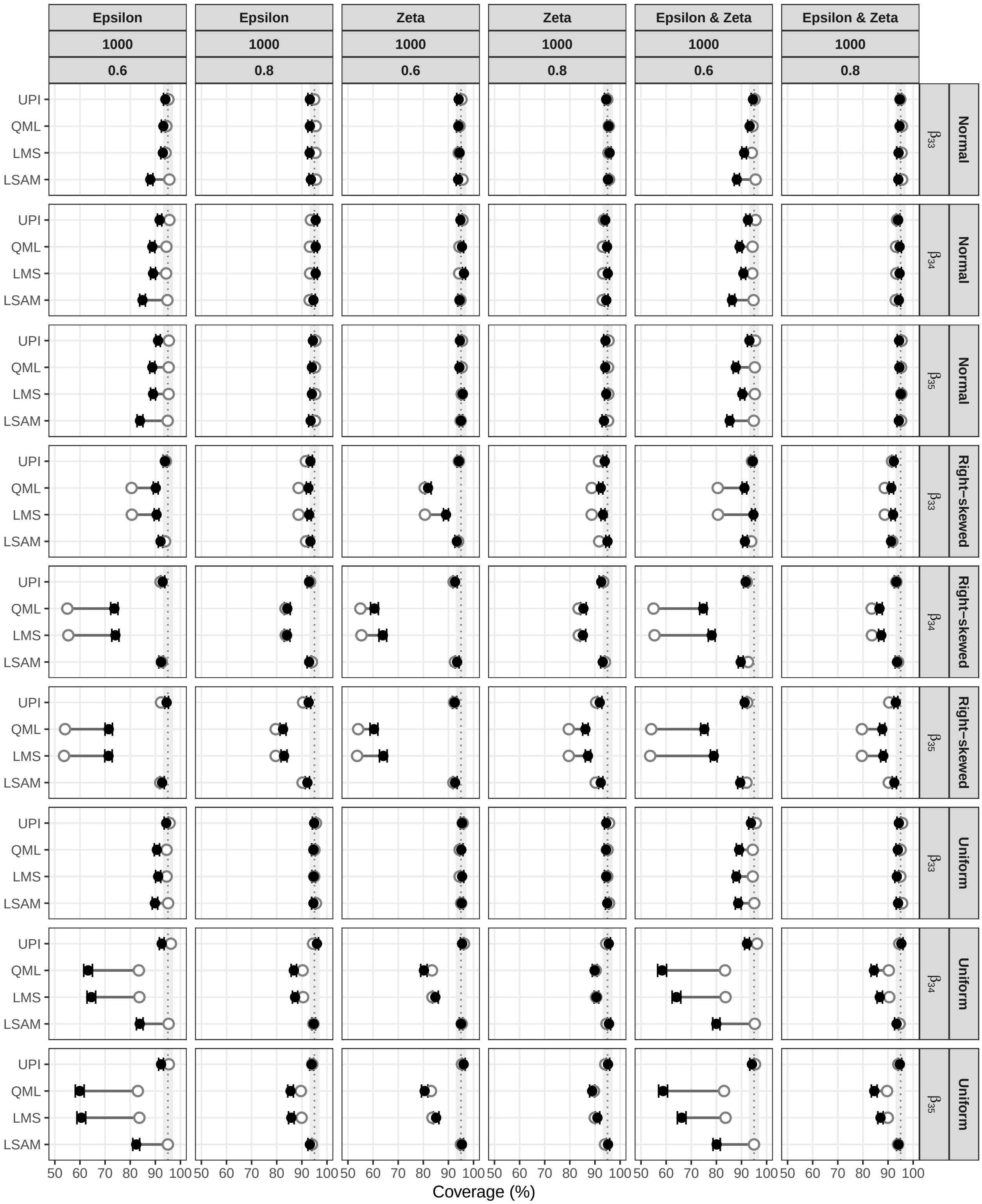
Additional distributional conditions: SE/SD Ratio for Simulation 1 (N = 400).



Note. White dots represent results under normal measurement error and structural disturbance, while black dots represent nonnormal conditions. Shaded region indicates acceptable range.

Figure 15

Additional distributional conditions: Coverage for Simulation 1 (N = 1000).



Note. White dots represent results under normal measurement error and structural disturbance, while black dots represent nonnormal conditions. Shaded region indicates acceptable range. Error bars show MCSE around nonnormal estimates.

Appendix A

Local Standard Errors for LSAM

To obtain two-step corrected standard errors for the parameters of the structural part of the model ($\boldsymbol{\theta}_2$), we use the familiar (sandwich) formula for robust standard errors (Savalei & Rosseel, 2021):

$$\text{Var}(\boldsymbol{\theta}_2) = \frac{1}{N}\left[(\boldsymbol{\Delta}^\top \mathcal{I}_1 \boldsymbol{\Delta})^{-1}(\boldsymbol{\Delta}^\top \mathcal{I}_1 \boldsymbol{\Gamma}_\eta \mathcal{I}_1 \boldsymbol{\Delta})(\boldsymbol{\Delta}^\top \mathcal{I}_1 \boldsymbol{\Delta})^{-1}\right]$$

where $\mathcal{I}_1$ is the unit expected information matrix of the unrestricted second step model (h1), $\boldsymbol{\Delta}$ is the Jacobian of the function that maps $\boldsymbol{\theta}_2$ to $\boldsymbol{\Sigma}(\boldsymbol{\theta}_2)$, and N is the number of observations. The critical ingredient is $\boldsymbol{\Gamma}_\eta$, which represents N times the asymptotic (co)variance matrix of the sufficient statistics that enter the second step. The sufficient statistics are (a subset of) the mean vector $\mathbb{E}(\boldsymbol{\eta}^\star \otimes \boldsymbol{\eta}^\star)$ and the non-redundant elements of the variance-covariance matrix $\text{Var}(\boldsymbol{\eta}^\star \otimes \boldsymbol{\eta}^\star)$. We collect these elements in the column vector $L(\boldsymbol{\eta}^\star \otimes \boldsymbol{\eta}^\star)$:

$$L(\boldsymbol{\eta}^\star \otimes \boldsymbol{\eta}^\star) = \left[\mathbb{E}(\boldsymbol{\eta}^\star \otimes \boldsymbol{\eta}^\star),\ \text{vech}(\text{Var}(\boldsymbol{\eta}^\star \otimes \boldsymbol{\eta}^\star))\right]^\top.$$

Let $l_i(\boldsymbol{\eta}^\star \otimes \boldsymbol{\eta}^\star)$ denote the casewise contribution to $L(\boldsymbol{\eta}^\star \otimes \boldsymbol{\eta}^\star)$ so that we can write

$$L(\boldsymbol{\eta}^\star \otimes \boldsymbol{\eta}^\star) = \frac{1}{N}\sum_i^N l_i(\boldsymbol{\eta}^\star \otimes \boldsymbol{\eta}^\star).$$

Then, if we would ignore the two-step process, a naive expression for $\boldsymbol{\Gamma}_\eta$ would be

$$\boldsymbol{\Gamma}_\eta^{(naive)} = \frac{1}{N}\sum_i^N l_i^{(c)}(\boldsymbol{\eta}^\star \otimes \boldsymbol{\eta}^\star)\, l_i^{(c)}(\boldsymbol{\eta}^\star \otimes \boldsymbol{\eta}^\star)^\top$$

where $l_i^{(c)}$ is the centered version of l_i; that is $l_i^{(c)} = l_i - L$. To obtain an estimate of $\boldsymbol{\Gamma}_\eta$ that takes into account the two-stage estimation procedure, we follow a procedure similar to Wall and Amemiya (2000). Assuming the measurement errors are normally distributed, $\boldsymbol{\Gamma}_\eta$ can then be expressed as follows:

$$\boldsymbol{\Gamma}_\eta = \boldsymbol{\Gamma}_\eta^{(naive)} + N \cdot \mathbf{C}\boldsymbol{\Sigma}_{11}\mathbf{C}^\top$$

where $\boldsymbol{\Sigma}_{11}$ is the variance-covariance matrix of the step one parameters ($\boldsymbol{\theta}_1$), and the matrix $\mathbf{C}$ can be constructed as

$$\mathbf{C} = \frac{1}{N}\sum_i^N \frac{\partial l_i(\boldsymbol{\theta}_1)}{\partial \boldsymbol{\theta}_1}$$

where we now consider $l_i(\boldsymbol{\theta}_1)$ to be a function of the step one parameters ($\boldsymbol{\theta}_1$).

Appendix B

Data Generation for Simulation Studies

Population models were first calibrated under full normality to achieve the desired reliabilities ($\rho \approx 0.4, 0.6, 0.8$) and $R^2 \approx 0.30$. The resulting structural residual variances (ψ_{kk}, where k indexes dependent latent variables) and measurement error variances (θ_{jj}, where j indexes observed indicators) were then held fixed across all distributional conditions. Using the fixed variance parameters from calibration, data were generated in three stages: (i) correlated exogenous latent predictors, (ii) dependent latent variables through the structural model, and (iii) observed indicators through the measurement model.

Stage 1: Exogenous Predictors. Correlated predictor latent variables were generated using the VITA method as implemented in the *covsim* package (version 1.1.0) (Grønneberg et al., 2022) with sampling via *rvinecopulib* (version 0.7.3.1.0) (Nagler & Vatter, 2025). Predictor latent variables were specified to follow covariance structures:

$$\boldsymbol{\Phi}_1 = \begin{bmatrix} 1 & 0.375 \\ 0.375 & 1 \end{bmatrix} \quad \text{(Simulation 1)}, \qquad \boldsymbol{\Phi}_2 = \begin{bmatrix} 1 & 0.3 & 0.2 \\ 0.3 & 1 & 0.3 \\ 0.2 & 0.3 & 1 \end{bmatrix} \quad \text{(Simulation 2)}.$$

Under the normal condition, predictors had Gaussian marginals with mean zero and the specified covariance structure. Under the right-skewed condition, gamma marginals with shape = 1 and rate = 1 were employed, producing natural skewness $\gamma_1 \approx 2$ and excess kurtosis $\gamma_2 \approx 6$. Gamma-distributed variables were shifted to have population mean zero by subtracting their natural mean (shape/rate = 1), ensuring comparability across distributional conditions. Under the uniform condition, predictors were generated from symmetric uniform distributions $U[-\sqrt{3\phi_{mm}}, \sqrt{3\phi_{mm}}]$, which naturally have mean zero and variance ϕ_{mm} (the diagonal element of $\boldsymbol{\Phi}$, where m indexes exogenous latent variables). Thus, all exogenous predictors were centered at zero across all distributional conditions, differing only in their shape characteristics. In all conditions, VITA constructed a vine copula to impose the specified dependence structure while preserving the marginal distributions.

Stage 2: Structural Model. Dependent latent variables were generated according to the structural equations with fixed residual variances ψ_{kk} (determined during calibration to yield $R^2 \approx 0.30$):

$$R^2_{\eta_k} = \frac{\mathrm{Var}(\eta_k^*)}{\mathrm{Var}(\eta_k^*) + \psi_{kk}} \approx 0.30,$$

where η_k^* denotes the deterministic portion of the dependent latent variable η_k. Structural disturbances (ζ_k) were sampled independently according to the factorial design: under the normal condition from $\mathcal{N}(0, \psi_{kk})$, and under the right-skewed condition from $\mathrm{Exp}^*(0, \psi_{kk})$, a centered and scaled exponential distribution with mean 0 and variance ψ_{kk}.

Stage 3: Measurement Model. Observed indicators were generated as

$$\mathbf{y} = \boldsymbol{\Lambda}\boldsymbol{\eta} + \boldsymbol{\epsilon},$$

with unit factor loadings ($\lambda_j = 1$) and fixed error variances θ_{jj} (determined during calibration) to achieve reliabilities

$$\rho_{y_j} = \frac{\lambda_j^2 \mathrm{Var}(\eta)}{\lambda_j^2 \mathrm{Var}(\eta) + \theta_{jj}} \approx \rho, \quad \rho \in \{0.4, 0.6, 0.8\},$$

where $\mathrm{Var}(\eta)$ is the variance of the latent variable that indicator y_j measures.
Each latent variable was measured by three indicators. Measurement errors (ϵ_j) were sampled independently according to the factorial design: under the normal condition from $\mathcal{N}(0, \theta_{jj})$, and under the right-skewed condition from $\mathrm{Exp}^*(0, \theta_{jj})$, a centered and scaled exponential distribution with mean 0 and variance θ_{jj}, defined as $\sqrt{\theta_{jj}}(X - 1)$ where $X \sim \mathrm{Exp}(1)$. Note that $\mathrm{Exp}(1) \equiv \mathrm{Gamma}(1, 1)$, which is the same base distribution used for the right-skewed exogenous latent predictors.

Appendix C

Convergence Issues and Outliers Statistics

Table C1

Convergence and Outlier Statistics (Simulation 1)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
1	LMS	Full	normal	normal	normal	400	0.4	997	99.7	981	13	1.3	906
	LSAM	Full	normal	normal	normal	400	0.4	1000	100.0	981	32	3.3	906
	QML	Full	normal	normal	normal	400	0.4	993	99.3	981	5	0.5	906
	UPI	Full	normal	normal	normal	400	0.4	988	98.8	981	54	5.5	906
2	LMS	Full	normal	normal	normal	1000	0.4	1000	100.0	1000	0	0.0	994
	LSAM	Full	normal	normal	normal	1000	0.4	1000	100.0	1000	3	0.3	994
	QML	Full	normal	normal	normal	1000	0.4	1000	100.0	1000	0	0.0	994
	UPI	Full	normal	normal	normal	1000	0.4	1000	100.0	1000	4	0.4	994
3	LMS	Full	normal	normal	normal	400	0.6	1000	100.0	1000	0	0.0	998
	LSAM	Full	normal	normal	normal	400	0.6	1000	100.0	1000	2	0.2	998
	QML	Full	normal	normal	normal	400	0.6	1000	100.0	1000	0	0.0	998
	UPI	Full	normal	normal	normal	400	0.6	1000	100.0	1000	0	0.0	998
4	LMS	Full	normal	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	1000
	LSAM	Full	normal	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	1000
	QML	Full	normal	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	1000
	UPI	Full	normal	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	1000
5	LMS	Full	normal	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Full	normal	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	normal	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	normal	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000
6	LMS	Full	normal	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Full	normal	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	normal	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	normal	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
7	LMS	Full	right-skewed	normal	normal	400	0.4	992	99.2	953	15	1.6	844
	LSAM	Full	right-skewed	normal	normal	400	0.4	1000	100.0	953	54	5.7	844
	QML	Full	right-skewed	normal	normal	400	0.4	991	99.1	953	15	1.6	844
	UPI	Full	right-skewed	normal	normal	400	0.4	965	96.5	953	75	7.9	844
8	LMS	Full	right-skewed	normal	normal	1000	0.4	999	99.9	997	3	0.3	980
	LSAM	Full	right-skewed	normal	normal	1000	0.4	1000	100.0	997	10	1.0	980
	QML	Full	right-skewed	normal	normal	1000	0.4	1000	100.0	997	1	0.1	980
	UPI	Full	right-skewed	normal	normal	1000	0.4	997	99.7	997	15	1.5	980

Table C1

Convergence and Outlier Statistics (Simulation 1) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
9	LMS	Full	right-skewed	normal	normal	400	0.6	1000	100.0	997	3	0.3	986
	LSAM	Full	right-skewed	normal	normal	400	0.6	1000	100.0	997	3	0.3	986
	QML	Full	right-skewed	normal	normal	400	0.6	1000	100.0	997	2	0.2	986
	UPI	Full	right-skewed	normal	normal	400	0.6	997	99.7	997	6	0.6	986
10	LMS	Full	right-skewed	normal	normal	1000	0.6	1000	100.0	1000	1	0.1	997
	LSAM	Full	right-skewed	normal	normal	1000	0.6	1000	100.0	1000	2	0.2	997
	QML	Full	right-skewed	normal	normal	1000	0.6	1000	100.0	1000	1	0.1	997
	UPI	Full	right-skewed	normal	normal	1000	0.6	1000	100.0	1000	1	0.1	997
11	LMS	Full	right-skewed	normal	normal	400	0.8	1000	100.0	1000	0	0.0	997
	LSAM	Full	right-skewed	normal	normal	400	0.8	1000	100.0	1000	1	0.1	997
	QML	Full	right-skewed	normal	normal	400	0.8	1000	100.0	1000	0	0.0	997
	UPI	Full	right-skewed	normal	normal	400	0.8	1000	100.0	1000	3	0.3	997
12	LMS	Full	right-skewed	normal	normal	1000	0.8	1000	100.0	1000	1	0.1	999
	LSAM	Full	right-skewed	normal	normal	1000	0.8	1000	100.0	1000	1	0.1	999
	QML	Full	right-skewed	normal	normal	1000	0.8	1000	100.0	1000	1	0.1	999
	UPI	Full	right-skewed	normal	normal	1000	0.8	1000	100.0	1000	1	0.1	999
13	LMS	Full	uniform	normal	normal	400	0.4	971	97.1	706	28	4.0	552
	LSAM	Full	uniform	normal	normal	400	0.4	1000	100.0	706	60	8.5	552
	QML	Full	uniform	normal	normal	400	0.4	922	92.2	706	16	2.3	552
	UPI	Full	uniform	normal	normal	400	0.4	751	75.1	706	103	14.6	552
14	LMS	Full	uniform	normal	normal	1000	0.4	999	99.9	970	6	0.6	915
	LSAM	Full	uniform	normal	normal	1000	0.4	1000	100.0	970	13	1.3	915
	QML	Full	uniform	normal	normal	1000	0.4	994	99.4	970	3	0.3	915
	UPI	Full	uniform	normal	normal	1000	0.4	975	97.5	970	50	5.2	915
15	LMS	Full	uniform	normal	normal	400	0.6	1000	100.0	999	1	0.1	990
	LSAM	Full	uniform	normal	normal	400	0.6	1000	100.0	999	3	0.3	990
	QML	Full	uniform	normal	normal	400	0.6	1000	100.0	999	2	0.2	990
	UPI	Full	uniform	normal	normal	400	0.6	999	99.9	999	7	0.7	990
16	LMS	Full	uniform	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	997
	LSAM	Full	uniform	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	997
	QML	Full	uniform	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	997
	UPI	Full	uniform	normal	normal	1000	0.6	1000	100.0	1000	3	0.3	997
17	LMS	Full	uniform	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Full	uniform	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	uniform	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	uniform	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000

Table C1

Convergence and Outlier Statistics (Simulation 1) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
18	LMS	Full	uniform	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Full	uniform	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	uniform	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	uniform	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
19	LMS	Full	normal	right-skewed	normal	400	0.4	927	92.7	318	61	19.2	201
	LSAM	Full	normal	right-skewed	normal	400	0.4	1000	100.0	318	19	6.0	201
	QML	Full	normal	right-skewed	normal	400	0.4	874	87.4	318	24	7.6	201
	UPI	Full	normal	right-skewed	normal	400	0.4	356	35.6	318	57	17.9	201
20	LMS	Full	normal	right-skewed	normal	1000	0.4	966	96.6	354	54	15.2	254
	LSAM	Full	normal	right-skewed	normal	1000	0.4	1000	100.0	354	0	0.0	254
	QML	Full	normal	right-skewed	normal	1000	0.4	932	93.2	354	24	6.8	254
	UPI	Full	normal	right-skewed	normal	1000	0.4	386	38.6	354	60	17.0	254
21	LMS	Full	normal	right-skewed	normal	400	0.6	1000	100.0	997	3	0.3	989
	LSAM	Full	normal	right-skewed	normal	400	0.6	1000	100.0	997	1	0.1	989
	QML	Full	normal	right-skewed	normal	400	0.6	999	99.9	997	1	0.1	989
	UPI	Full	normal	right-skewed	normal	400	0.6	998	99.8	997	4	0.4	989
22	LMS	Full	normal	right-skewed	normal	1000	0.6	1000	100.0	1000	0	0.0	1000
	LSAM	Full	normal	right-skewed	normal	1000	0.6	1000	100.0	1000	0	0.0	1000
	QML	Full	normal	right-skewed	normal	1000	0.6	1000	100.0	1000	0	0.0	1000
	UPI	Full	normal	right-skewed	normal	1000	0.6	1000	100.0	1000	0	0.0	1000
23	LMS	Full	normal	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Full	normal	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	normal	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	normal	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
24	LMS	Full	normal	right-skewed	normal	1000	0.8	1000	100.0	1000	1	0.1	999
	LSAM	Full	normal	right-skewed	normal	1000	0.8	1000	100.0	1000	1	0.1	999
	QML	Full	normal	right-skewed	normal	1000	0.8	1000	100.0	1000	1	0.1	999
	UPI	Full	normal	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	999
25	LMS	Full	right-skewed	right-skewed	normal	400	0.4	995	99.5	936	56	6.0	596
	LSAM	Full	right-skewed	right-skewed	normal	400	0.4	1000	100.0	936	185	19.8	596
	QML	Full	right-skewed	right-skewed	normal	400	0.4	991	99.1	936	28	3.0	596
	UPI	Full	right-skewed	right-skewed	normal	400	0.4	946	94.6	936	138	14.7	596
26	LMS	Full	right-skewed	right-skewed	normal	1000	0.4	1000	100.0	980	30	3.1	639
	LSAM	Full	right-skewed	right-skewed	normal	1000	0.4	1000	100.0	980	246	25.1	639
	QML	Full	right-skewed	right-skewed	normal	1000	0.4	991	99.1	980	14	1.4	639
	UPI	Full	right-skewed	right-skewed	normal	1000	0.4	989	98.9	980	117	11.9	639

Table C1

Convergence and Outlier Statistics (Simulation 1) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
27	LMS	Full	right-skewed	right-skewed	normal	400	0.6	1000	100.0	977	39	4.0	883
	LSAM	Full	right-skewed	right-skewed	normal	400	0.6	1000	100.0	977	9	0.9	883
	QML	Full	right-skewed	right-skewed	normal	400	0.6	983	98.3	977	14	1.4	883
	UPI	Full	right-skewed	right-skewed	normal	400	0.6	994	99.4	977	54	5.5	883
28	LMS	Full	right-skewed	right-skewed	normal	1000	0.6	1000	100.0	989	30	3.0	946
	LSAM	Full	right-skewed	right-skewed	normal	1000	0.6	1000	100.0	989	4	0.4	946
	QML	Full	right-skewed	right-skewed	normal	1000	0.6	989	98.9	989	9	0.9	946
	UPI	Full	right-skewed	right-skewed	normal	1000	0.6	1000	100.0	989	14	1.4	946
29	LMS	Full	right-skewed	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Full	right-skewed	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	right-skewed	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	right-skewed	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
30	LMS	Full	right-skewed	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Full	right-skewed	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	right-skewed	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	right-skewed	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
31	LMS	Full	uniform	right-skewed	normal	400	0.4	832	83.2	120	26	21.7	60
	LSAM	Full	uniform	right-skewed	normal	400	0.4	1000	100.0	120	9	7.5	60
	QML	Full	uniform	right-skewed	normal	400	0.4	711	71.1	120	10	8.3	60
	UPI	Full	uniform	right-skewed	normal	400	0.4	159	15.9	120	42	35.0	60
32	LMS	Full	uniform	right-skewed	normal	1000	0.4	903	90.3	111	16	14.4	50
	LSAM	Full	uniform	right-skewed	normal	1000	0.4	1000	100.0	111	2	1.8	50
	QML	Full	uniform	right-skewed	normal	1000	0.4	808	80.8	111	6	5.4	50
	UPI	Full	uniform	right-skewed	normal	1000	0.4	141	14.1	111	50	45.0	50
33	LMS	Full	uniform	right-skewed	normal	400	0.6	996	99.6	751	9	1.2	700
	LSAM	Full	uniform	right-skewed	normal	400	0.6	1000	100.0	751	3	0.4	700
	QML	Full	uniform	right-skewed	normal	400	0.6	977	97.7	751	4	0.5	700
	UPI	Full	uniform	right-skewed	normal	400	0.6	767	76.7	751	41	5.5	700
34	LMS	Full	uniform	right-skewed	normal	1000	0.6	1000	100.0	800	5	0.6	778
	LSAM	Full	uniform	right-skewed	normal	1000	0.6	1000	100.0	800	0	0.0	778
	QML	Full	uniform	right-skewed	normal	1000	0.6	998	99.8	800	3	0.4	778
	UPI	Full	uniform	right-skewed	normal	1000	0.6	800	80.0	800	18	2.2	778
35	LMS	Full	uniform	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Full	uniform	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	uniform	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	uniform	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000

Table C1

Convergence and Outlier Statistics (Simulation 1) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
36	LMS	Full	uniform	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Full	uniform	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	uniform	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	uniform	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
37	LMS	Full	normal	normal	right-skewed	400	0.4	997	99.7	979	71	7.2	841
	LSAM	Full	normal	normal	right-skewed	400	0.4	1000	100.0	979	36	3.7	841
	QML	Full	normal	normal	right-skewed	400	0.4	994	99.4	979	10	1.0	841
	UPI	Full	normal	normal	right-skewed	400	0.4	983	98.3	979	64	6.5	841
38	LMS	Full	normal	normal	right-skewed	1000	0.4	1000	100.0	1000	95	9.5	901
	LSAM	Full	normal	normal	right-skewed	1000	0.4	1000	100.0	1000	4	0.4	901
	QML	Full	normal	normal	right-skewed	1000	0.4	1000	100.0	1000	1	0.1	901
	UPI	Full	normal	normal	right-skewed	1000	0.4	1000	100.0	1000	9	0.9	901
39	LMS	Full	normal	normal	right-skewed	400	0.6	1000	100.0	1000	19	1.9	979
	LSAM	Full	normal	normal	right-skewed	400	0.6	1000	100.0	1000	1	0.1	979
	QML	Full	normal	normal	right-skewed	400	0.6	1000	100.0	1000	2	0.2	979
	UPI	Full	normal	normal	right-skewed	400	0.6	1000	100.0	1000	3	0.3	979
40	LMS	Full	normal	normal	right-skewed	1000	0.6	1000	100.0	1000	4	0.4	996
	LSAM	Full	normal	normal	right-skewed	1000	0.6	1000	100.0	1000	0	0.0	996
	QML	Full	normal	normal	right-skewed	1000	0.6	1000	100.0	1000	0	0.0	996
	UPI	Full	normal	normal	right-skewed	1000	0.6	1000	100.0	1000	1	0.1	996
41	LMS	Full	normal	normal	right-skewed	400	0.8	1000	100.0	1000	1	0.1	999
	LSAM	Full	normal	normal	right-skewed	400	0.8	1000	100.0	1000	0	0.0	999
	QML	Full	normal	normal	right-skewed	400	0.8	1000	100.0	1000	0	0.0	999
	UPI	Full	normal	normal	right-skewed	400	0.8	1000	100.0	1000	0	0.0	999
42	LMS	Full	normal	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Full	normal	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	normal	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	normal	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
43	LMS	Full	right-skewed	normal	right-skewed	400	0.4	992	99.2	956	50	5.2	810
	LSAM	Full	right-skewed	normal	right-skewed	400	0.4	1000	100.0	956	66	6.9	810
	QML	Full	right-skewed	normal	right-skewed	400	0.4	990	99.0	956	14	1.5	810
	UPI	Full	right-skewed	normal	right-skewed	400	0.4	965	96.5	956	86	9.0	810
44	LMS	Full	right-skewed	normal	right-skewed	1000	0.4	1000	100.0	997	54	5.4	930
	LSAM	Full	right-skewed	normal	right-skewed	1000	0.4	1000	100.0	997	9	0.9	930
	QML	Full	right-skewed	normal	right-skewed	1000	0.4	1000	100.0	997	7	0.7	930
	UPI	Full	right-skewed	normal	right-skewed	1000	0.4	997	99.7	997	16	1.6	930

Table C1

Convergence and Outlier Statistics (Simulation 1) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
45	LMS	Full	right-skewed	normal	right-skewed	400	0.6	1000	100.0	997	37	3.7	951
	LSAM	Full	right-skewed	normal	right-skewed	400	0.6	1000	100.0	997	6	0.6	951
	QML	Full	right-skewed	normal	right-skewed	400	0.6	1000	100.0	997	9	0.9	951
	UPI	Full	right-skewed	normal	right-skewed	400	0.6	997	99.7	997	10	1.0	951
46	LMS	Full	right-skewed	normal	right-skewed	1000	0.6	1000	100.0	1000	4	0.4	994
	LSAM	Full	right-skewed	normal	right-skewed	1000	0.6	1000	100.0	1000	1	0.1	994
	QML	Full	right-skewed	normal	right-skewed	1000	0.6	1000	100.0	1000	1	0.1	994
	UPI	Full	right-skewed	normal	right-skewed	1000	0.6	1000	100.0	1000	2	0.2	994
47	LMS	Full	right-skewed	normal	right-skewed	400	0.8	1000	100.0	1000	3	0.3	995
	LSAM	Full	right-skewed	normal	right-skewed	400	0.8	1000	100.0	1000	4	0.4	995
	QML	Full	right-skewed	normal	right-skewed	400	0.8	1000	100.0	1000	3	0.3	995
	UPI	Full	right-skewed	normal	right-skewed	400	0.8	1000	100.0	1000	3	0.3	995
48	LMS	Full	right-skewed	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Full	right-skewed	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	right-skewed	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	right-skewed	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
49	LMS	Full	uniform	normal	right-skewed	400	0.4	970	97.0	735	86	11.7	535
	LSAM	Full	uniform	normal	right-skewed	400	0.4	1000	100.0	735	62	8.4	535
	QML	Full	uniform	normal	right-skewed	400	0.4	927	92.7	735	21	2.9	535
	UPI	Full	uniform	normal	right-skewed	400	0.4	775	77.5	735	121	16.5	535
50	LMS	Full	uniform	normal	right-skewed	1000	0.4	999	99.9	976	80	8.2	861
	LSAM	Full	uniform	normal	right-skewed	1000	0.4	1000	100.0	976	20	2.0	861
	QML	Full	uniform	normal	right-skewed	1000	0.4	992	99.2	976	4	0.4	861
	UPI	Full	uniform	normal	right-skewed	1000	0.4	985	98.5	976	48	4.9	861
51	LMS	Full	uniform	normal	right-skewed	400	0.6	1000	100.0	995	22	2.2	970
	LSAM	Full	uniform	normal	right-skewed	400	0.6	1000	100.0	995	2	0.2	970
	QML	Full	uniform	normal	right-skewed	400	0.6	1000	100.0	995	4	0.4	970
	UPI	Full	uniform	normal	right-skewed	400	0.6	995	99.5	995	3	0.3	970
52	LMS	Full	uniform	normal	right-skewed	1000	0.6	1000	100.0	1000	11	1.1	987
	LSAM	Full	uniform	normal	right-skewed	1000	0.6	1000	100.0	1000	0	0.0	987
	QML	Full	uniform	normal	right-skewed	1000	0.6	1000	100.0	1000	0	0.0	987
	UPI	Full	uniform	normal	right-skewed	1000	0.6	1000	100.0	1000	2	0.2	987
53	LMS	Full	uniform	normal	right-skewed	400	0.8	1000	100.0	1000	1	0.1	999
	LSAM	Full	uniform	normal	right-skewed	400	0.8	1000	100.0	1000	1	0.1	999
	QML	Full	uniform	normal	right-skewed	400	0.8	1000	100.0	1000	1	0.1	999
	UPI	Full	uniform	normal	right-skewed	400	0.8	1000	100.0	1000	1	0.1	999

Table C1

Convergence and Outlier Statistics (Simulation 1) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
54	LMS	Full	uniform	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	999
	LSAM	Full	uniform	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	999
	QML	Full	uniform	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	999
	UPI	Full	uniform	normal	right-skewed	1000	0.8	1000	100.0	1000	1	0.1	999
55	LMS	Full	normal	right-skewed	right-skewed	400	0.4	936	93.6	325	65	20.0	216
	LSAM	Full	normal	right-skewed	right-skewed	400	0.4	1000	100.0	325	14	4.3	216
	QML	Full	normal	right-skewed	right-skewed	400	0.4	854	85.4	325	19	5.8	216
	UPI	Full	normal	right-skewed	right-skewed	400	0.4	374	37.4	325	50	15.4	216
56	LMS	Full	normal	right-skewed	right-skewed	1000	0.4	961	96.1	323	73	22.6	207
	LSAM	Full	normal	right-skewed	right-skewed	1000	0.4	1000	100.0	323	2	0.6	207
	QML	Full	normal	right-skewed	right-skewed	1000	0.4	915	91.5	323	10	3.1	207
	UPI	Full	normal	right-skewed	right-skewed	1000	0.4	350	35.0	323	57	17.6	207
57	LMS	Full	normal	right-skewed	right-skewed	400	0.6	1000	100.0	995	23	2.3	970
	LSAM	Full	normal	right-skewed	right-skewed	400	0.6	1000	100.0	995	1	0.1	970
	QML	Full	normal	right-skewed	right-skewed	400	0.6	998	99.8	995	3	0.3	970
	UPI	Full	normal	right-skewed	right-skewed	400	0.6	997	99.7	995	3	0.3	970
58	LMS	Full	normal	right-skewed	right-skewed	1000	0.6	1000	100.0	1000	5	0.5	994
	LSAM	Full	normal	right-skewed	right-skewed	1000	0.6	1000	100.0	1000	1	0.1	994
	QML	Full	normal	right-skewed	right-skewed	1000	0.6	1000	100.0	1000	2	0.2	994
	UPI	Full	normal	right-skewed	right-skewed	1000	0.6	1000	100.0	1000	1	0.1	994
59	LMS	Full	normal	right-skewed	right-skewed	400	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Full	normal	right-skewed	right-skewed	400	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	normal	right-skewed	right-skewed	400	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	normal	right-skewed	right-skewed	400	0.8	1000	100.0	1000	0	0.0	1000
60	LMS	Full	normal	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Full	normal	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	normal	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	normal	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
61	LMS	Full	right-skewed	right-skewed	right-skewed	400	0.4	993	99.3	930	57	6.1	595
	LSAM	Full	right-skewed	right-skewed	right-skewed	400	0.4	1000	100.0	930	183	19.7	595
	QML	Full	right-skewed	right-skewed	right-skewed	400	0.4	978	97.8	930	20	2.1	595
	UPI	Full	right-skewed	right-skewed	right-skewed	400	0.4	949	94.9	930	143	15.4	595
62	LMS	Full	right-skewed	right-skewed	right-skewed	1000	0.4	1000	100.0	978	139	14.2	568
	LSAM	Full	right-skewed	right-skewed	right-skewed	1000	0.4	1000	100.0	978	241	24.6	568
	QML	Full	right-skewed	right-skewed	right-skewed	1000	0.4	990	99.0	978	17	1.7	568
	UPI	Full	right-skewed	right-skewed	right-skewed	1000	0.4	988	98.8	978	111	11.3	568

Table C1

Convergence and Outlier Statistics (Simulation 1) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
63	LMS	Full	right-skewed	right-skewed	right-skewed	400	0.6	999	99.9	982	58	5.9	873
	LSAM	Full	right-skewed	right-skewed	right-skewed	400	0.6	1000	100.0	982	13	1.3	873
	QML	Full	right-skewed	right-skewed	right-skewed	400	0.6	983	98.3	982	9	0.9	873
	UPI	Full	right-skewed	right-skewed	right-skewed	400	0.6	999	99.9	982	53	5.4	873
64	LMS	Full	right-skewed	right-skewed	right-skewed	1000	0.6	1000	100.0	991	30	3.0	951
	LSAM	Full	right-skewed	right-skewed	right-skewed	1000	0.6	1000	100.0	991	1	0.1	951
	QML	Full	right-skewed	right-skewed	right-skewed	1000	0.6	991	99.1	991	7	0.7	951
	UPI	Full	right-skewed	right-skewed	right-skewed	1000	0.6	1000	100.0	991	10	1.0	951
65	LMS	Full	right-skewed	right-skewed	right-skewed	400	0.8	1000	100.0	1000	6	0.6	992
	LSAM	Full	right-skewed	right-skewed	right-skewed	400	0.8	1000	100.0	1000	5	0.5	992
	QML	Full	right-skewed	right-skewed	right-skewed	400	0.8	1000	100.0	1000	6	0.6	992
	UPI	Full	right-skewed	right-skewed	right-skewed	400	0.8	1000	100.0	1000	2	0.2	992
66	LMS	Full	right-skewed	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Full	right-skewed	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	right-skewed	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	right-skewed	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
67	LMS	Full	uniform	right-skewed	right-skewed	400	0.4	867	86.7	104	31	29.8	38
	LSAM	Full	uniform	right-skewed	right-skewed	400	0.4	1000	100.0	104	15	14.4	38
	QML	Full	uniform	right-skewed	right-skewed	400	0.4	717	71.7	104	9	8.7	38
	UPI	Full	uniform	right-skewed	right-skewed	400	0.4	130	13.0	104	40	38.5	38
68	LMS	Full	uniform	right-skewed	right-skewed	1000	0.4	894	89.4	100	24	24.0	40
	LSAM	Full	uniform	right-skewed	right-skewed	1000	0.4	1000	100.0	100	2	2.0	40
	QML	Full	uniform	right-skewed	right-skewed	1000	0.4	819	81.9	100	3	3.0	40
	UPI	Full	uniform	right-skewed	right-skewed	1000	0.4	120	12.0	100	49	49.0	40
69	LMS	Full	uniform	right-skewed	right-skewed	400	0.6	997	99.7	759	31	4.1	689
	LSAM	Full	uniform	right-skewed	right-skewed	400	0.6	1000	100.0	759	1	0.1	689
	QML	Full	uniform	right-skewed	right-skewed	400	0.6	972	97.2	759	6	0.8	689
	UPI	Full	uniform	right-skewed	right-skewed	400	0.6	773	77.3	759	41	5.4	689
70	LMS	Full	uniform	right-skewed	right-skewed	1000	0.6	1000	100.0	804	6	0.8	769
	LSAM	Full	uniform	right-skewed	right-skewed	1000	0.6	1000	100.0	804	1	0.1	769
	QML	Full	uniform	right-skewed	right-skewed	1000	0.6	990	99.0	804	2	0.2	769
	UPI	Full	uniform	right-skewed	right-skewed	1000	0.6	812	81.2	804	31	3.9	769
71	LMS	Full	uniform	right-skewed	right-skewed	400	0.8	1000	100.0	1000	1	0.1	999
	LSAM	Full	uniform	right-skewed	right-skewed	400	0.8	1000	100.0	1000	0	0.0	999
	QML	Full	uniform	right-skewed	right-skewed	400	0.8	1000	100.0	1000	0	0.0	999
	UPI	Full	uniform	right-skewed	right-skewed	400	0.8	1000	100.0	1000	0	0.0	999

Table C1

Convergence and Outlier Statistics (Simulation 1) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
72	LMS	Full	uniform	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Full	uniform	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	uniform	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	uniform	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
73	LMS	Linear	normal	normal	normal	400	0.4	998	99.8	974	11	1.1	901
	LSAM	Linear	normal	normal	normal	400	0.4	1000	100.0	974	31	3.2	901
	QML	Linear	normal	normal	normal	400	0.4	989	98.9	974	9	0.9	901
	UPI	Linear	normal	normal	normal	400	0.4	983	98.3	974	53	5.4	901
74	LMS	Linear	normal	normal	normal	1000	0.4	1000	100.0	1000	1	0.1	997
	LSAM	Linear	normal	normal	normal	1000	0.4	1000	100.0	1000	1	0.1	997
	QML	Linear	normal	normal	normal	1000	0.4	1000	100.0	1000	1	0.1	997
	UPI	Linear	normal	normal	normal	1000	0.4	1000	100.0	1000	3	0.3	997
75	LMS	Linear	normal	normal	normal	400	0.6	1000	100.0	1000	0	0.0	997
	LSAM	Linear	normal	normal	normal	400	0.6	1000	100.0	1000	1	0.1	997
	QML	Linear	normal	normal	normal	400	0.6	1000	100.0	1000	0	0.0	997
	UPI	Linear	normal	normal	normal	400	0.6	1000	100.0	1000	3	0.3	997
76	LMS	Linear	normal	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	1000
	LSAM	Linear	normal	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	1000
	QML	Linear	normal	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	1000
	UPI	Linear	normal	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	1000
77	LMS	Linear	normal	normal	normal	400	0.8	1000	100.0	1000	1	0.1	999
	LSAM	Linear	normal	normal	normal	400	0.8	1000	100.0	1000	0	0.0	999
	QML	Linear	normal	normal	normal	400	0.8	1000	100.0	1000	1	0.1	999
	UPI	Linear	normal	normal	normal	400	0.8	1000	100.0	1000	1	0.1	999
78	LMS	Linear	normal	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Linear	normal	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	normal	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	normal	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
79	LMS	Linear	right-skewed	normal	normal	400	0.4	994	99.4	964	21	2.2	842
	LSAM	Linear	right-skewed	normal	normal	400	0.4	1000	100.0	964	60	6.2	842
	QML	Linear	right-skewed	normal	normal	400	0.4	994	99.4	964	16	1.7	842
	UPI	Linear	right-skewed	normal	normal	400	0.4	972	97.2	964	92	9.5	842
80	LMS	Linear	right-skewed	normal	normal	1000	0.4	999	99.9	996	1	0.1	973
	LSAM	Linear	right-skewed	normal	normal	1000	0.4	1000	100.0	996	8	0.8	973
	QML	Linear	right-skewed	normal	normal	1000	0.4	1000	100.0	996	1	0.1	973
	UPI	Linear	right-skewed	normal	normal	1000	0.4	996	99.6	996	18	1.8	973

Table C1

Convergence and Outlier Statistics (Simulation 1) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
81	LMS	Linear	right-skewed	normal	normal	400	0.6	999	99.9	998	1	0.1	985
	LSAM	Linear	right-skewed	normal	normal	400	0.6	1000	100.0	998	4	0.4	985
	QML	Linear	right-skewed	normal	normal	400	0.6	1000	100.0	998	1	0.1	985
	UPI	Linear	right-skewed	normal	normal	400	0.6	998	99.8	998	11	1.1	985
82	LMS	Linear	right-skewed	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	999
	LSAM	Linear	right-skewed	normal	normal	1000	0.6	1000	100.0	1000	1	0.1	999
	QML	Linear	right-skewed	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	999
	UPI	Linear	right-skewed	normal	normal	1000	0.6	1000	100.0	1000	1	0.1	999
83	LMS	Linear	right-skewed	normal	normal	400	0.8	1000	100.0	1000	0	0.0	999
	LSAM	Linear	right-skewed	normal	normal	400	0.8	1000	100.0	1000	1	0.1	999
	QML	Linear	right-skewed	normal	normal	400	0.8	1000	100.0	1000	0	0.0	999
	UPI	Linear	right-skewed	normal	normal	400	0.8	1000	100.0	1000	1	0.1	999
84	LMS	Linear	right-skewed	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Linear	right-skewed	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	right-skewed	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	right-skewed	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
85	LMS	Linear	uniform	normal	normal	400	0.4	968	96.8	675	30	4.4	517
	LSAM	Linear	uniform	normal	normal	400	0.4	1000	100.0	675	60	8.9	517
	QML	Linear	uniform	normal	normal	400	0.4	904	90.4	675	16	2.4	517
	UPI	Linear	uniform	normal	normal	400	0.4	730	73.0	675	112	16.6	517
86	LMS	Linear	uniform	normal	normal	1000	0.4	999	99.9	955	11	1.1	899
	LSAM	Linear	uniform	normal	normal	1000	0.4	1000	100.0	955	10	1.0	899
	QML	Linear	uniform	normal	normal	1000	0.4	995	99.5	955	6	0.6	899
	UPI	Linear	uniform	normal	normal	1000	0.4	960	96.0	955	43	4.5	899
87	LMS	Linear	uniform	normal	normal	400	0.6	1000	100.0	996	0	0.0	991
	LSAM	Linear	uniform	normal	normal	400	0.6	1000	100.0	996	0	0.0	991
	QML	Linear	uniform	normal	normal	400	0.6	1000	100.0	996	0	0.0	991
	UPI	Linear	uniform	normal	normal	400	0.6	996	99.6	996	5	0.5	991
88	LMS	Linear	uniform	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	999
	LSAM	Linear	uniform	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	999
	QML	Linear	uniform	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	999
	UPI	Linear	uniform	normal	normal	1000	0.6	1000	100.0	1000	1	0.1	999
89	LMS	Linear	uniform	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Linear	uniform	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	uniform	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	uniform	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000

Table C1

Convergence and Outlier Statistics (Simulation 1) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
90	LMS	Linear	uniform	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Linear	uniform	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	uniform	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	uniform	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
91	LMS	Linear	normal	right-skewed	normal	400	0.4	933	93.3	337	54	16.0	231
	LSAM	Linear	normal	right-skewed	normal	400	0.4	1000	100.0	337	13	3.9	231
	QML	Linear	normal	right-skewed	normal	400	0.4	827	82.7	337	17	5.0	231
	UPI	Linear	normal	right-skewed	normal	400	0.4	402	40.2	337	56	16.6	231
92	LMS	Linear	normal	right-skewed	normal	1000	0.4	955	95.5	366	47	12.8	244
	LSAM	Linear	normal	right-skewed	normal	1000	0.4	1000	100.0	366	1	0.3	244
	QML	Linear	normal	right-skewed	normal	1000	0.4	903	90.3	366	16	4.4	244
	UPI	Linear	normal	right-skewed	normal	1000	0.4	408	40.8	366	81	22.1	244
93	LMS	Linear	normal	right-skewed	normal	400	0.6	1000	100.0	994	2	0.2	989
	LSAM	Linear	normal	right-skewed	normal	400	0.6	1000	100.0	994	1	0.1	989
	QML	Linear	normal	right-skewed	normal	400	0.6	997	99.7	994	1	0.1	989
	UPI	Linear	normal	right-skewed	normal	400	0.6	997	99.7	994	4	0.4	989
94	LMS	Linear	normal	right-skewed	normal	1000	0.6	1000	100.0	1000	0	0.0	999
	LSAM	Linear	normal	right-skewed	normal	1000	0.6	1000	100.0	1000	1	0.1	999
	QML	Linear	normal	right-skewed	normal	1000	0.6	1000	100.0	1000	0	0.0	999
	UPI	Linear	normal	right-skewed	normal	1000	0.6	1000	100.0	1000	1	0.1	999
95	LMS	Linear	normal	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	999
	LSAM	Linear	normal	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	999
	QML	Linear	normal	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	999
	UPI	Linear	normal	right-skewed	normal	400	0.8	1000	100.0	1000	1	0.1	999
96	LMS	Linear	normal	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	999
	LSAM	Linear	normal	right-skewed	normal	1000	0.8	1000	100.0	1000	1	0.1	999
	QML	Linear	normal	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	999
	UPI	Linear	normal	right-skewed	normal	1000	0.8	1000	100.0	1000	1	0.1	999
97	LMS	Linear	right-skewed	right-skewed	normal	400	0.4	991	99.1	882	90	10.2	499
	LSAM	Linear	right-skewed	right-skewed	normal	400	0.4	1000	100.0	882	173	19.6	499
	QML	Linear	right-skewed	right-skewed	normal	400	0.4	963	96.3	882	38	4.3	499
	UPI	Linear	right-skewed	right-skewed	normal	400	0.4	914	91.4	882	182	20.6	499
98	LMS	Linear	right-skewed	right-skewed	normal	1000	0.4	999	99.9	954	73	7.7	575
	LSAM	Linear	right-skewed	right-skewed	normal	1000	0.4	1000	100.0	954	223	23.4	575
	QML	Linear	right-skewed	right-skewed	normal	1000	0.4	987	98.7	954	32	3.4	575
	UPI	Linear	right-skewed	right-skewed	normal	1000	0.4	967	96.7	954	143	15.0	575

Table C1

Convergence and Outlier Statistics (Simulation 1) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
99	LMS	Linear	right-skewed	right-skewed	normal	400	0.6	1000	100.0	983	41	4.2	893
	LSAM	Linear	right-skewed	right-skewed	normal	400	0.6	1000	100.0	983	19	1.9	893
	QML	Linear	right-skewed	right-skewed	normal	400	0.6	987	98.7	983	15	1.5	893
	UPI	Linear	right-skewed	right-skewed	normal	400	0.6	996	99.6	983	50	5.1	893
100	LMS	Linear	right-skewed	right-skewed	normal	1000	0.6	1000	100.0	991	27	2.7	951
	LSAM	Linear	right-skewed	right-skewed	normal	1000	0.6	1000	100.0	991	3	0.3	951
	QML	Linear	right-skewed	right-skewed	normal	1000	0.6	991	99.1	991	14	1.4	951
	UPI	Linear	right-skewed	right-skewed	normal	1000	0.6	1000	100.0	991	14	1.4	951
101	LMS	Linear	right-skewed	right-skewed	normal	400	0.8	1000	100.0	1000	2	0.2	996
	LSAM	Linear	right-skewed	right-skewed	normal	400	0.8	1000	100.0	1000	1	0.1	996
	QML	Linear	right-skewed	right-skewed	normal	400	0.8	1000	100.0	1000	2	0.2	996
	UPI	Linear	right-skewed	right-skewed	normal	400	0.8	1000	100.0	1000	3	0.3	996
102	LMS	Linear	right-skewed	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Linear	right-skewed	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	right-skewed	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	right-skewed	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
103	LMS	Linear	uniform	right-skewed	normal	400	0.4	825	82.5	113	19	16.8	66
	LSAM	Linear	uniform	right-skewed	normal	400	0.4	1000	100.0	113	8	7.1	66
	QML	Linear	uniform	right-skewed	normal	400	0.4	719	71.9	113	10	8.8	66
	UPI	Linear	uniform	right-skewed	normal	400	0.4	160	16.0	113	35	31.0	66
104	LMS	Linear	uniform	right-skewed	normal	1000	0.4	893	89.3	112	9	8.0	52
	LSAM	Linear	uniform	right-skewed	normal	1000	0.4	1000	100.0	112	3	2.7	52
	QML	Linear	uniform	right-skewed	normal	1000	0.4	786	78.6	112	4	3.6	52
	UPI	Linear	uniform	right-skewed	normal	1000	0.4	132	13.2	112	52	46.4	52
105	LMS	Linear	uniform	right-skewed	normal	400	0.6	989	98.9	740	14	1.9	689
	LSAM	Linear	uniform	right-skewed	normal	400	0.6	1000	100.0	740	4	0.5	689
	QML	Linear	uniform	right-skewed	normal	400	0.6	960	96.0	740	5	0.7	689
	UPI	Linear	uniform	right-skewed	normal	400	0.6	759	75.9	740	39	5.3	689
106	LMS	Linear	uniform	right-skewed	normal	1000	0.6	1000	100.0	792	3	0.4	756
	LSAM	Linear	uniform	right-skewed	normal	1000	0.6	1000	100.0	792	0	0.0	756
	QML	Linear	uniform	right-skewed	normal	1000	0.6	992	99.2	792	2	0.2	756
	UPI	Linear	uniform	right-skewed	normal	1000	0.6	796	79.6	792	34	4.3	756
107	LMS	Linear	uniform	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	999
	LSAM	Linear	uniform	right-skewed	normal	400	0.8	1000	100.0	1000	1	0.1	999
	QML	Linear	uniform	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	999
	UPI	Linear	uniform	right-skewed	normal	400	0.8	1000	100.0	1000	1	0.1	999

Table C1

Convergence and Outlier Statistics (Simulation 1) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
108	LMS	Linear	uniform	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Linear	uniform	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	uniform	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	uniform	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
109	LMS	Linear	normal	normal	right-skewed	400	0.4	999	99.9	968	110	11.4	811
	LSAM	Linear	normal	normal	right-skewed	400	0.4	1000	100.0	968	30	3.1	811
	QML	Linear	normal	normal	right-skewed	400	0.4	993	99.3	968	7	0.7	811
	UPI	Linear	normal	normal	right-skewed	400	0.4	975	97.5	968	56	5.8	811
110	LMS	Linear	normal	normal	right-skewed	1000	0.4	1000	100.0	1000	67	6.7	930
	LSAM	Linear	normal	normal	right-skewed	1000	0.4	1000	100.0	1000	0	0.0	930
	QML	Linear	normal	normal	right-skewed	1000	0.4	1000	100.0	1000	3	0.3	930
	UPI	Linear	normal	normal	right-skewed	1000	0.4	1000	100.0	1000	3	0.3	930
111	LMS	Linear	normal	normal	right-skewed	400	0.6	1000	100.0	1000	11	1.1	988
	LSAM	Linear	normal	normal	right-skewed	400	0.6	1000	100.0	1000	1	0.1	988
	QML	Linear	normal	normal	right-skewed	400	0.6	1000	100.0	1000	1	0.1	988
	UPI	Linear	normal	normal	right-skewed	400	0.6	1000	100.0	1000	2	0.2	988
112	LMS	Linear	normal	normal	right-skewed	1000	0.6	1000	100.0	1000	4	0.4	996
	LSAM	Linear	normal	normal	right-skewed	1000	0.6	1000	100.0	1000	1	0.1	996
	QML	Linear	normal	normal	right-skewed	1000	0.6	1000	100.0	1000	1	0.1	996
	UPI	Linear	normal	normal	right-skewed	1000	0.6	1000	100.0	1000	0	0.0	996
113	LMS	Linear	normal	normal	right-skewed	400	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Linear	normal	normal	right-skewed	400	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	normal	normal	right-skewed	400	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	normal	normal	right-skewed	400	0.8	1000	100.0	1000	0	0.0	1000
114	LMS	Linear	normal	normal	right-skewed	1000	0.8	1000	100.0	1000	1	0.1	999
	LSAM	Linear	normal	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	999
	QML	Linear	normal	normal	right-skewed	1000	0.8	1000	100.0	1000	1	0.1	999
	UPI	Linear	normal	normal	right-skewed	1000	0.8	1000	100.0	1000	1	0.1	999
115	LMS	Linear	right-skewed	normal	right-skewed	400	0.4	993	99.3	949	125	13.2	759
	LSAM	Linear	right-skewed	normal	right-skewed	400	0.4	1000	100.0	949	56	5.9	759
	QML	Linear	right-skewed	normal	right-skewed	400	0.4	991	99.1	949	18	1.9	759
	UPI	Linear	right-skewed	normal	right-skewed	400	0.4	961	96.1	949	75	7.9	759
116	LMS	Linear	right-skewed	normal	right-skewed	1000	0.4	999	99.9	996	85	8.5	903
	LSAM	Linear	right-skewed	normal	right-skewed	1000	0.4	1000	100.0	996	6	0.6	903
	QML	Linear	right-skewed	normal	right-skewed	1000	0.4	1000	100.0	996	0	0.0	903
	UPI	Linear	right-skewed	normal	right-skewed	1000	0.4	997	99.7	996	8	0.8	903

Table C1

Convergence and Outlier Statistics (Simulation 1) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
117	LMS	Linear	right-skewed	normal	right-skewed	400	0.6	1000	100.0	996	10	1.0	972
	LSAM	Linear	right-skewed	normal	right-skewed	400	0.6	1000	100.0	996	5	0.5	972
	QML	Linear	right-skewed	normal	right-skewed	400	0.6	1000	100.0	996	4	0.4	972
	UPI	Linear	right-skewed	normal	right-skewed	400	0.6	996	99.6	996	15	1.5	972
118	LMS	Linear	right-skewed	normal	right-skewed	1000	0.6	1000	100.0	1000	9	0.9	989
	LSAM	Linear	right-skewed	normal	right-skewed	1000	0.6	1000	100.0	1000	1	0.1	989
	QML	Linear	right-skewed	normal	right-skewed	1000	0.6	1000	100.0	1000	4	0.4	989
	UPI	Linear	right-skewed	normal	right-skewed	1000	0.6	1000	100.0	1000	3	0.3	989
119	LMS	Linear	right-skewed	normal	right-skewed	400	0.8	1000	100.0	1000	6	0.6	991
	LSAM	Linear	right-skewed	normal	right-skewed	400	0.8	1000	100.0	1000	2	0.2	991
	QML	Linear	right-skewed	normal	right-skewed	400	0.8	1000	100.0	1000	4	0.4	991
	UPI	Linear	right-skewed	normal	right-skewed	400	0.8	1000	100.0	1000	3	0.3	991
120	LMS	Linear	right-skewed	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Linear	right-skewed	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	right-skewed	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	right-skewed	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
121	LMS	Linear	uniform	normal	right-skewed	400	0.4	966	96.6	700	90	12.9	513
	LSAM	Linear	uniform	normal	right-skewed	400	0.4	1000	100.0	700	63	9.0	513
	QML	Linear	uniform	normal	right-skewed	400	0.4	909	90.9	700	23	3.3	513
	UPI	Linear	uniform	normal	right-skewed	400	0.4	760	76.0	700	99	14.1	513
122	LMS	Linear	uniform	normal	right-skewed	1000	0.4	999	99.9	964	84	8.7	843
	LSAM	Linear	uniform	normal	right-skewed	1000	0.4	1000	100.0	964	19	2.0	843
	QML	Linear	uniform	normal	right-skewed	1000	0.4	993	99.3	964	6	0.6	843
	UPI	Linear	uniform	normal	right-skewed	1000	0.4	971	97.1	964	44	4.6	843
123	LMS	Linear	uniform	normal	right-skewed	400	0.6	1000	100.0	997	17	1.7	976
	LSAM	Linear	uniform	normal	right-skewed	400	0.6	1000	100.0	997	0	0.0	976
	QML	Linear	uniform	normal	right-skewed	400	0.6	1000	100.0	997	1	0.1	976
	UPI	Linear	uniform	normal	right-skewed	400	0.6	997	99.7	997	4	0.4	976
124	LMS	Linear	uniform	normal	right-skewed	1000	0.6	1000	100.0	1000	5	0.5	995
	LSAM	Linear	uniform	normal	right-skewed	1000	0.6	1000	100.0	1000	0	0.0	995
	QML	Linear	uniform	normal	right-skewed	1000	0.6	1000	100.0	1000	0	0.0	995
	UPI	Linear	uniform	normal	right-skewed	1000	0.6	1000	100.0	1000	0	0.0	995
125	LMS	Linear	uniform	normal	right-skewed	400	0.8	1000	100.0	1000	3	0.3	997
	LSAM	Linear	uniform	normal	right-skewed	400	0.8	1000	100.0	1000	0	0.0	997
	QML	Linear	uniform	normal	right-skewed	400	0.8	1000	100.0	1000	0	0.0	997
	UPI	Linear	uniform	normal	right-skewed	400	0.8	1000	100.0	1000	0	0.0	997

Table C1

Convergence and Outlier Statistics (Simulation 1) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
126	LMS	Linear	uniform	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Linear	uniform	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	uniform	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	uniform	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
127	LMS	Linear	normal	right-skewed	right-skewed	400	0.4	919	91.9	313	93	29.7	162
	LSAM	Linear	normal	right-skewed	right-skewed	400	0.4	1000	100.0	313	16	5.1	162
	QML	Linear	normal	right-skewed	right-skewed	400	0.4	814	81.4	313	17	5.4	162
	UPI	Linear	normal	right-skewed	right-skewed	400	0.4	381	38.1	313	77	24.6	162
128	LMS	Linear	normal	right-skewed	right-skewed	1000	0.4	966	96.6	351	105	29.9	194
	LSAM	Linear	normal	right-skewed	right-skewed	1000	0.4	1000	100.0	351	2	0.6	194
	QML	Linear	normal	right-skewed	right-skewed	1000	0.4	897	89.7	351	19	5.4	194
	UPI	Linear	normal	right-skewed	right-skewed	1000	0.4	394	39.4	351	76	21.6	194
129	LMS	Linear	normal	right-skewed	right-skewed	400	0.6	1000	100.0	993	24	2.4	966
	LSAM	Linear	normal	right-skewed	right-skewed	400	0.6	1000	100.0	993	2	0.2	966
	QML	Linear	normal	right-skewed	right-skewed	400	0.6	998	99.8	993	5	0.5	966
	UPI	Linear	normal	right-skewed	right-skewed	400	0.6	995	99.5	993	5	0.5	966
130	LMS	Linear	normal	right-skewed	right-skewed	1000	0.6	1000	100.0	1000	5	0.5	995
	LSAM	Linear	normal	right-skewed	right-skewed	1000	0.6	1000	100.0	1000	0	0.0	995
	QML	Linear	normal	right-skewed	right-skewed	1000	0.6	1000	100.0	1000	2	0.2	995
	UPI	Linear	normal	right-skewed	right-skewed	1000	0.6	1000	100.0	1000	0	0.0	995
131	LMS	Linear	normal	right-skewed	right-skewed	400	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Linear	normal	right-skewed	right-skewed	400	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	normal	right-skewed	right-skewed	400	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	normal	right-skewed	right-skewed	400	0.8	1000	100.0	1000	0	0.0	1000
132	LMS	Linear	normal	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Linear	normal	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	normal	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	normal	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
133	LMS	Linear	right-skewed	right-skewed	right-skewed	400	0.4	990	99.0	892	201	22.5	476
	LSAM	Linear	right-skewed	right-skewed	right-skewed	400	0.4	1000	100.0	892	159	17.8	476
	QML	Linear	right-skewed	right-skewed	right-skewed	400	0.4	955	95.5	892	42	4.7	476
	UPI	Linear	right-skewed	right-skewed	right-skewed	400	0.4	931	93.1	892	165	18.5	476
134	LMS	Linear	right-skewed	right-skewed	right-skewed	1000	0.4	1000	100.0	953	180	18.9	518
	LSAM	Linear	right-skewed	right-skewed	right-skewed	1000	0.4	1000	100.0	953	221	23.2	518
	QML	Linear	right-skewed	right-skewed	right-skewed	1000	0.4	985	98.5	953	30	3.1	518
	UPI	Linear	right-skewed	right-skewed	right-skewed	1000	0.4	967	96.7	953	142	14.9	518

Table C1

Convergence and Outlier Statistics (Simulation 1) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
135	LMS	Linear	right-skewed	right-skewed	right-skewed	400	0.6	1000	100.0	977	52	5.3	872
	LSAM	Linear	right-skewed	right-skewed	right-skewed	400	0.6	1000	100.0	977	23	2.4	872
	QML	Linear	right-skewed	right-skewed	right-skewed	400	0.6	981	98.1	977	20	2.0	872
	UPI	Linear	right-skewed	right-skewed	right-skewed	400	0.6	995	99.5	977	61	6.2	872
136	LMS	Linear	right-skewed	right-skewed	right-skewed	1000	0.6	1000	100.0	974	37	3.8	922
	LSAM	Linear	right-skewed	right-skewed	right-skewed	1000	0.6	1000	100.0	974	1	0.1	922
	QML	Linear	right-skewed	right-skewed	right-skewed	1000	0.6	974	97.4	974	16	1.6	922
	UPI	Linear	right-skewed	right-skewed	right-skewed	1000	0.6	1000	100.0	974	15	1.5	922
137	LMS	Linear	right-skewed	right-skewed	right-skewed	400	0.8	1000	100.0	1000	8	0.8	988
	LSAM	Linear	right-skewed	right-skewed	right-skewed	400	0.8	1000	100.0	1000	8	0.8	988
	QML	Linear	right-skewed	right-skewed	right-skewed	400	0.8	1000	100.0	1000	7	0.7	988
	UPI	Linear	right-skewed	right-skewed	right-skewed	400	0.8	1000	100.0	1000	10	1.0	988
138	LMS	Linear	right-skewed	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	1	0.1	998
	LSAM	Linear	right-skewed	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	1	0.1	998
	QML	Linear	right-skewed	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	998
	UPI	Linear	right-skewed	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	1	0.1	998
139	LMS	Linear	uniform	right-skewed	right-skewed	400	0.4	820	82.0	118	42	35.6	52
	LSAM	Linear	uniform	right-skewed	right-skewed	400	0.4	1000	100.0	118	11	9.3	52
	QML	Linear	uniform	right-skewed	right-skewed	400	0.4	689	68.9	118	12	10.2	52
	UPI	Linear	uniform	right-skewed	right-skewed	400	0.4	154	15.4	118	40	33.9	52
140	LMS	Linear	uniform	right-skewed	right-skewed	1000	0.4	875	87.5	115	40	34.8	41
	LSAM	Linear	uniform	right-skewed	right-skewed	1000	0.4	1000	100.0	115	3	2.6	41
	QML	Linear	uniform	right-skewed	right-skewed	1000	0.4	791	79.1	115	9	7.8	41
	UPI	Linear	uniform	right-skewed	right-skewed	1000	0.4	140	14.0	115	56	48.7	41
141	LMS	Linear	uniform	right-skewed	right-skewed	400	0.6	994	99.4	721	38	5.3	636
	LSAM	Linear	uniform	right-skewed	right-skewed	400	0.6	1000	100.0	721	4	0.6	636
	QML	Linear	uniform	right-skewed	right-skewed	400	0.6	975	97.5	721	10	1.4	636
	UPI	Linear	uniform	right-skewed	right-skewed	400	0.6	735	73.5	721	49	6.8	636
142	LMS	Linear	uniform	right-skewed	right-skewed	1000	0.6	1000	100.0	785	8	1.0	744
	LSAM	Linear	uniform	right-skewed	right-skewed	1000	0.6	1000	100.0	785	2	0.2	744
	QML	Linear	uniform	right-skewed	right-skewed	1000	0.6	996	99.6	785	3	0.4	744
	UPI	Linear	uniform	right-skewed	right-skewed	1000	0.6	788	78.8	785	33	4.2	744
143	LMS	Linear	uniform	right-skewed	right-skewed	400	0.8	1000	100.0	1000	2	0.2	998
	LSAM	Linear	uniform	right-skewed	right-skewed	400	0.8	1000	100.0	1000	0	0.0	998
	QML	Linear	uniform	right-skewed	right-skewed	400	0.8	1000	100.0	1000	2	0.2	998
	UPI	Linear	uniform	right-skewed	right-skewed	400	0.8	1000	100.0	1000	0	0.0	998

Table C1

Convergence and Outlier Statistics (Simulation 1) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
144	LMS	Linear	uniform	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	LSAM	Linear	uniform	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	uniform	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	uniform	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000

Note. Distr. = distribution of exogenous latent variables. Epsilon = distribution of measurement errors. Zeta = distribution of structural residuals. N Conv. = number of replications that converged for this method. Conv. Rate = percentage of replications that converged (%). N Global Conv. = sample size after entire non-convergent iterations across approaches excluded. N Outliers = number of outliers detected for this method. Outlier Rate = percentage of converged cases flagged as outliers (%). N Final = final sample size after global outlier removal across methods.

Table C2

Convergence and Outlier Statistics (Simulation 2)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
1	LSAM	Full	normal	normal	normal	400	0.4	1000	100.0	886	18	2.0	722
	QML	Full	normal	normal	normal	400	0.4	940	94.0	886	9	1.0	722
	UPI	Full	normal	normal	normal	400	0.4	938	93.8	886	147	16.6	722
2	LSAM	Full	normal	normal	normal	1000	0.4	1000	100.0	999	4	0.4	984
	QML	Full	normal	normal	normal	1000	0.4	999	99.9	999	0	0.0	984
	UPI	Full	normal	normal	normal	1000	0.4	1000	100.0	999	15	1.5	984
3	LSAM	Full	normal	normal	normal	400	0.6	1000	100.0	1000	1	0.1	992
	QML	Full	normal	normal	normal	400	0.6	1000	100.0	1000	0	0.0	992
	UPI	Full	normal	normal	normal	400	0.6	1000	100.0	1000	7	0.7	992
4	LSAM	Full	normal	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	1000
	QML	Full	normal	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	1000
	UPI	Full	normal	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	1000
5	LSAM	Full	normal	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	normal	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	normal	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000
6	LSAM	Full	normal	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	normal	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	normal	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
7	LSAM	Full	right-skewed	normal	normal	400	0.4	1000	100.0	881	54	6.1	676
	QML	Full	right-skewed	normal	normal	400	0.4	940	94.0	881	30	3.4	676
	UPI	Full	right-skewed	normal	normal	400	0.4	925	92.5	881	164	18.6	676
8	LSAM	Full	right-skewed	normal	normal	1000	0.4	1000	100.0	986	13	1.3	945
	QML	Full	right-skewed	normal	normal	1000	0.4	993	99.3	986	2	0.2	945
	UPI	Full	right-skewed	normal	normal	1000	0.4	991	99.1	986	31	3.1	945

Table C2

Convergence and Outlier Statistics (Simulation 2) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
9	LSAM	Full	right-skewed	normal	normal	400	0.6	1000	100.0	998	13	1.3	972
	QML	Full	right-skewed	normal	normal	400	0.6	1000	100.0	998	2	0.2	972
	UPI	Full	right-skewed	normal	normal	400	0.6	998	99.8	998	16	1.6	972
10	LSAM	Full	right-skewed	normal	normal	1000	0.6	1000	100.0	1000	6	0.6	990
	QML	Full	right-skewed	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	990
	UPI	Full	right-skewed	normal	normal	1000	0.6	1000	100.0	1000	8	0.8	990
11	LSAM	Full	right-skewed	normal	normal	400	0.8	1000	100.0	1000	6	0.6	990
	QML	Full	right-skewed	normal	normal	400	0.8	1000	100.0	1000	5	0.5	990
	UPI	Full	right-skewed	normal	normal	400	0.8	1000	100.0	1000	8	0.8	990
12	LSAM	Full	right-skewed	normal	normal	1000	0.8	1000	100.0	1000	1	0.1	999
	QML	Full	right-skewed	normal	normal	1000	0.8	1000	100.0	1000	1	0.1	999
	UPI	Full	right-skewed	normal	normal	1000	0.8	1000	100.0	1000	1	0.1	999
13	LSAM	Full	uniform	normal	normal	400	0.4	1000	100.0	487	28	5.8	265
	QML	Full	uniform	normal	normal	400	0.4	797	79.7	487	23	4.7	265
	UPI	Full	uniform	normal	normal	400	0.4	596	59.6	487	189	38.8	265
14	LSAM	Full	uniform	normal	normal	1000	0.4	1000	100.0	881	16	1.8	730
	QML	Full	uniform	normal	normal	1000	0.4	965	96.5	881	11	1.2	730
	UPI	Full	uniform	normal	normal	1000	0.4	909	90.9	881	131	14.9	730
15	LSAM	Full	uniform	normal	normal	400	0.6	1000	100.0	978	4	0.4	947
	QML	Full	uniform	normal	normal	400	0.6	1000	100.0	978	1	0.1	947
	UPI	Full	uniform	normal	normal	400	0.6	978	97.8	978	29	3.0	947
16	LSAM	Full	uniform	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	999
	QML	Full	uniform	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	999
	UPI	Full	uniform	normal	normal	1000	0.6	1000	100.0	1000	1	0.1	999
17	LSAM	Full	uniform	normal	normal	400	0.8	1000	100.0	1000	0	0.0	999
	QML	Full	uniform	normal	normal	400	0.8	1000	100.0	1000	0	0.0	999
	UPI	Full	uniform	normal	normal	400	0.8	1000	100.0	1000	1	0.1	999
18	LSAM	Full	uniform	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	uniform	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	uniform	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
19	LSAM	Full	normal	right-skewed	normal	400	0.4	1000	100.0	164	3	1.8	104
	QML	Full	normal	right-skewed	normal	400	0.4	635	63.5	164	9	5.5	104
	UPI	Full	normal	right-skewed	normal	400	0.4	208	20.8	164	52	31.7	104
20	LSAM	Full	normal	right-skewed	normal	1000	0.4	1000	100.0	175	2	1.1	123
	QML	Full	normal	right-skewed	normal	1000	0.4	842	84.2	175	4	2.3	123
	UPI	Full	normal	right-skewed	normal	1000	0.4	190	19.0	175	46	26.3	123

Table C2

Convergence and Outlier Statistics (Simulation 2) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
21	LSAM	Full	normal	right-skewed	normal	400	0.6	1000	100.0	995	0	0.0	989
	QML	Full	normal	right-skewed	normal	400	0.6	999	99.9	995	0	0.0	989
	UPI	Full	normal	right-skewed	normal	400	0.6	996	99.6	995	6	0.6	989
22	LSAM	Full	normal	right-skewed	normal	1000	0.6	1000	100.0	1000	1	0.1	998
	QML	Full	normal	right-skewed	normal	1000	0.6	1000	100.0	1000	1	0.1	998
	UPI	Full	normal	right-skewed	normal	1000	0.6	1000	100.0	1000	1	0.1	998
23	LSAM	Full	normal	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	normal	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	normal	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
24	LSAM	Full	normal	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	normal	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	normal	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
25	LSAM	Full	right-skewed	right-skewed	normal	400	0.4	1000	100.0	891	58	6.5	615
	QML	Full	right-skewed	right-skewed	normal	400	0.4	939	93.9	891	15	1.7	615
	UPI	Full	right-skewed	right-skewed	normal	400	0.4	939	93.9	891	236	26.5	615
26	LSAM	Full	right-skewed	right-skewed	normal	1000	0.4	1000	100.0	980	37	3.8	852
	QML	Full	right-skewed	right-skewed	normal	1000	0.4	991	99.1	980	0	0.0	852
	UPI	Full	right-skewed	right-skewed	normal	1000	0.4	987	98.7	980	96	9.8	852
27	LSAM	Full	right-skewed	right-skewed	normal	400	0.6	1000	100.0	992	19	1.9	914
	QML	Full	right-skewed	right-skewed	normal	400	0.6	993	99.3	992	8	0.8	914
	UPI	Full	right-skewed	right-skewed	normal	400	0.6	999	99.9	992	69	7.0	914
28	LSAM	Full	right-skewed	right-skewed	normal	1000	0.6	1000	100.0	998	0	0.0	992
	QML	Full	right-skewed	right-skewed	normal	1000	0.6	998	99.8	998	2	0.2	992
	UPI	Full	right-skewed	right-skewed	normal	1000	0.6	1000	100.0	998	4	0.4	992
29	LSAM	Full	right-skewed	right-skewed	normal	400	0.8	1000	100.0	1000	2	0.2	997
	QML	Full	right-skewed	right-skewed	normal	400	0.8	1000	100.0	1000	1	0.1	997
	UPI	Full	right-skewed	right-skewed	normal	400	0.8	1000	100.0	1000	2	0.2	997
30	LSAM	Full	right-skewed	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	999
	QML	Full	right-skewed	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	999
	UPI	Full	right-skewed	right-skewed	normal	1000	0.8	1000	100.0	1000	1	0.1	999
31	LSAM	Full	uniform	right-skewed	normal	400	0.4	1000	100.0	31	2	6.4	15
	QML	Full	uniform	right-skewed	normal	400	0.4	401	40.1	31	4	12.9	15
	UPI	Full	uniform	right-skewed	normal	400	0.4	48	4.8	31	14	45.2	15
32	LSAM	Full	uniform	right-skewed	normal	1000	0.4	1000	100.0	22	2	9.1	12
	QML	Full	uniform	right-skewed	normal	1000	0.4	481	48.1	22	2	9.1	12
	UPI	Full	uniform	right-skewed	normal	1000	0.4	36	3.6	22	7	31.8	12

Table C2

Convergence and Outlier Statistics (Simulation 2) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
33	LSAM	Full	uniform	right-skewed	normal	400	0.6	1000	100.0	679	4	0.6	592
	QML	Full	uniform	right-skewed	normal	400	0.6	956	95.6	679	5	0.7	592
	UPI	Full	uniform	right-skewed	normal	400	0.6	699	69.9	679	79	11.6	592
34	LSAM	Full	uniform	right-skewed	normal	1000	0.6	1000	100.0	718	0	0.0	667
	QML	Full	uniform	right-skewed	normal	1000	0.6	992	99.2	718	2	0.3	667
	UPI	Full	uniform	right-skewed	normal	1000	0.6	724	72.4	718	49	6.8	667
35	LSAM	Full	uniform	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	999
	QML	Full	uniform	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	999
	UPI	Full	uniform	right-skewed	normal	400	0.8	1000	100.0	1000	1	0.1	999
36	LSAM	Full	uniform	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	uniform	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	uniform	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
37	LSAM	Full	normal	normal	right-skewed	400	0.4	1000	100.0	869	27	3.1	650
	QML	Full	normal	normal	right-skewed	400	0.4	939	93.9	869	27	3.1	650
	UPI	Full	normal	normal	right-skewed	400	0.4	925	92.5	869	192	22.1	650
38	LSAM	Full	normal	normal	right-skewed	1000	0.4	1000	100.0	999	4	0.4	971
	QML	Full	normal	normal	right-skewed	1000	0.4	999	99.9	999	3	0.3	971
	UPI	Full	normal	normal	right-skewed	1000	0.4	1000	100.0	999	23	2.3	971
39	LSAM	Full	normal	normal	right-skewed	400	0.6	1000	100.0	1000	4	0.4	988
	QML	Full	normal	normal	right-skewed	400	0.6	1000	100.0	1000	2	0.2	988
	UPI	Full	normal	normal	right-skewed	400	0.6	1000	100.0	1000	12	1.2	988
40	LSAM	Full	normal	normal	right-skewed	1000	0.6	1000	100.0	1000	0	0.0	1000
	QML	Full	normal	normal	right-skewed	1000	0.6	1000	100.0	1000	0	0.0	1000
	UPI	Full	normal	normal	right-skewed	1000	0.6	1000	100.0	1000	0	0.0	1000
41	LSAM	Full	normal	normal	right-skewed	400	0.8	1000	100.0	1000	1	0.1	998
	QML	Full	normal	normal	right-skewed	400	0.8	1000	100.0	1000	0	0.0	998
	UPI	Full	normal	normal	right-skewed	400	0.8	1000	100.0	1000	2	0.2	998
42	LSAM	Full	normal	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	normal	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	normal	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
43	LSAM	Full	right-skewed	normal	right-skewed	400	0.4	1000	100.0	870	41	4.7	644
	QML	Full	right-skewed	normal	right-skewed	400	0.4	932	93.2	870	28	3.2	644
	UPI	Full	right-skewed	normal	right-skewed	400	0.4	917	91.7	870	193	22.2	644
44	LSAM	Full	right-skewed	normal	right-skewed	1000	0.4	1000	100.0	985	9	0.9	933
	QML	Full	right-skewed	normal	right-skewed	1000	0.4	995	99.5	985	3	0.3	933
	UPI	Full	right-skewed	normal	right-skewed	1000	0.4	988	98.8	985	44	4.5	933

Table C2

Convergence and Outlier Statistics (Simulation 2) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
45	LSAM	Full	right-skewed	normal	right-skewed	400	0.6	1000	100.0	998	13	1.3	953
	QML	Full	right-skewed	normal	right-skewed	400	0.6	1000	100.0	998	11	1.1	953
	UPI	Full	right-skewed	normal	right-skewed	400	0.6	998	99.8	998	33	3.3	953
46	LSAM	Full	right-skewed	normal	right-skewed	1000	0.6	1000	100.0	1000	1	0.1	997
	QML	Full	right-skewed	normal	right-skewed	1000	0.6	1000	100.0	1000	1	0.1	997
	UPI	Full	right-skewed	normal	right-skewed	1000	0.6	1000	100.0	1000	1	0.1	997
47	LSAM	Full	right-skewed	normal	right-skewed	400	0.8	1000	100.0	1000	8	0.8	984
	QML	Full	right-skewed	normal	right-skewed	400	0.8	1000	100.0	1000	10	1.0	984
	UPI	Full	right-skewed	normal	right-skewed	400	0.8	1000	100.0	1000	8	0.8	984
48	LSAM	Full	right-skewed	normal	right-skewed	1000	0.8	1000	100.0	1000	1	0.1	996
	QML	Full	right-skewed	normal	right-skewed	1000	0.8	1000	100.0	1000	1	0.1	996
	UPI	Full	right-skewed	normal	right-skewed	1000	0.8	1000	100.0	1000	3	0.3	996
49	LSAM	Full	uniform	normal	right-skewed	400	0.4	1000	100.0	485	26	5.4	256
	QML	Full	uniform	normal	right-skewed	400	0.4	764	76.4	485	36	7.4	256
	UPI	Full	uniform	normal	right-skewed	400	0.4	596	59.6	485	201	41.4	256
50	LSAM	Full	uniform	normal	right-skewed	1000	0.4	1000	100.0	876	13	1.5	694
	QML	Full	uniform	normal	right-skewed	1000	0.4	957	95.7	876	17	1.9	694
	UPI	Full	uniform	normal	right-skewed	1000	0.4	916	91.6	876	169	19.3	694
51	LSAM	Full	uniform	normal	right-skewed	400	0.6	1000	100.0	979	2	0.2	930
	QML	Full	uniform	normal	right-skewed	400	0.6	995	99.5	979	3	0.3	930
	UPI	Full	uniform	normal	right-skewed	400	0.6	983	98.3	979	45	4.6	930
52	LSAM	Full	uniform	normal	right-skewed	1000	0.6	1000	100.0	1000	0	0.0	996
	QML	Full	uniform	normal	right-skewed	1000	0.6	1000	100.0	1000	0	0.0	996
	UPI	Full	uniform	normal	right-skewed	1000	0.6	1000	100.0	1000	4	0.4	996
53	LSAM	Full	uniform	normal	right-skewed	400	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	uniform	normal	right-skewed	400	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	uniform	normal	right-skewed	400	0.8	1000	100.0	1000	0	0.0	1000
54	LSAM	Full	uniform	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	999
	QML	Full	uniform	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	999
	UPI	Full	uniform	normal	right-skewed	1000	0.8	1000	100.0	1000	1	0.1	999
55	LSAM	Full	normal	right-skewed	right-skewed	400	0.4	1000	100.0	177	9	5.1	99
	QML	Full	normal	right-skewed	right-skewed	400	0.4	650	65.0	177	14	7.9	99
	UPI	Full	normal	right-skewed	right-skewed	400	0.4	217	21.7	177	73	41.2	99
56	LSAM	Full	normal	right-skewed	right-skewed	1000	0.4	1000	100.0	183	1	0.6	130
	QML	Full	normal	right-skewed	right-skewed	1000	0.4	824	82.4	183	8	4.4	130
	UPI	Full	normal	right-skewed	right-skewed	1000	0.4	197	19.7	183	47	25.7	130

Table C2

Convergence and Outlier Statistics (Simulation 2) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
57	LSAM	Full	normal	right-skewed	right-skewed	400	0.6	1000	100.0	995	7	0.7	971
	QML	Full	normal	right-skewed	right-skewed	400	0.6	997	99.7	995	8	0.8	971
	UPI	Full	normal	right-skewed	right-skewed	400	0.6	998	99.8	995	12	1.2	971
58	LSAM	Full	normal	right-skewed	right-skewed	1000	0.6	1000	100.0	1000	1	0.1	997
	QML	Full	normal	right-skewed	right-skewed	1000	0.6	1000	100.0	1000	3	0.3	997
	UPI	Full	normal	right-skewed	right-skewed	1000	0.6	1000	100.0	1000	0	0.0	997
59	LSAM	Full	normal	right-skewed	right-skewed	400	0.8	1000	100.0	1000	1	0.1	996
	QML	Full	normal	right-skewed	right-skewed	400	0.8	1000	100.0	1000	3	0.3	996
	UPI	Full	normal	right-skewed	right-skewed	400	0.8	1000	100.0	1000	2	0.2	996
60	LSAM	Full	normal	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	normal	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	normal	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
61	LSAM	Full	right-skewed	right-skewed	right-skewed	400	0.4	1000	100.0	900	55	6.1	622
	QML	Full	right-skewed	right-skewed	right-skewed	400	0.4	956	95.6	900	17	1.9	622
	UPI	Full	right-skewed	right-skewed	right-skewed	400	0.4	930	93.0	900	233	25.9	622
62	LSAM	Full	right-skewed	right-skewed	right-skewed	1000	0.4	1000	100.0	988	44	4.4	852
	QML	Full	right-skewed	right-skewed	right-skewed	1000	0.4	997	99.7	988	4	0.4	852
	UPI	Full	right-skewed	right-skewed	right-skewed	1000	0.4	991	99.1	988	93	9.4	852
63	LSAM	Full	right-skewed	right-skewed	right-skewed	400	0.6	1000	100.0	989	20	2.0	905
	QML	Full	right-skewed	right-skewed	right-skewed	400	0.6	992	99.2	989	14	1.4	905
	UPI	Full	right-skewed	right-skewed	right-skewed	400	0.6	997	99.7	989	67	6.8	905
64	LSAM	Full	right-skewed	right-skewed	right-skewed	1000	0.6	1000	100.0	1000	5	0.5	976
	QML	Full	right-skewed	right-skewed	right-skewed	1000	0.6	1000	100.0	1000	7	0.7	976
	UPI	Full	right-skewed	right-skewed	right-skewed	1000	0.6	1000	100.0	1000	16	1.6	976
65	LSAM	Full	right-skewed	right-skewed	right-skewed	400	0.8	1000	100.0	1000	6	0.6	988
	QML	Full	right-skewed	right-skewed	right-skewed	400	0.8	1000	100.0	1000	6	0.6	988
	UPI	Full	right-skewed	right-skewed	right-skewed	400	0.8	1000	100.0	1000	9	0.9	988
66	LSAM	Full	right-skewed	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	3	0.3	996
	QML	Full	right-skewed	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	4	0.4	996
	UPI	Full	right-skewed	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	2	0.2	996
67	LSAM	Full	uniform	right-skewed	right-skewed	400	0.4	1000	100.0	28	8	28.6	5
	QML	Full	uniform	right-skewed	right-skewed	400	0.4	394	39.4	28	6	21.4	5
	UPI	Full	uniform	right-skewed	right-skewed	400	0.4	50	5.0	28	14	50.0	5
68	LSAM	Full	uniform	right-skewed	right-skewed	1000	0.4	1000	100.0	34	1	2.9	11
	QML	Full	uniform	right-skewed	right-skewed	1000	0.4	489	48.9	34	6	17.6	11
	UPI	Full	uniform	right-skewed	right-skewed	1000	0.4	44	4.4	34	21	61.8	11

Table C2

Convergence and Outlier Statistics (Simulation 2) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
69	LSAM	Full	uniform	right-skewed	right-skewed	400	0.6	1000	100.0	665	5	0.8	545
	QML	Full	uniform	right-skewed	right-skewed	400	0.6	943	94.3	665	9	1.4	545
	UPI	Full	uniform	right-skewed	right-skewed	400	0.6	699	69.9	665	113	17.0	545
70	LSAM	Full	uniform	right-skewed	right-skewed	1000	0.6	1000	100.0	712	1	0.1	633
	QML	Full	uniform	right-skewed	right-skewed	1000	0.6	986	98.6	712	3	0.4	633
	UPI	Full	uniform	right-skewed	right-skewed	1000	0.6	718	71.8	712	75	10.5	633
71	LSAM	Full	uniform	right-skewed	right-skewed	400	0.8	1000	100.0	1000	1	0.1	999
	QML	Full	uniform	right-skewed	right-skewed	400	0.8	1000	100.0	1000	1	0.1	999
	UPI	Full	uniform	right-skewed	right-skewed	400	0.8	1000	100.0	1000	1	0.1	999
72	LSAM	Full	uniform	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Full	uniform	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Full	uniform	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
73	LSAM	Linear	normal	normal	normal	400	0.4	1000	100.0	879	44	5.0	692
	QML	Linear	normal	normal	normal	400	0.4	945	94.5	879	15	1.7	692
	UPI	Linear	normal	normal	normal	400	0.4	926	92.6	879	162	18.4	692
74	LSAM	Linear	normal	normal	normal	1000	0.4	1000	100.0	997	4	0.4	982
	QML	Linear	normal	normal	normal	1000	0.4	998	99.8	997	1	0.1	982
	UPI	Linear	normal	normal	normal	1000	0.4	999	99.9	997	13	1.3	982
75	LSAM	Linear	normal	normal	normal	400	0.6	1000	100.0	1000	2	0.2	993
	QML	Linear	normal	normal	normal	400	0.6	1000	100.0	1000	1	0.1	993
	UPI	Linear	normal	normal	normal	400	0.6	1000	100.0	1000	7	0.7	993
76	LSAM	Linear	normal	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	1000
	QML	Linear	normal	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	1000
	UPI	Linear	normal	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	1000
77	LSAM	Linear	normal	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	normal	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	normal	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000
78	LSAM	Linear	normal	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	normal	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	normal	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
79	LSAM	Linear	right-skewed	normal	normal	400	0.4	1000	100.0	811	61	7.5	559
	QML	Linear	right-skewed	normal	normal	400	0.4	914	91.4	811	20	2.5	559
	UPI	Linear	right-skewed	normal	normal	400	0.4	866	86.6	811	220	27.1	559
80	LSAM	Linear	right-skewed	normal	normal	1000	0.4	1000	100.0	982	20	2.0	934
	QML	Linear	right-skewed	normal	normal	1000	0.4	996	99.6	982	2	0.2	934
	UPI	Linear	right-skewed	normal	normal	1000	0.4	985	98.5	982	40	4.1	934

Table C2

Convergence and Outlier Statistics (Simulation 2) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
81	LSAM	Linear	right-skewed	normal	normal	400	0.6	1000	100.0	992	16	1.6	955
	QML	Linear	right-skewed	normal	normal	400	0.6	998	99.8	992	5	0.5	955
	UPI	Linear	right-skewed	normal	normal	400	0.6	994	99.4	992	31	3.1	955
82	LSAM	Linear	right-skewed	normal	normal	1000	0.6	1000	100.0	999	0	0.0	996
	QML	Linear	right-skewed	normal	normal	1000	0.6	1000	100.0	999	1	0.1	996
	UPI	Linear	right-skewed	normal	normal	1000	0.6	999	99.9	999	2	0.2	996
83	LSAM	Linear	right-skewed	normal	normal	400	0.8	1000	100.0	1000	2	0.2	994
	QML	Linear	right-skewed	normal	normal	400	0.8	1000	100.0	1000	1	0.1	994
	UPI	Linear	right-skewed	normal	normal	400	0.8	1000	100.0	1000	4	0.4	994
84	LSAM	Linear	right-skewed	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	999
	QML	Linear	right-skewed	normal	normal	1000	0.8	1000	100.0	1000	1	0.1	999
	UPI	Linear	right-skewed	normal	normal	1000	0.8	1000	100.0	1000	1	0.1	999
85	LSAM	Linear	uniform	normal	normal	400	0.4	1000	100.0	500	18	3.6	299
	QML	Linear	uniform	normal	normal	400	0.4	809	80.9	500	30	6.0	299
	UPI	Linear	uniform	normal	normal	400	0.4	594	59.4	500	181	36.2	299
86	LSAM	Linear	uniform	normal	normal	1000	0.4	1000	100.0	873	19	2.2	728
	QML	Linear	uniform	normal	normal	1000	0.4	951	95.1	873	7	0.8	728
	UPI	Linear	uniform	normal	normal	1000	0.4	914	91.4	873	134	15.3	728
87	LSAM	Linear	uniform	normal	normal	400	0.6	1000	100.0	980	3	0.3	961
	QML	Linear	uniform	normal	normal	400	0.6	996	99.6	980	2	0.2	961
	UPI	Linear	uniform	normal	normal	400	0.6	984	98.4	980	17	1.7	961
88	LSAM	Linear	uniform	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	999
	QML	Linear	uniform	normal	normal	1000	0.6	1000	100.0	1000	0	0.0	999
	UPI	Linear	uniform	normal	normal	1000	0.6	1000	100.0	1000	1	0.1	999
89	LSAM	Linear	uniform	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	uniform	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	uniform	normal	normal	400	0.8	1000	100.0	1000	0	0.0	1000
90	LSAM	Linear	uniform	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	uniform	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	uniform	normal	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
91	LSAM	Linear	normal	right-skewed	normal	400	0.4	1000	100.0	184	4	2.2	110
	QML	Linear	normal	right-skewed	normal	400	0.4	615	61.5	184	13	7.1	110
	UPI	Linear	normal	right-skewed	normal	400	0.4	245	24.5	184	65	35.3	110
92	LSAM	Linear	normal	right-skewed	normal	1000	0.4	1000	100.0	216	2	0.9	150
	QML	Linear	normal	right-skewed	normal	1000	0.4	836	83.6	216	8	3.7	150
	UPI	Linear	normal	right-skewed	normal	1000	0.4	243	24.3	216	61	28.2	150

Table C2

Convergence and Outlier Statistics (Simulation 2) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
93	LSAM	Linear	normal	right-skewed	normal	400	0.6	1000	100.0	998	3	0.3	989
	QML	Linear	normal	right-skewed	normal	400	0.6	999	99.9	998	0	0.0	989
	UPI	Linear	normal	right-skewed	normal	400	0.6	999	99.9	998	8	0.8	989
94	LSAM	Linear	normal	right-skewed	normal	1000	0.6	1000	100.0	1000	1	0.1	999
	QML	Linear	normal	right-skewed	normal	1000	0.6	1000	100.0	1000	0	0.0	999
	UPI	Linear	normal	right-skewed	normal	1000	0.6	1000	100.0	1000	1	0.1	999
95	LSAM	Linear	normal	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	normal	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	normal	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
96	LSAM	Linear	normal	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	normal	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	normal	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
97	LSAM	Linear	right-skewed	right-skewed	normal	400	0.4	1000	100.0	787	62	7.9	473
	QML	Linear	right-skewed	right-skewed	normal	400	0.4	894	89.4	787	27	3.4	473
	UPI	Linear	right-skewed	right-skewed	normal	400	0.4	859	85.9	787	261	33.2	473
98	LSAM	Linear	right-skewed	right-skewed	normal	1000	0.4	1000	100.0	965	94	9.7	727
	QML	Linear	right-skewed	right-skewed	normal	1000	0.4	992	99.2	965	12	1.2	727
	UPI	Linear	right-skewed	right-skewed	normal	1000	0.4	971	97.1	965	148	15.3	727
99	LSAM	Linear	right-skewed	right-skewed	normal	400	0.6	1000	100.0	973	19	2.0	884
	QML	Linear	right-skewed	right-skewed	normal	400	0.6	979	97.9	973	5	0.5	884
	UPI	Linear	right-skewed	right-skewed	normal	400	0.6	994	99.4	973	80	8.2	884
100	LSAM	Linear	right-skewed	right-skewed	normal	1000	0.6	1000	100.0	997	7	0.7	974
	QML	Linear	right-skewed	right-skewed	normal	1000	0.6	997	99.7	997	2	0.2	974
	UPI	Linear	right-skewed	right-skewed	normal	1000	0.6	1000	100.0	997	21	2.1	974
101	LSAM	Linear	right-skewed	right-skewed	normal	400	0.8	1000	100.0	1000	10	1.0	987
	QML	Linear	right-skewed	right-skewed	normal	400	0.8	1000	100.0	1000	4	0.4	987
	UPI	Linear	right-skewed	right-skewed	normal	400	0.8	1000	100.0	1000	7	0.7	987
102	LSAM	Linear	right-skewed	right-skewed	normal	1000	0.8	1000	100.0	1000	1	0.1	999
	QML	Linear	right-skewed	right-skewed	normal	1000	0.8	1000	100.0	1000	1	0.1	999
	UPI	Linear	right-skewed	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	999
103	LSAM	Linear	uniform	right-skewed	normal	400	0.4	1000	100.0	28	2	7.1	13
	QML	Linear	uniform	right-skewed	normal	400	0.4	403	40.3	28	5	17.9	13
	UPI	Linear	uniform	right-skewed	normal	400	0.4	42	4.2	28	12	42.9	13
104	LSAM	Linear	uniform	right-skewed	normal	1000	0.4	1000	100.0	28	0	0.0	10
	QML	Linear	uniform	right-skewed	normal	1000	0.4	505	50.5	28	5	17.9	10
	UPI	Linear	uniform	right-skewed	normal	1000	0.4	43	4.3	28	16	57.1	10

Table C2

Convergence and Outlier Statistics (Simulation 2) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
105	LSAM	Linear	uniform	right-skewed	normal	400	0.6	1000	100.0	682	1	0.1	595
	QML	Linear	uniform	right-skewed	normal	400	0.6	955	95.5	682	3	0.4	595
	UPI	Linear	uniform	right-skewed	normal	400	0.6	712	71.2	682	84	12.3	595
106	LSAM	Linear	uniform	right-skewed	normal	1000	0.6	1000	100.0	753	0	0.0	679
	QML	Linear	uniform	right-skewed	normal	1000	0.6	991	99.1	753	1	0.1	679
	UPI	Linear	uniform	right-skewed	normal	1000	0.6	759	75.9	753	74	9.8	679
107	LSAM	Linear	uniform	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	uniform	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	uniform	right-skewed	normal	400	0.8	1000	100.0	1000	0	0.0	1000
108	LSAM	Linear	uniform	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	uniform	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	uniform	right-skewed	normal	1000	0.8	1000	100.0	1000	0	0.0	1000
109	LSAM	Linear	normal	normal	right-skewed	400	0.4	1000	100.0	840	50	6.0	582
	QML	Linear	normal	normal	right-skewed	400	0.4	915	91.5	840	26	3.1	582
	UPI	Linear	normal	normal	right-skewed	400	0.4	902	90.2	840	226	26.9	582
110	LSAM	Linear	normal	normal	right-skewed	1000	0.4	1000	100.0	998	6	0.6	967
	QML	Linear	normal	normal	right-skewed	1000	0.4	998	99.8	998	1	0.1	967
	UPI	Linear	normal	normal	right-skewed	1000	0.4	1000	100.0	998	28	2.8	967
111	LSAM	Linear	normal	normal	right-skewed	400	0.6	1000	100.0	1000	2	0.2	994
	QML	Linear	normal	normal	right-skewed	400	0.6	1000	100.0	1000	2	0.2	994
	UPI	Linear	normal	normal	right-skewed	400	0.6	1000	100.0	1000	3	0.3	994
112	LSAM	Linear	normal	normal	right-skewed	1000	0.6	1000	100.0	1000	3	0.3	997
	QML	Linear	normal	normal	right-skewed	1000	0.6	1000	100.0	1000	3	0.3	997
	UPI	Linear	normal	normal	right-skewed	1000	0.6	1000	100.0	1000	1	0.1	997
113	LSAM	Linear	normal	normal	right-skewed	400	0.8	1000	100.0	1000	1	0.1	996
	QML	Linear	normal	normal	right-skewed	400	0.8	1000	100.0	1000	4	0.4	996
	UPI	Linear	normal	normal	right-skewed	400	0.8	1000	100.0	1000	2	0.2	996
114	LSAM	Linear	normal	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	normal	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	normal	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
115	LSAM	Linear	right-skewed	normal	right-skewed	400	0.4	1000	100.0	784	62	7.9	489
	QML	Linear	right-skewed	normal	right-skewed	400	0.4	893	89.3	784	39	5.0	489
	UPI	Linear	right-skewed	normal	right-skewed	400	0.4	862	86.2	784	252	32.1	489
116	LSAM	Linear	right-skewed	normal	right-skewed	1000	0.4	1000	100.0	983	18	1.8	908
	QML	Linear	right-skewed	normal	right-skewed	1000	0.4	996	99.6	983	9	0.9	908
	UPI	Linear	right-skewed	normal	right-skewed	1000	0.4	987	98.7	983	62	6.3	908

Table C2

Convergence and Outlier Statistics (Simulation 2) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
117	LSAM	Linear	right-skewed	normal	right-skewed	400	0.6	1000	100.0	992	27	2.7	923
	QML	Linear	right-skewed	normal	right-skewed	400	0.6	997	99.7	992	21	2.1	923
	UPI	Linear	right-skewed	normal	right-skewed	400	0.6	994	99.4	992	47	4.7	923
118	LSAM	Linear	right-skewed	normal	right-skewed	1000	0.6	1000	100.0	1000	5	0.5	988
	QML	Linear	right-skewed	normal	right-skewed	1000	0.6	1000	100.0	1000	3	0.3	988
	UPI	Linear	right-skewed	normal	right-skewed	1000	0.6	1000	100.0	1000	8	0.8	988
119	LSAM	Linear	right-skewed	normal	right-skewed	400	0.8	1000	100.0	1000	12	1.2	978
	QML	Linear	right-skewed	normal	right-skewed	400	0.8	1000	100.0	1000	15	1.5	978
	UPI	Linear	right-skewed	normal	right-skewed	400	0.8	1000	100.0	1000	13	1.3	978
120	LSAM	Linear	right-skewed	normal	right-skewed	1000	0.8	1000	100.0	1000	2	0.2	998
	QML	Linear	right-skewed	normal	right-skewed	1000	0.8	1000	100.0	1000	1	0.1	998
	UPI	Linear	right-skewed	normal	right-skewed	1000	0.8	1000	100.0	1000	1	0.1	998
121	LSAM	Linear	uniform	normal	right-skewed	400	0.4	1000	100.0	427	27	6.3	247
	QML	Linear	uniform	normal	right-skewed	400	0.4	759	75.9	427	37	8.7	247
	UPI	Linear	uniform	normal	right-skewed	400	0.4	544	54.4	427	154	36.1	247
122	LSAM	Linear	uniform	normal	right-skewed	1000	0.4	1000	100.0	836	17	2.0	658
	QML	Linear	uniform	normal	right-skewed	1000	0.4	944	94.4	836	16	1.9	658
	UPI	Linear	uniform	normal	right-skewed	1000	0.4	883	88.3	836	167	20.0	658
123	LSAM	Linear	uniform	normal	right-skewed	400	0.6	1000	100.0	971	4	0.4	923
	QML	Linear	uniform	normal	right-skewed	400	0.6	990	99.0	971	5	0.5	923
	UPI	Linear	uniform	normal	right-skewed	400	0.6	980	98.0	971	43	4.4	923
124	LSAM	Linear	uniform	normal	right-skewed	1000	0.6	1000	100.0	1000	1	0.1	998
	QML	Linear	uniform	normal	right-skewed	1000	0.6	1000	100.0	1000	0	0.0	998
	UPI	Linear	uniform	normal	right-skewed	1000	0.6	1000	100.0	1000	2	0.2	998
125	LSAM	Linear	uniform	normal	right-skewed	400	0.8	1000	100.0	1000	1	0.1	999
	QML	Linear	uniform	normal	right-skewed	400	0.8	1000	100.0	1000	0	0.0	999
	UPI	Linear	uniform	normal	right-skewed	400	0.8	1000	100.0	1000	1	0.1	999
126	LSAM	Linear	uniform	normal	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	999
	QML	Linear	uniform	normal	right-skewed	1000	0.8	1000	100.0	1000	1	0.1	999
	UPI	Linear	uniform	normal	right-skewed	1000	0.8	1000	100.0	1000	1	0.1	999
127	LSAM	Linear	normal	right-skewed	right-skewed	400	0.4	1000	100.0	175	11	6.3	94
	QML	Linear	normal	right-skewed	right-skewed	400	0.4	607	60.7	175	18	10.3	94
	UPI	Linear	normal	right-skewed	right-skewed	400	0.4	235	23.5	175	67	38.3	94
128	LSAM	Linear	normal	right-skewed	right-skewed	1000	0.4	1000	100.0	168	1	0.6	112
	QML	Linear	normal	right-skewed	right-skewed	1000	0.4	766	76.6	168	6	3.6	112
	UPI	Linear	normal	right-skewed	right-skewed	1000	0.4	194	19.4	168	52	31.0	112

Table C2

Convergence and Outlier Statistics (Simulation 2) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
129	LSAM	Linear	normal	right-skewed	right-skewed	400	0.6	1000	100.0	987	3	0.3	964
	QML	Linear	normal	right-skewed	right-skewed	400	0.6	992	99.2	987	4	0.4	964
	UPI	Linear	normal	right-skewed	right-skewed	400	0.6	995	99.5	987	18	1.8	964
130	LSAM	Linear	normal	right-skewed	right-skewed	1000	0.6	1000	100.0	1000	1	0.1	998
	QML	Linear	normal	right-skewed	right-skewed	1000	0.6	1000	100.0	1000	0	0.0	998
	UPI	Linear	normal	right-skewed	right-skewed	1000	0.6	1000	100.0	1000	2	0.2	998
131	LSAM	Linear	normal	right-skewed	right-skewed	400	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	normal	right-skewed	right-skewed	400	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	normal	right-skewed	right-skewed	400	0.8	1000	100.0	1000	0	0.0	1000
132	LSAM	Linear	normal	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	normal	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	normal	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
133	LSAM	Linear	right-skewed	right-skewed	right-skewed	400	0.4	1000	100.0	779	89	11.4	445
	QML	Linear	right-skewed	right-skewed	right-skewed	400	0.4	890	89.0	779	35	4.5	445
	UPI	Linear	right-skewed	right-skewed	right-skewed	400	0.4	857	85.7	779	261	33.5	445
134	LSAM	Linear	right-skewed	right-skewed	right-skewed	1000	0.4	1000	100.0	957	103	10.8	711
	QML	Linear	right-skewed	right-skewed	right-skewed	1000	0.4	986	98.6	957	17	1.8	711
	UPI	Linear	right-skewed	right-skewed	right-skewed	1000	0.4	971	97.1	957	151	15.8	711
135	LSAM	Linear	right-skewed	right-skewed	right-skewed	400	0.6	1000	100.0	951	30	3.1	801
	QML	Linear	right-skewed	right-skewed	right-skewed	400	0.6	960	96.0	951	27	2.8	801
	UPI	Linear	right-skewed	right-skewed	right-skewed	400	0.6	989	98.9	951	121	12.7	801
136	LSAM	Linear	right-skewed	right-skewed	right-skewed	1000	0.6	1000	100.0	994	2	0.2	971
	QML	Linear	right-skewed	right-skewed	right-skewed	1000	0.6	994	99.4	994	2	0.2	971
	UPI	Linear	right-skewed	right-skewed	right-skewed	1000	0.6	1000	100.0	994	19	1.9	971
137	LSAM	Linear	right-skewed	right-skewed	right-skewed	400	0.8	1000	100.0	1000	12	1.2	986
	QML	Linear	right-skewed	right-skewed	right-skewed	400	0.8	1000	100.0	1000	7	0.7	986
	UPI	Linear	right-skewed	right-skewed	right-skewed	400	0.8	1000	100.0	1000	9	0.9	986
138	LSAM	Linear	right-skewed	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	999
	QML	Linear	right-skewed	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	1	0.1	999
	UPI	Linear	right-skewed	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	1	0.1	999
139	LSAM	Linear	uniform	right-skewed	right-skewed	400	0.4	1000	100.0	35	1	2.9	14
	QML	Linear	uniform	right-skewed	right-skewed	400	0.4	369	36.9	35	4	11.4	14
	UPI	Linear	uniform	right-skewed	right-skewed	400	0.4	52	5.2	35	19	54.3	14
140	LSAM	Linear	uniform	right-skewed	right-skewed	1000	0.4	1000	100.0	28	1	3.6	14
	QML	Linear	uniform	right-skewed	right-skewed	1000	0.4	495	49.5	28	4	14.3	14
	UPI	Linear	uniform	right-skewed	right-skewed	1000	0.4	40	4.0	28	13	46.4	14

Table C2

Convergence and Outlier Statistics (Simulation 2) (continued)

								Convergence			Outliers		
Cond.	Method	Model	Distr.	Epsilon	Zeta	N	Rel.	N Conv.	Conv. Rate	N Global Conv.	N Outliers	Outlier Rate	N Final
141	LSAM	Linear	uniform	right-skewed	right-skewed	400	0.6	1000	100.0	654	2	0.3	540
	QML	Linear	uniform	right-skewed	right-skewed	400	0.6	916	91.6	654	4	0.6	540
	UPI	Linear	uniform	right-skewed	right-skewed	400	0.6	703	70.3	654	112	17.1	540
142	LSAM	Linear	uniform	right-skewed	right-skewed	1000	0.6	1000	100.0	732	3	0.4	632
	QML	Linear	uniform	right-skewed	right-skewed	1000	0.6	974	97.4	732	6	0.8	632
	UPI	Linear	uniform	right-skewed	right-skewed	1000	0.6	752	75.2	732	94	12.8	632
143	LSAM	Linear	uniform	right-skewed	right-skewed	400	0.8	1000	100.0	1000	1	0.1	999
	QML	Linear	uniform	right-skewed	right-skewed	400	0.8	1000	100.0	1000	1	0.1	999
	UPI	Linear	uniform	right-skewed	right-skewed	400	0.8	1000	100.0	1000	1	0.1	999
144	LSAM	Linear	uniform	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	QML	Linear	uniform	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000
	UPI	Linear	uniform	right-skewed	right-skewed	1000	0.8	1000	100.0	1000	0	0.0	1000

Note. Distr. = distribution of exogenous latent variables. Epsilon = distribution of measurement errors. Zeta = distribution of structural residuals. N Conv. = number of replications that converged for this method. Conv. Rate = percentage of replications that converged (%). N Global Conv. = sample size after requiring all methods to converge. N Outliers = number of outliers detected for this method. Outlier Rate = percentage of converged cases flagged as outliers (%). N Final = final sample size after global outlier removal across methods.